

AI for Science in the Era of Large Language Models

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About the Tutor



Xuan Wang, Assistant Professor,
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Research Interests: natural
language processing, data mining, AI
for sciences, and AI for healthcare.

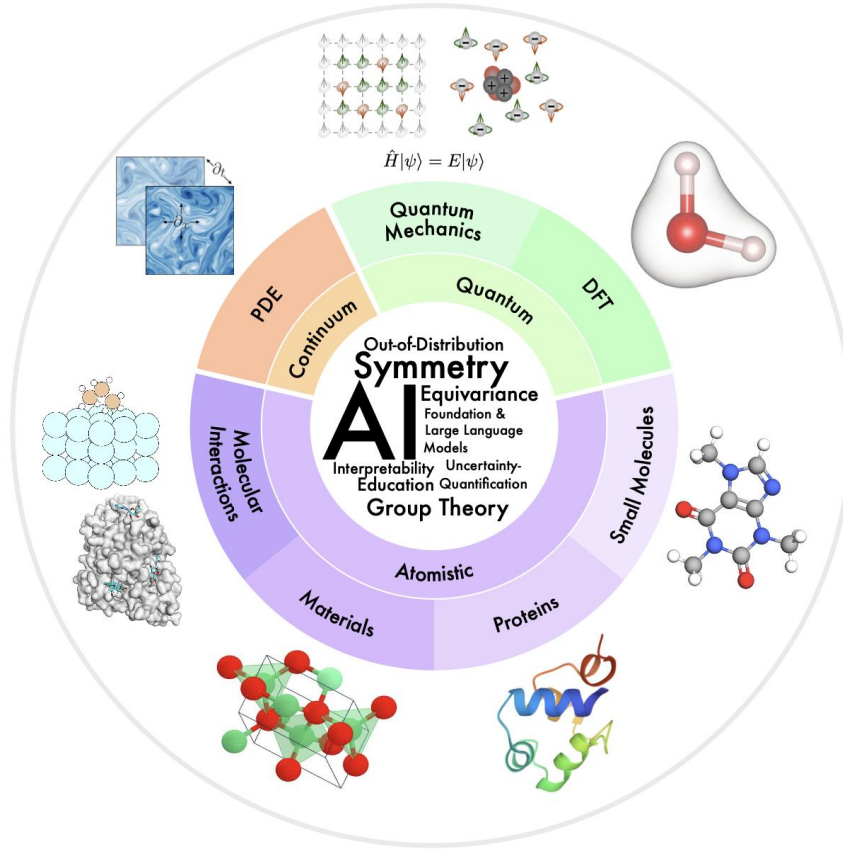
Website:

<https://xuanwang91.github.io/>

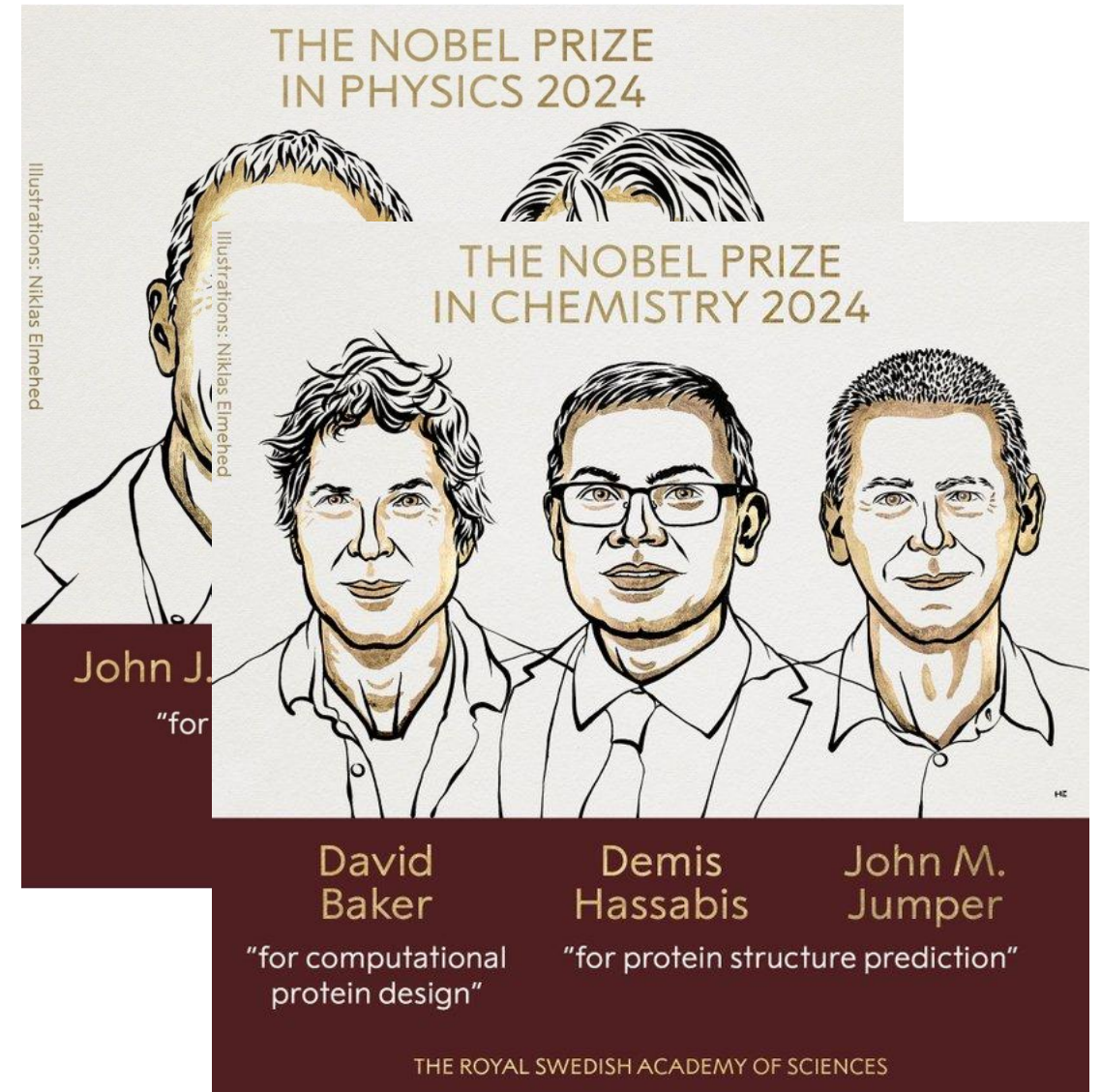
Tutorial Outline

- **8:30 am – 9:00 am: Introduction** 
- **9:00 am – 10:00 am: Part I: Scientific Text**
- **10:00 am – 10:30 am: Part II: Brain Signals**
- **10:30 am – 11:00 am: Break**
- **11:00 am – 12:00 am: Part III: Biological Sequences**
- **12:00 pm – 12:30 pm: Summary and Q&A**

AI for Sciences



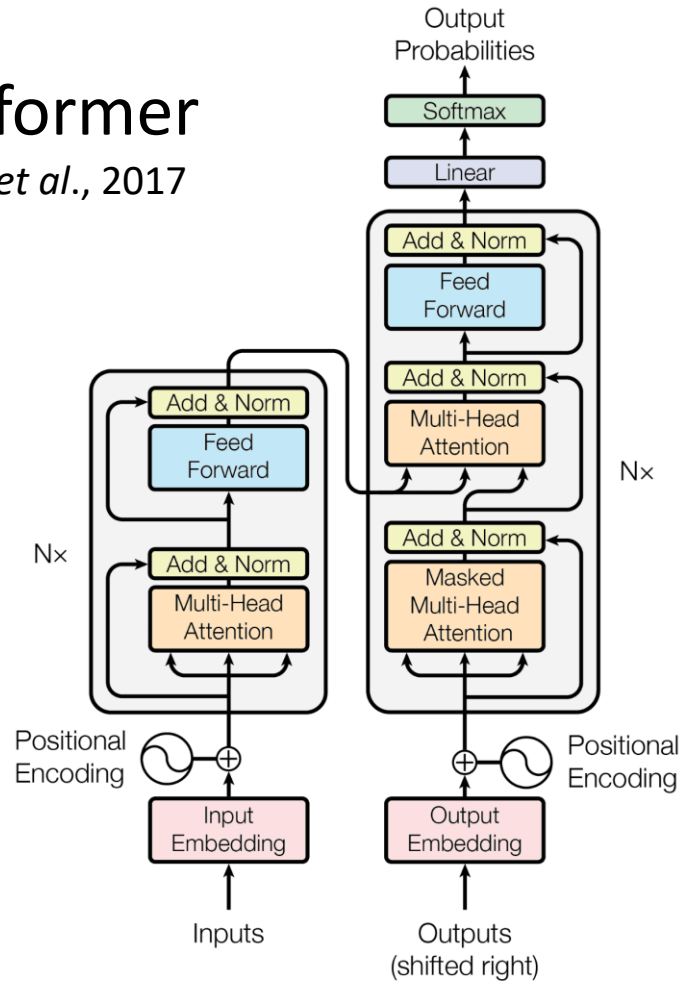
Zhang et al., "Artificial Intelligence for Science in Quantum, Atomistic, and Continuum Systems", *arXiv*, 2023



Large Language Models (LLMs)

Transformer

Vaswani *et al.*, 2017



BERT

Kenton *et al.*, 2019



GPT

Brown *et al.*, 2019



T5

Raffel *et al.*, 2020

Machine Translation

Google Research Philosophy Research Areas **Publications** People Tools & Downloads Outreach Careers Blog

PUBLICATIONS

Google's Neural Machine Translation System: Bridging the Gap between Human and Machine Translation

Yonghui Wu, Mike Schuster, Jeff Klingner, Apurva Nishanik, George Kurian, Nishanth

CoRR, vol. abs/1609.0

[Download](#)

Google Translate

Text Documents

DETECT LANGUAGE ENGLISH SPANISH FRENCH **ENGLISH** SPANISH ARABIC

Search languages

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Albanian	Dutch	Hungarian	Luxembourgish	Russian	Thai
Amharic	English	Icelandic	Macedonian	Samoan	Turkish
Arabic	Esperanto	Igbo	Malagasy	Scots Gaelic	Turkmen
Armenian	Estonian	Indonesian	Malay	Serbian	Ukrainian
Azerbaijani	Filipino	Irish	Malayalam	Sesotho	Urdu
Basque	Finnish	Italian	Maltese	Shona	Uyghur
Belarusian	French	Japanese	Maori	Sindhi	Uzbek
Bengali	Frisian	Javanese	Marathi	Sinhala	Vietnamese
Bosnian	Galician	Kannada	Mongolian	Slovak	Welsh
Bulgarian	Georgian	Kazakh	Myanmar (Burmese)	Slovenian	Xhosa
Catalan	German	Khmer	Nepali	Somali	Yiddish
Cebuano	Greek	Kinyarwanda	Norwegian	Spanish	Yoruba
Chichewa	Gujarati	Korean	Odia (Oriya)	Sundanese	Zulu



Dialog Systems, Chatbots, Digital Assistants



"Hey Siri, what's the best sushi place in town?"

Persona	Persona
I bought my first home. I love to barbecue. I live in Springfield. I'm a writer.	I weight 300 pounds. I am not healthy. I am a man. I like The Godfather.



Hello how are you, I am new to the Springfield area.



Hi! Seen any good movies lately?



I have been to the movies.



I love The Godfather, one of my favorites! Was that filmed?



I don't believe so. I don't watch movies more of a writer.



What do you write? Any diet books ? I am not very healthy.

Get help online

Xfinity Assistant and online communities are always available to help you.



FIND AN XFINITY LOCATION

Check the status and hours of locations near you.

Store Locator

Help Communities

Join our community of users to get help with any questions you have.

Ask Xfinity

Connect with Us

Xfinity Support Forum

Xfinity Assistant

Having internet issues?
I can troubleshoot your connection if you need help.

Having internet issues?

- Manage my account
- Troubleshooting help
- Billing and payments
- Xfinity Privacy Center

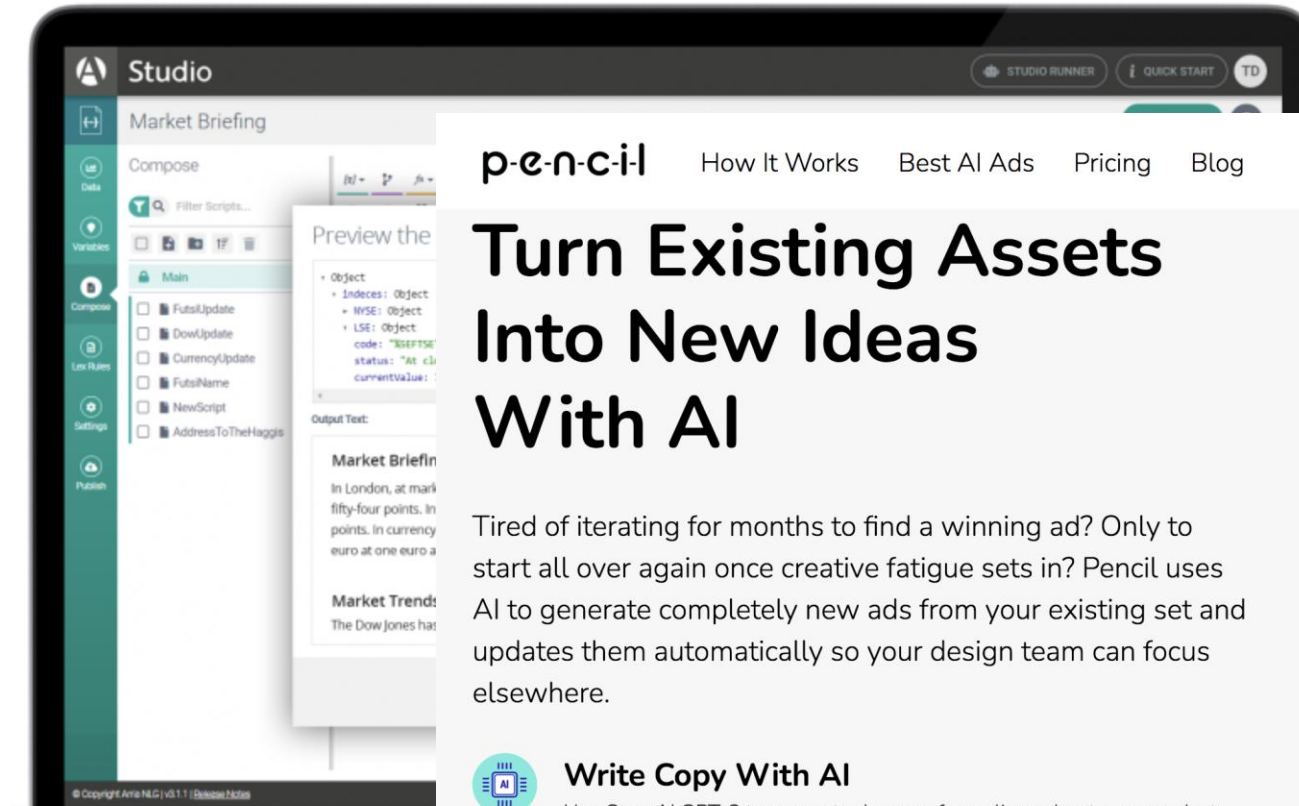
Troubleshooting help

Which Xfinity service do you need help with?

- TV
- Internet
- Something else

Ask a new question...

Natural Language Generation



pencil How It Works Best AI Ads Pricing Blog

Turn Existing Assets Into New Ideas With AI

Tired of iterating for months to find a winning ad? Only to start all over again once creative fatigue sets in? Pencil uses AI to generate completely new ads from your existing set and updates them automatically so your design team can focus elsewhere.



Write Copy With AI

Use OpenAI GPT-3 to generate dozens of new lines about your product



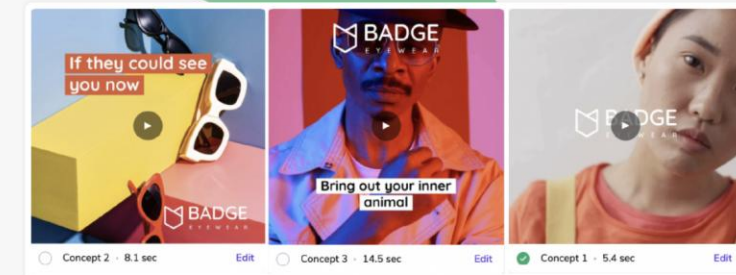
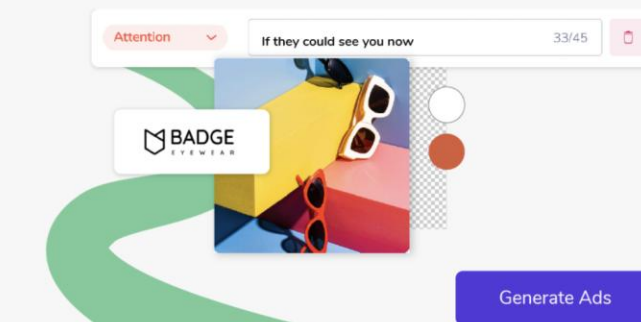
Automate Video Creation

Automatically cut images and video into whole new ad concepts.

Your work email

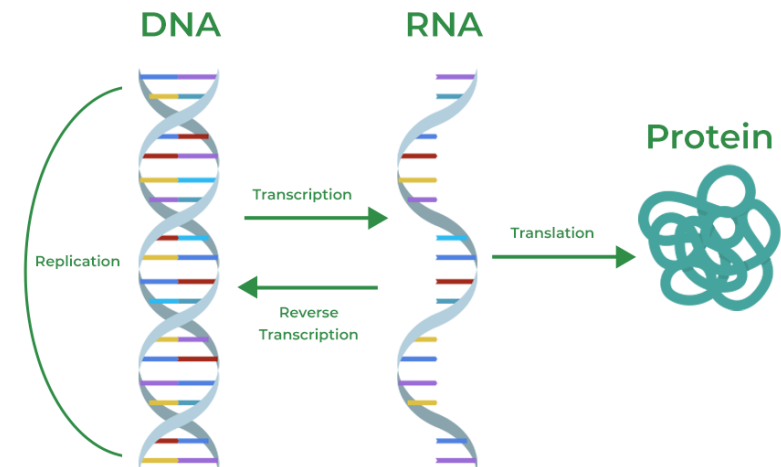
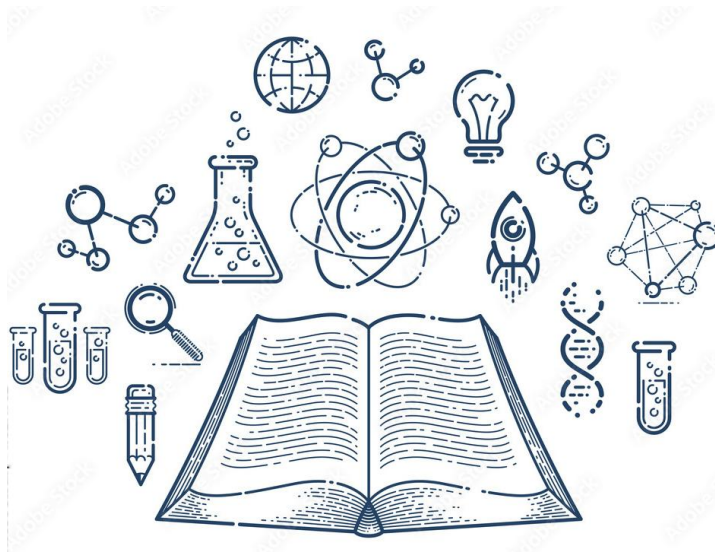
Book Demo

Login



Resemblance Between Scientific Data and Language

- **Sequences!**
- **Scientific Textual Data:** Scientific Literature, Electronic Health Record
- **Sensor Data:** Brain Electroencephalogram (EEG) Signals
- **Biological Sequences:** DNA, RNA, protein



This Tutorial: Can we harness the power of these **recent LLMs** to drive **scientific progress**?

Scientific Large Language Models: Challenges and Opportunities

Xuan Wang

Assistant Professor

Department of Computer Science

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Tutorial Outline

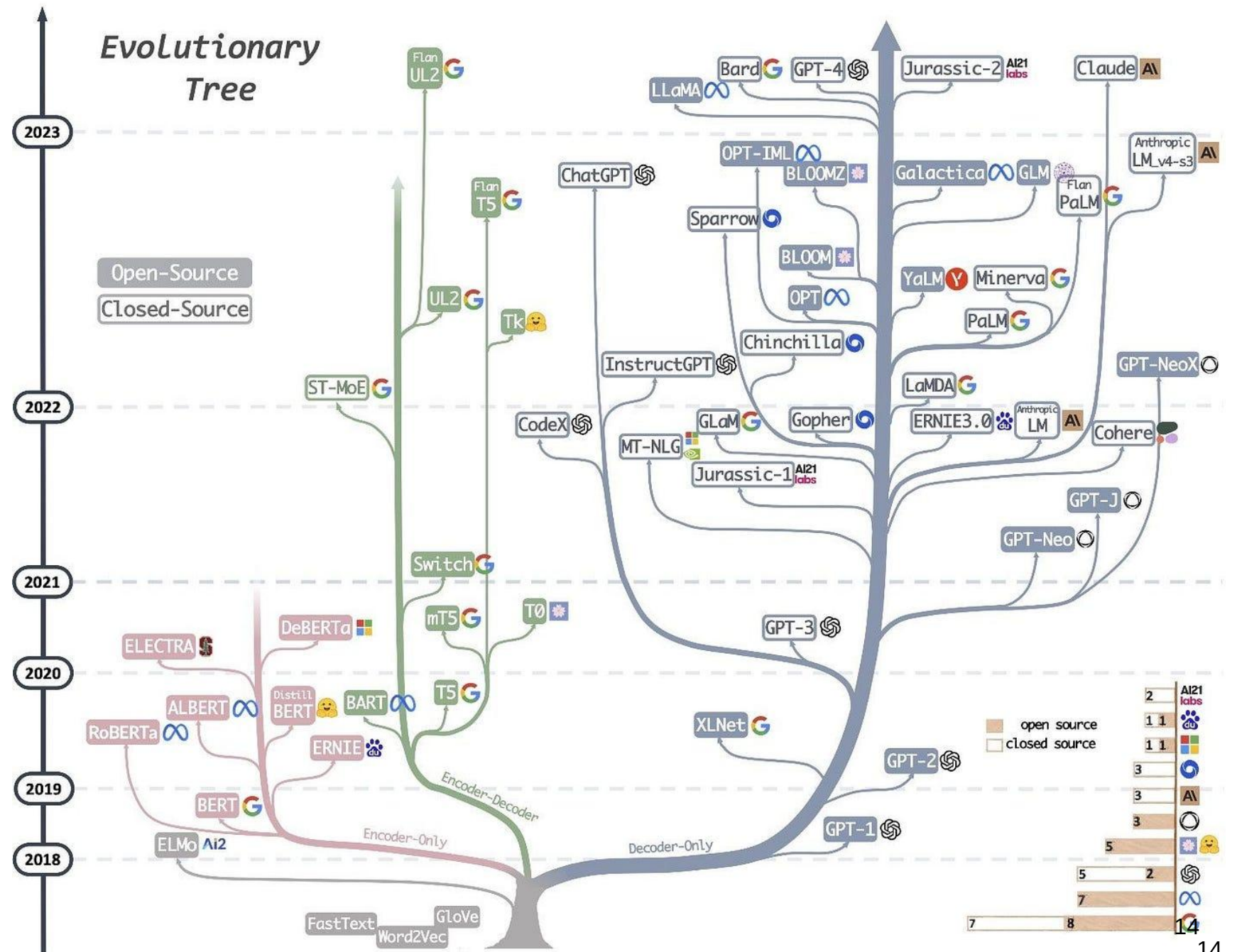
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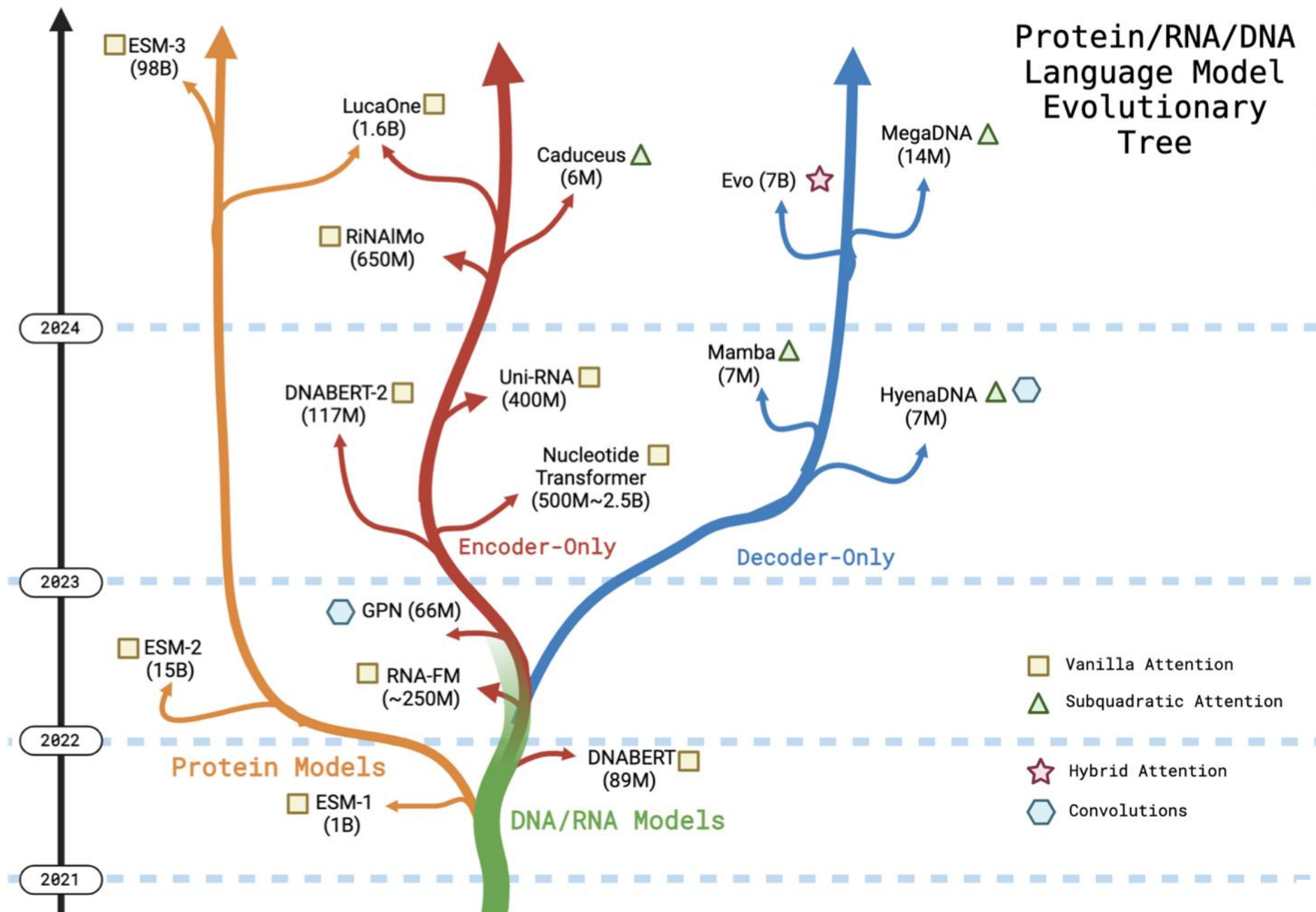
Outline

- **Scientific Large Language Models**
- Future Directions:
 - Complex Reasoning and Planning
 - Multi-modal Learning
 - Trustworthiness of LLMs

Large Language Models (LLMs)

Yang, J., Jin, H., Tang, R., Han, X., Feng, Q., Jiang, H., ... & Hu, X. (2024). Harnessing the Power of LLMs in Practice: A Survey on ChatGPT and Beyond. *ACM Transactions on Knowledge Discovery from Data*, 18(6), 1-32.





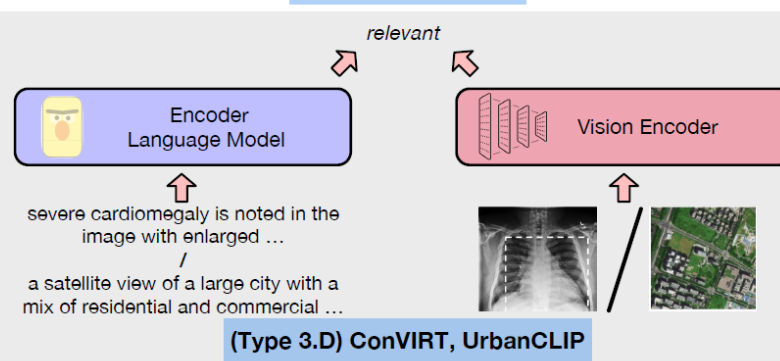
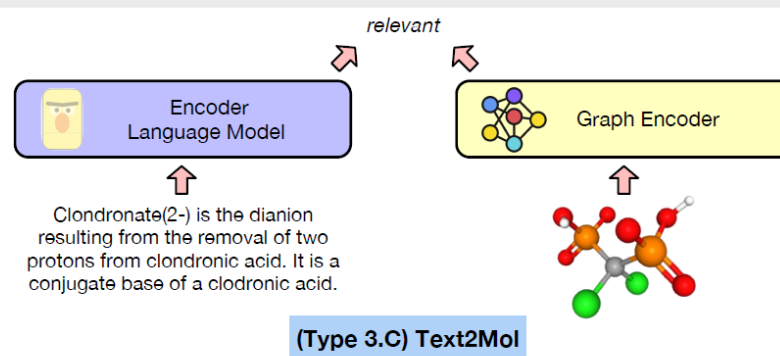
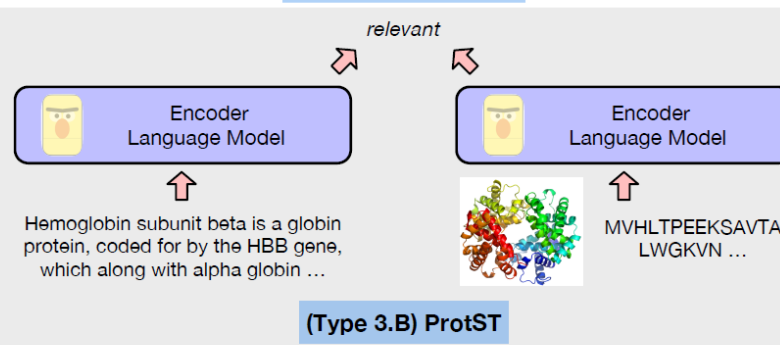
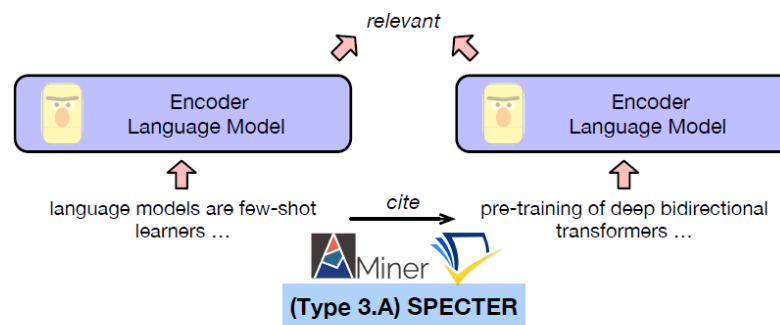
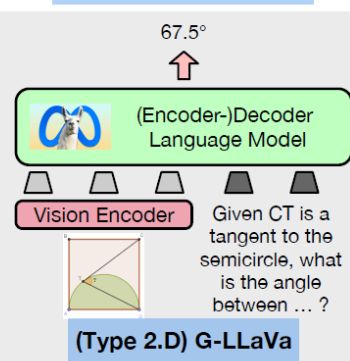
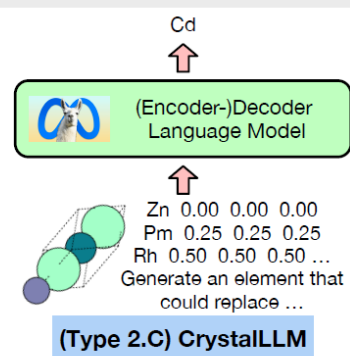
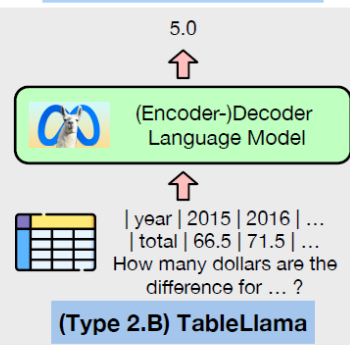
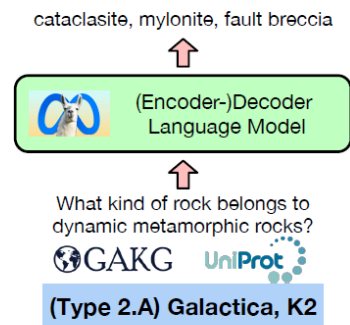
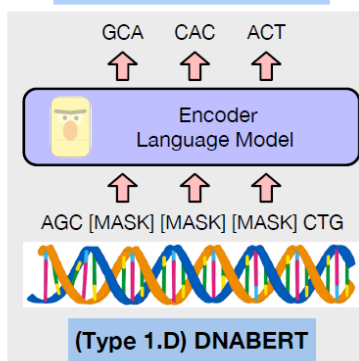
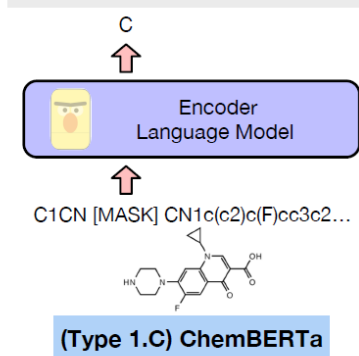
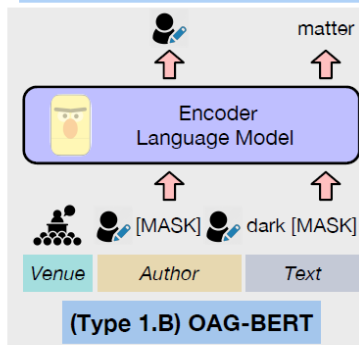
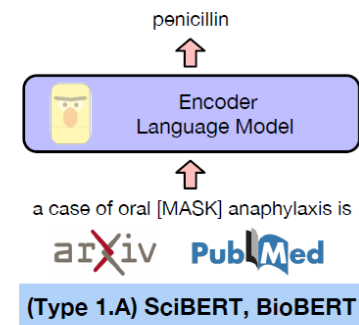
Jian Ma, "Large Language Models in Computational Biology– A Primer (2024 Update)", 2024



Nicholas Ho

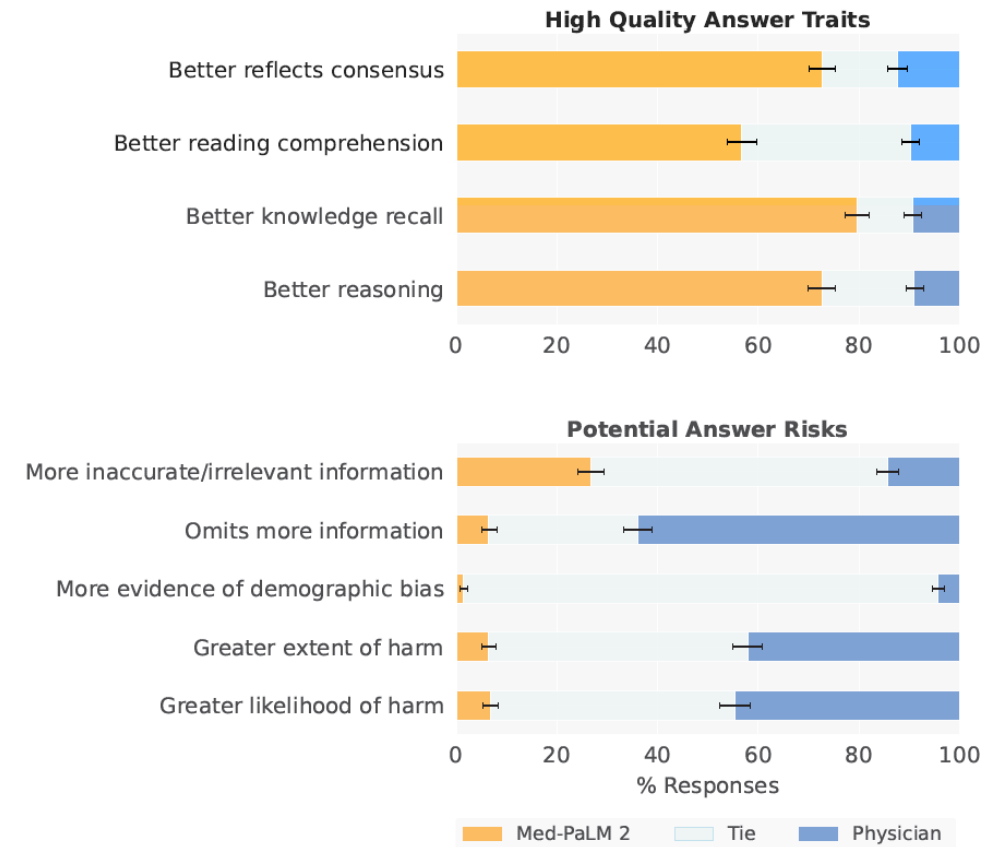
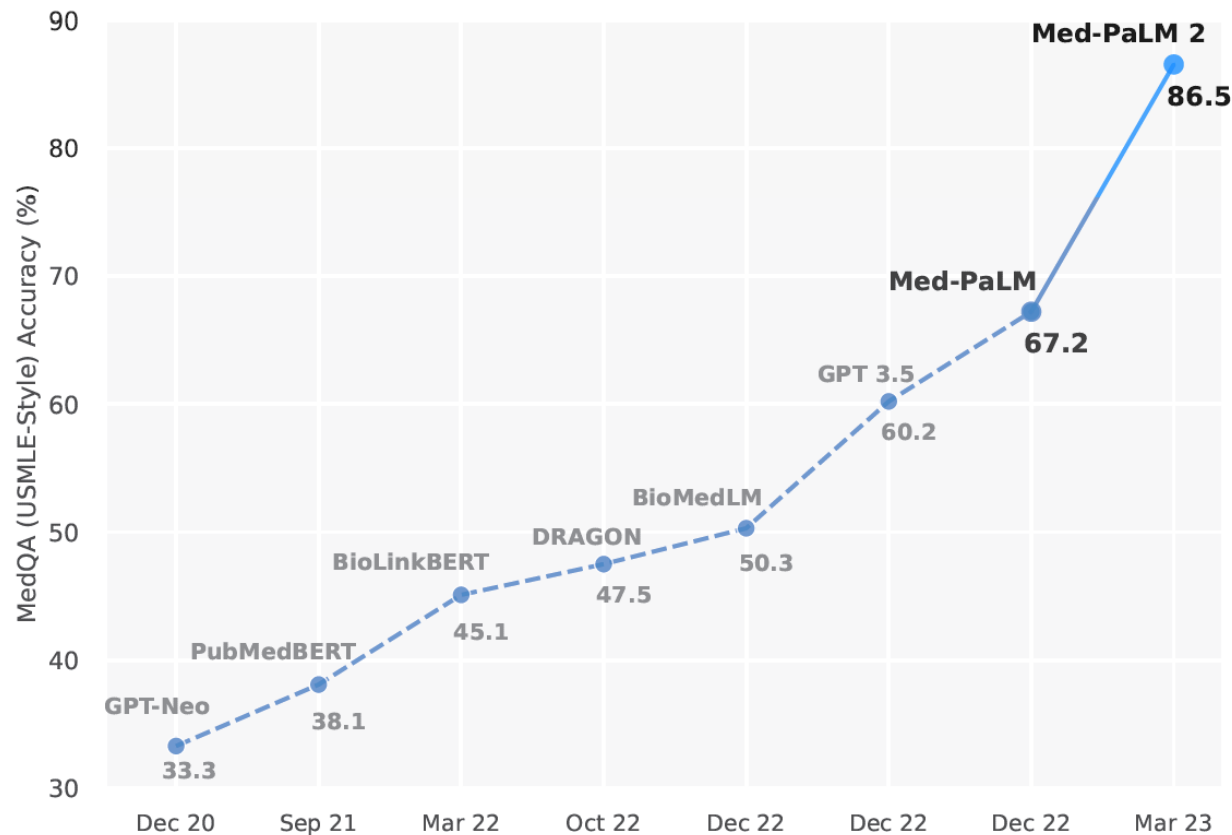
A Comprehensive Survey of Scientific Large Language Models and Their Applications in Scientific Discovery (Zhang *et al.*, EMNLP 2024)

- Survey over **260 scientific LLMs**
- **Across fields:** 1) general science, 2) mathematics, 3) physics, 4) chemistry and material science, 5) biology and medicine, and 6) geography, geology, and environmental science
- **Across modalities:** 1) text, 2) graph, 3) vision, and 4) time series
- **Website:** <https://github.com/yuzhimanhua/Awesome-Scientific-Language-Models>



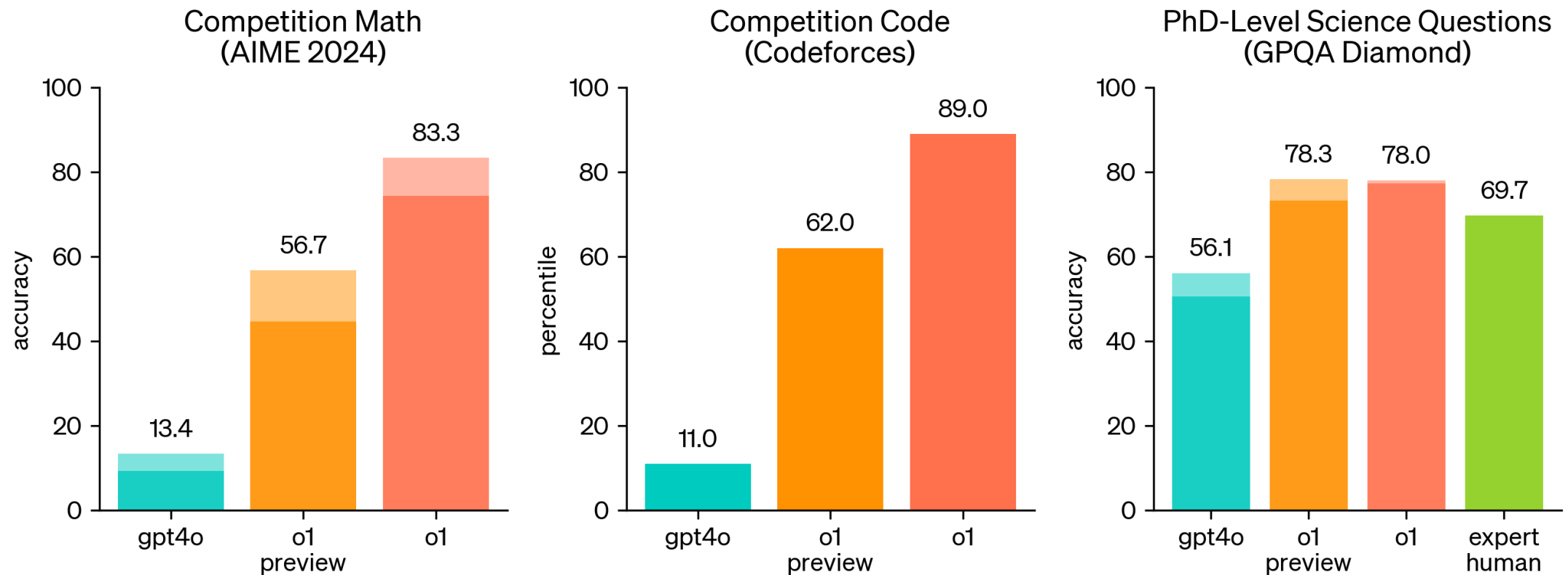
A Comprehensive Survey of Scientific Large Language Models and Their Applications in Scientific Discovery (Zhang et al., EMNLP 2024)

Towards Expert-Level Medical Question Answering with Large Language Models (Med-PaLM-2, Google, 2024)



Singhal, K., Tu, T., Gottweis, J., Sayres, R., Wulczyn, E., Hou, L., ... & Natarajan, V. (2023). Towards expert-level medical question answering with large language models. arXiv preprint arXiv:2305.09617.

OpenAI o1 Surpasses Human Performance on PhD-Level Science Questions (OpenAI, 2024)

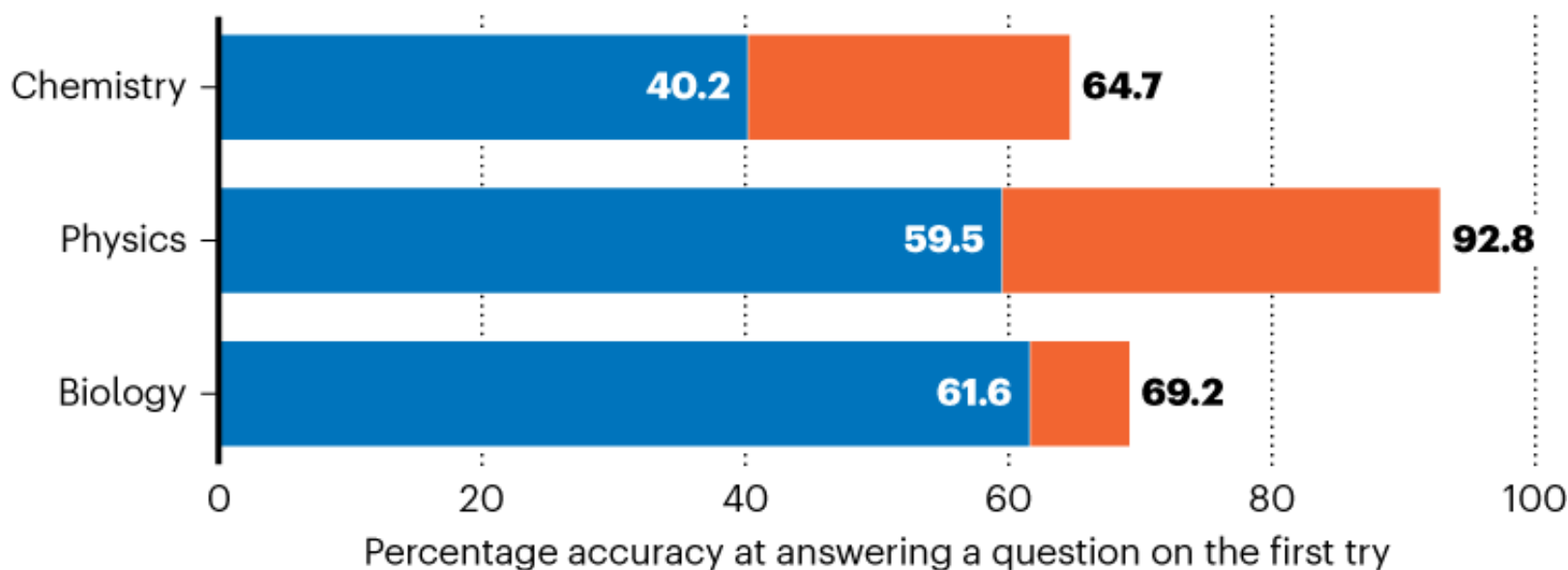


OpenAI o1 Surpasses Human Performance on PhD-Level Science Questions (OpenAI, 2024)

NEXT LEVEL

OpenAI's latest large language model, o1, outperforms its previous best model, GPT-4o, on a standardized test* aimed at PhD-level scholars. When answering the hardest 'diamond' level questions, o1 scored best at physics and least well at chemistry.

■ GPT-4o ■ OpenAI o1



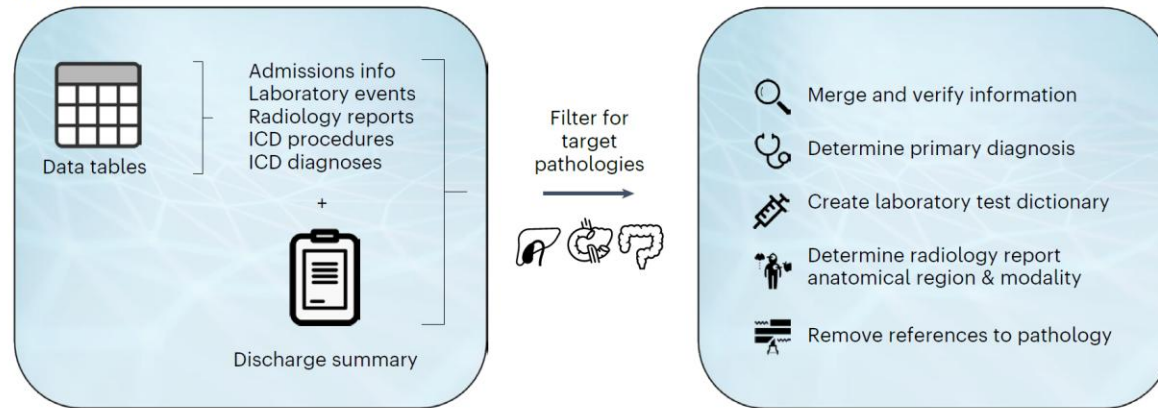
*Graduate-Level Google-Proof Q&A Benchmark test.

Outline

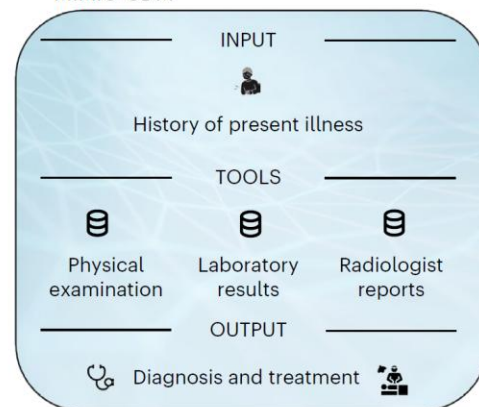
- Scientific Large Language Models
- **Future Directions:**
 - **Complex Reasoning and Planning**
 - Multi-modal Learning
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Evaluation and Mitigation of the Limitations of Large Language Models in Clinical Decision-Making (Hager *et al.*, Nature Medicine 2024)

a MIMIC-IV



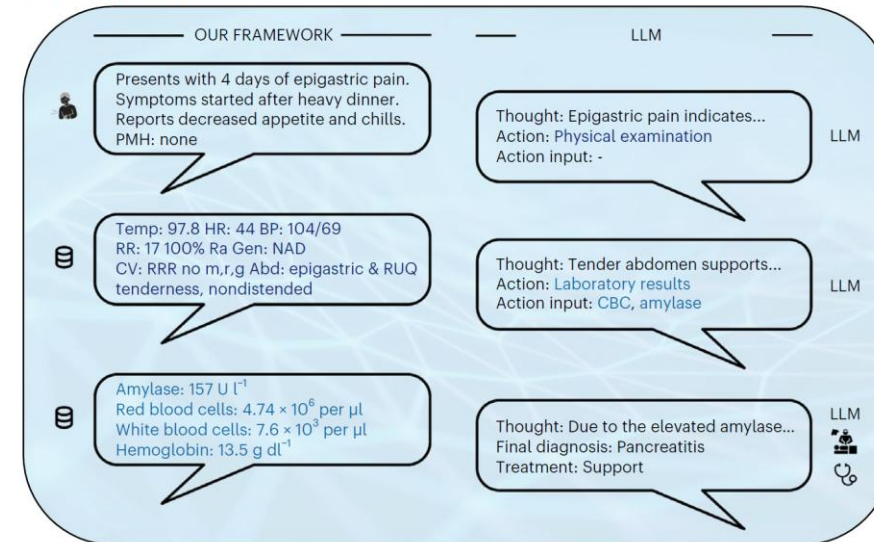
MIMIC-CDM



MIMIC-CDM statistics

2,400 cases			
Appendicitis	957	CT abdomen	1,836
Cholecystitis	648	Radiograph chest	1,728
Diverticulitis	257	US abdomen	1,325
Pancreatitis	538	Radiograph abdomen	342
		MRCP abdomen	227
		Other	501
2,400 physical examinations		143,191 laboratory tests	5,959 radiologist reports

b EVALUATION



- Diagnostic accuracy
- Clinical guidelines
- Instruction following
- Lab test interpretation
- Robustness

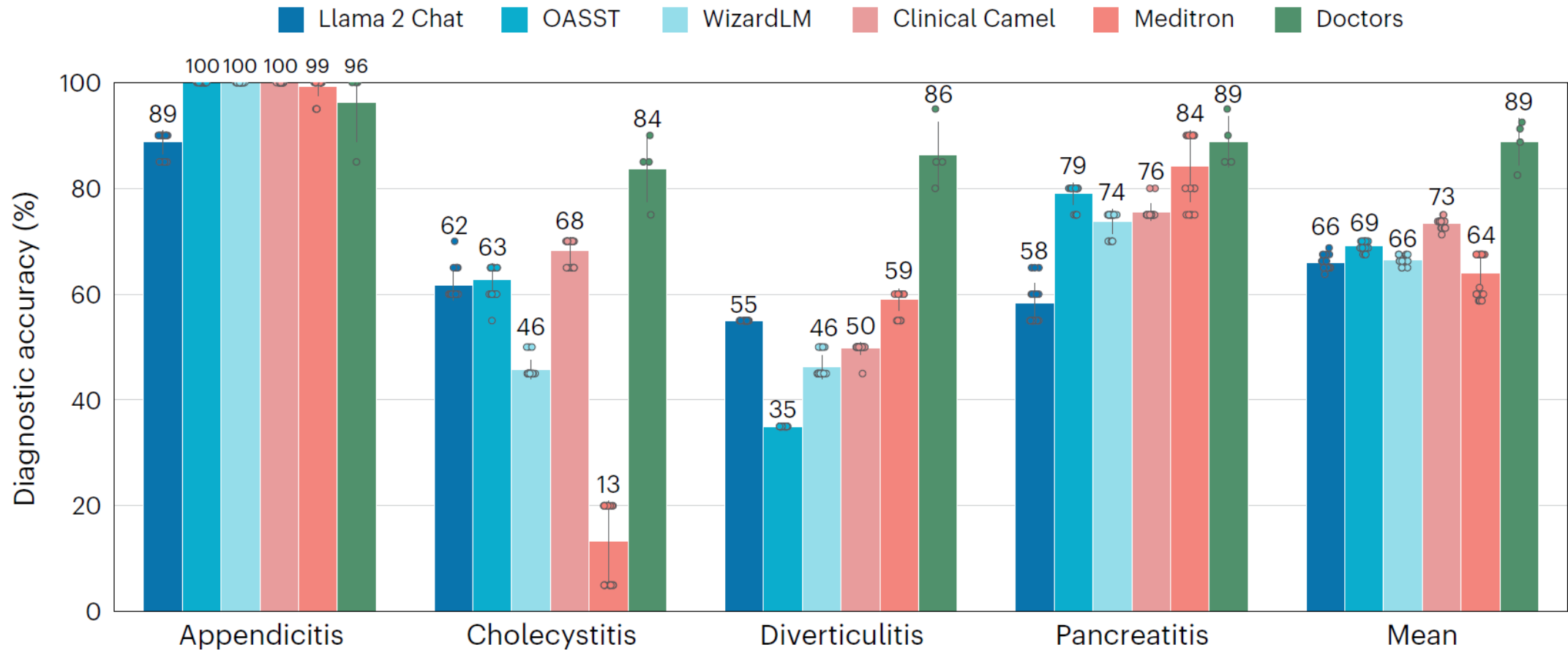
Evaluation and Mitigation of the Limitations of Large Language Models in Clinical Decision-Making (Hager *et al.*, Nature Medicine 2024)

Table 1 | An overview of the considered LLMs and their properties

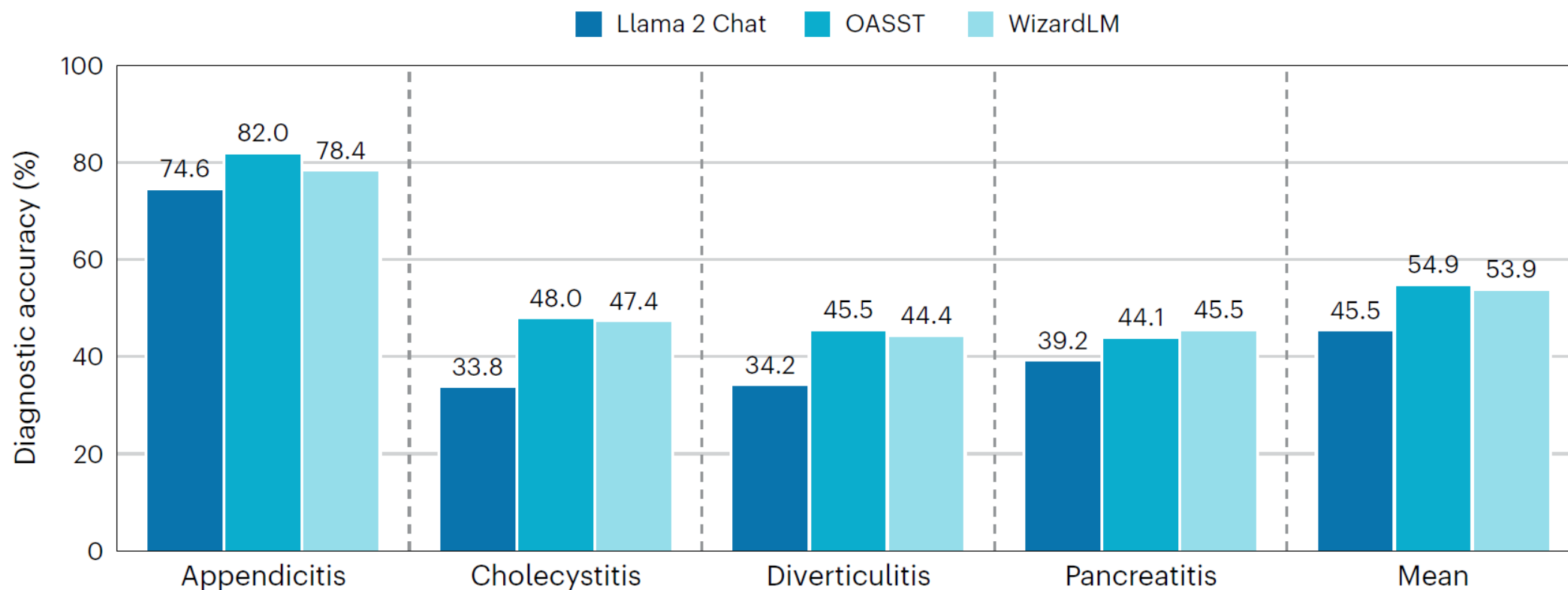
Model	Base	Parameters	Training dataset	Downloadable
Llama 2 Chat ³²	Llama 2 (ref. 32)	70B	Public data ^a	✓
OASST ³³	Llama 2 (ref. 32)	70B	Public data ^a , https://huggingface.co/OpenAssistant/llama2-70b-oasst-sft-v10/ , open-source data	✓
WizardLM ³⁴	Llama 2 (ref. 32)	70B	Public data ^a , Evol-Instruct generated ³⁴	✓
Clinical Camel ¹⁹	Llama 2 (ref. 32)	70B	Public data ^a , https://sharegpt.com/ ; ShareGPT; PubMed articles (before 2021) ¹⁹ , MedQA ¹³	✓
Meditron ³⁵	Llama 2 (ref. 32)	70B	Public data ^a , https://huggingface.co/datasets/epfl-llm/guidelines/ ; clinical guidelines, public PubMed abstracts ³⁵ , public PubMed papers ³⁵ , RedPajama ⁵⁸	✓
Chat-GPT ⁵⁹	GPT3.5 (ref. 60)	???	User conversations ^b , Common Crawl ⁶¹ , WebText2 (ref. 62), Books1 (ref. 63), Books2 (ref. 63), Wikipedia	✗
GPT-4 (ref. 64)	???	???	???	✗
Med-PaLM ⁹	Flan-PaLM ⁶⁵	540B	Webpages ^b , Wikipedia ^b , social media ^b , GitHub ^b , news articles ^b , books ^b , 473 instruction fine-tuning datasets ⁶⁵ , HealthSearchQA ⁹ , MedicationQA ⁶⁶ , LiveQA ⁶⁷	✗
Med-PaLM 2 (ref. 8)	PaLM 2 (ref. 68)	340B	Web Documents ^b , books ^b , code ^b , mathematics ^b , conversational data ^b , MedQA ¹³ , HealthSearchQA ⁹ , MedicationQA ⁶⁶ , LiveQA ⁶⁷	✗

Due to the data usage agreement of MIMIC-IV, only open-access models that can be downloaded can be used with the data; thus, only LLMs based on Llama 2 were used in this study. ??? indicates no information has been made public. ^aMeta defines 'public data' as a 'mix of data from publicly available sources'. ^bNo further information provided.

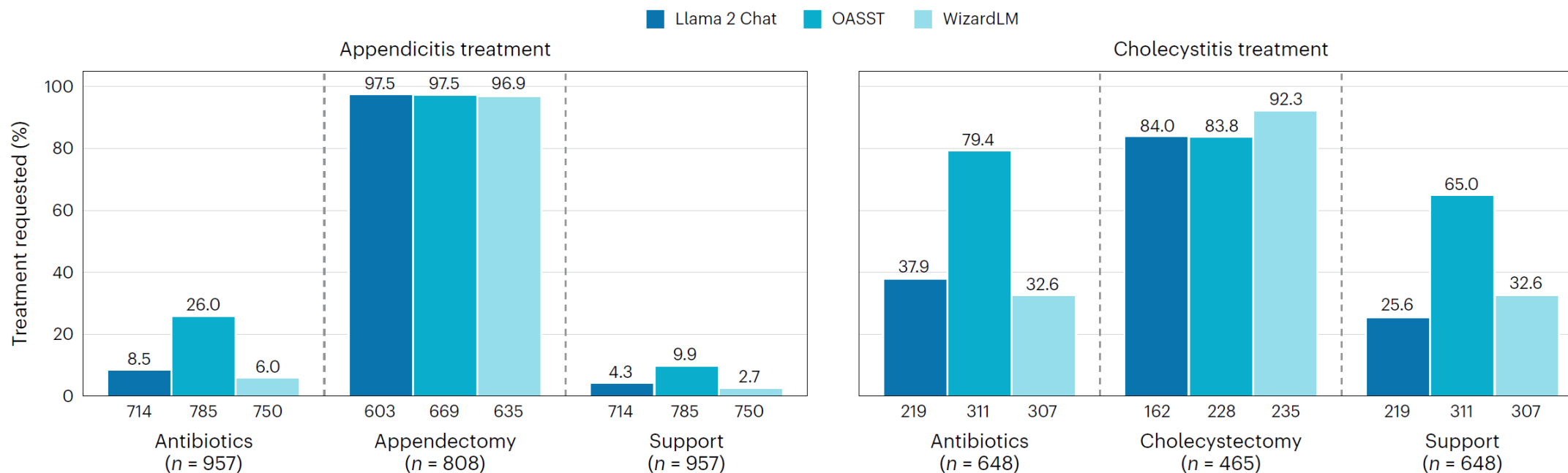
LLMs Diagnose Significantly Worse than Clinicians



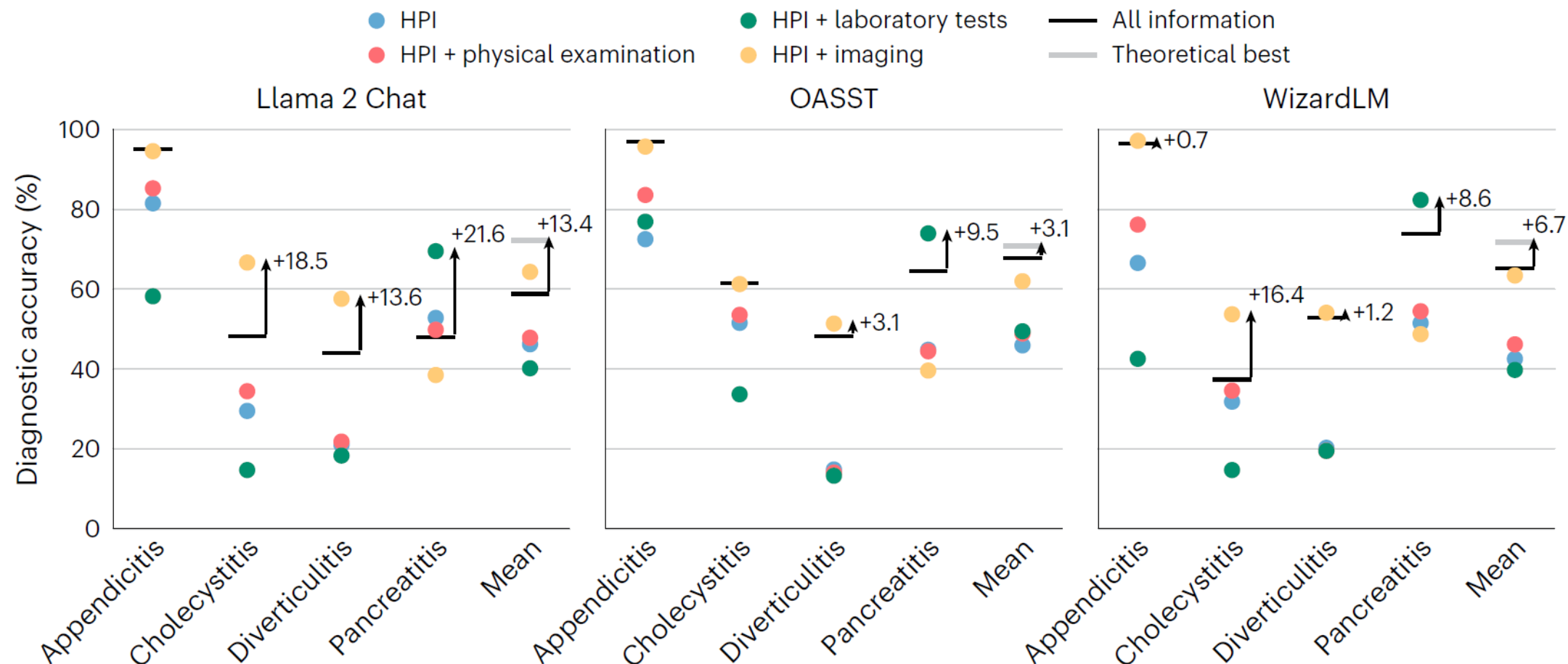
Diagnostic Accuracy of LLMs Decreased in an Autonomous Clinical Decision-Making Scenario



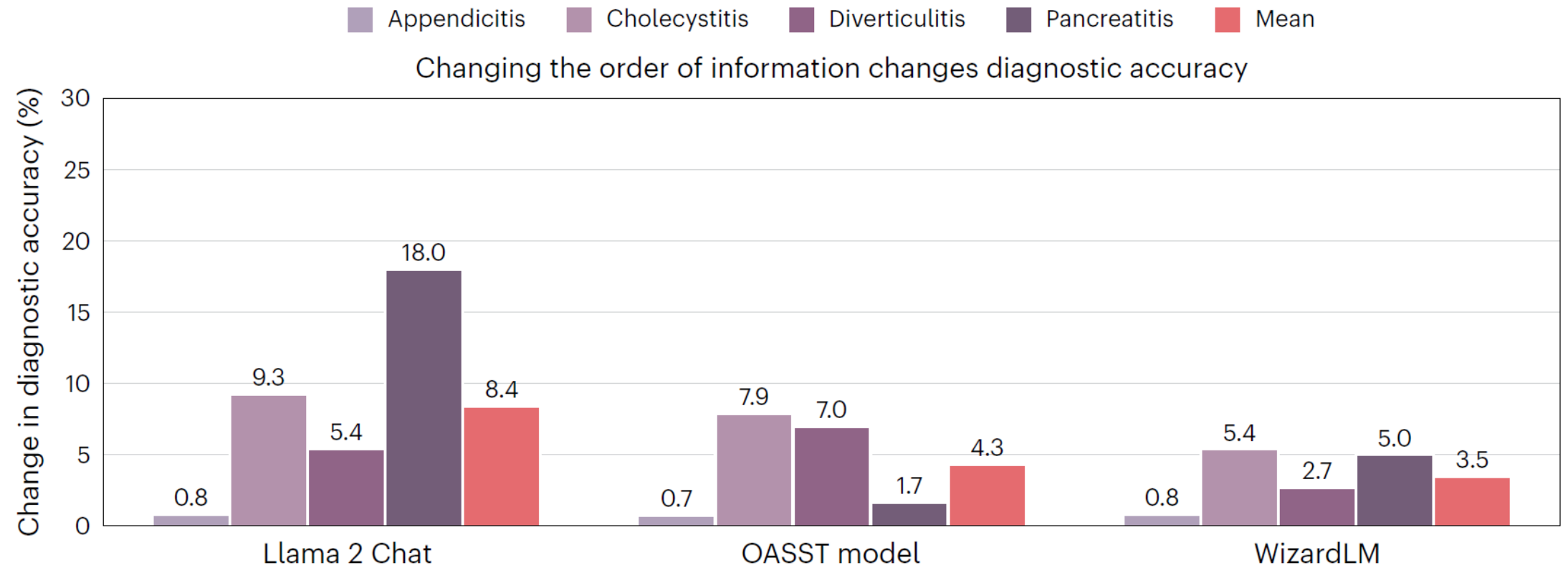
LLMs Do Not Consistently Recommend Essential and Patient-Specific Treatment



LLMs Are Sensitive to the Quantity of Information Provided



LLMs Are Sensitive to the Order of Information



TriageAgent: Towards Better Multi-Agent Collaboration for Large Language Model-Based Clinical Triage

(Lu *et al.*, EMNLP 2024)



Clinical Document: EMS arrives with a 28-year -old male who was stabbed in the left side of his neck during an altercation. You notice a large hematoma around the wound, and the patient is moaning he can't breathe. HR 110, RR 36, SpO2 89%.



External Tools: ESI handbook PubMed Wikipedia

Step 1. Allocating Documents



Doctor A: High ESI Level, with **90% confidence**.
Doctor B: Initial classification: **ESI level 2**
Confidence score: **80%**
After debating, I revised the classification: **ESI level 1**, Revised confidence score: **95%**

Step 2. Group-Based Classification Analysis



Doctor C: Based on the provided clinical record, as an expert medical doctor, **Confidence score** of **ESI2: 90%**. Upon further reflection and considering... **Confidence score after second revision for ESI-1: 95%**

Step 3. Confidence Report Summarization

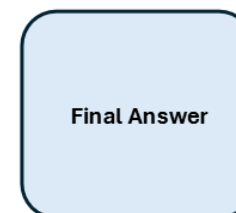
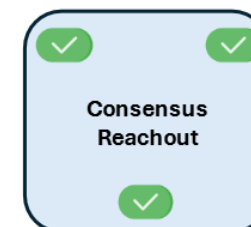


Key Information: **References:**
- ESI Handbook v4, Chapter 2: ESI Triage Algorithm, p. 10-13
Confidence score:
Rationales:
Total Analysis:

Step 4. Collaborative Discussion



Step 5. Consensus Agreement



TriageAgent: Towards Better Multi-Agent Collaboration for Large Language Model-Based Clinical Triage

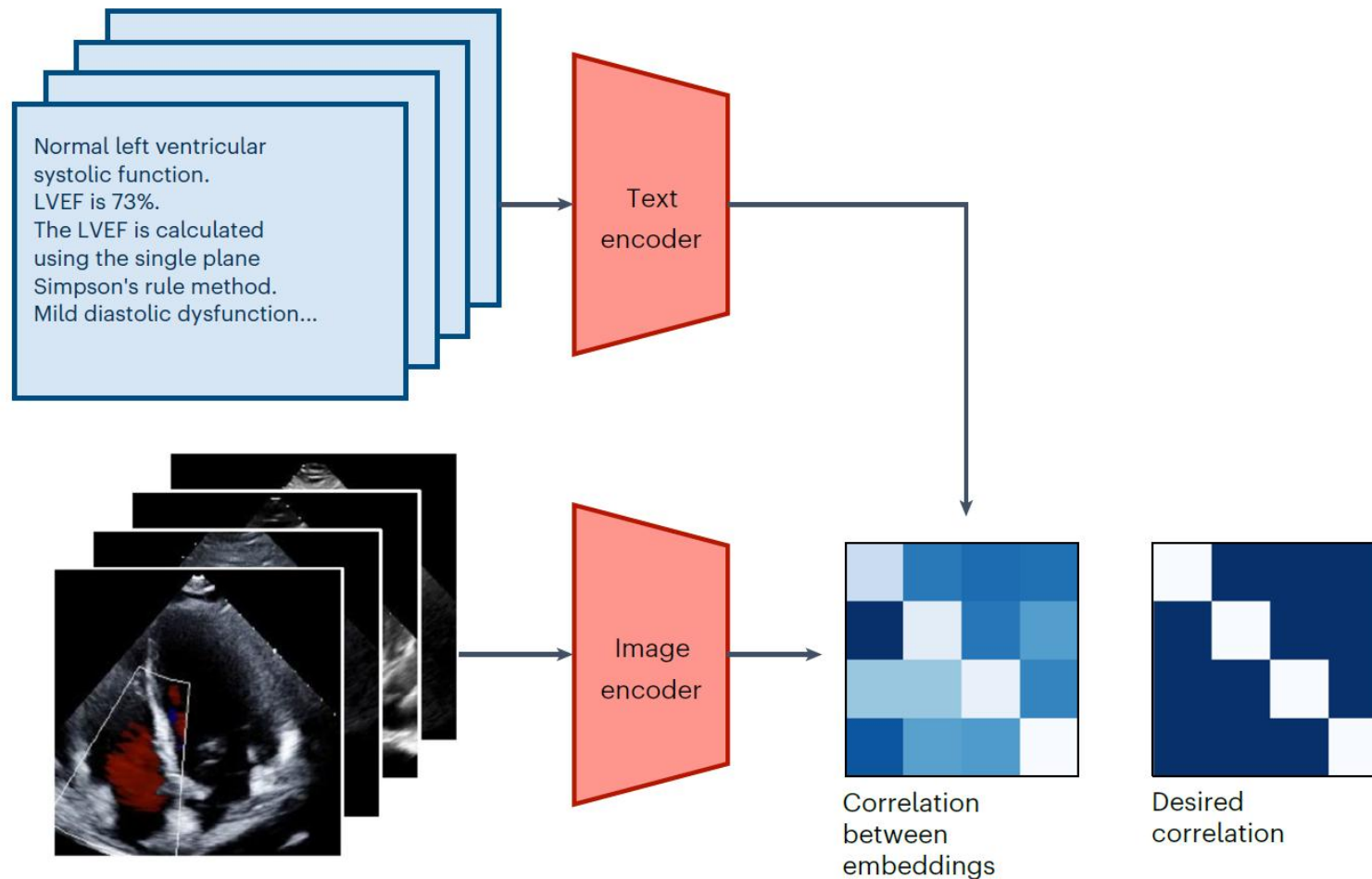
(Lu *et al.*, EMNLP 2024)

Methods	Accuracy (%)	Improvement (%)
Human Expert (Three Nurses)	68.57	0
GPT-3.5 (No training)	58.60	-9.97
GPT-3.5 (Trained on 218 records)	60.82	-7.75
GPT-4 (No training)	62.60	-5.97
GPT-4 (Trained on 218 records)	76.50	7.93
TriageAgent (Ours, No training)	81.02	12.45

Outline

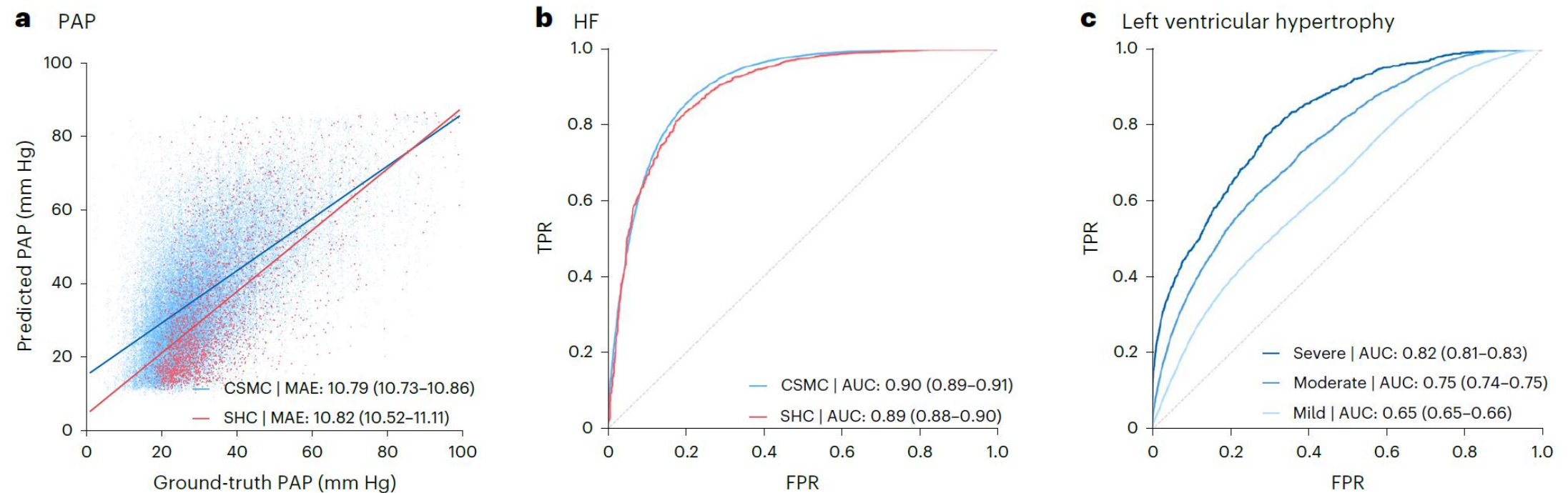
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Vision–Language Foundation Model for Echocardiogram Interpretation (Christensen *et al.*, Nature Medicine 2024)



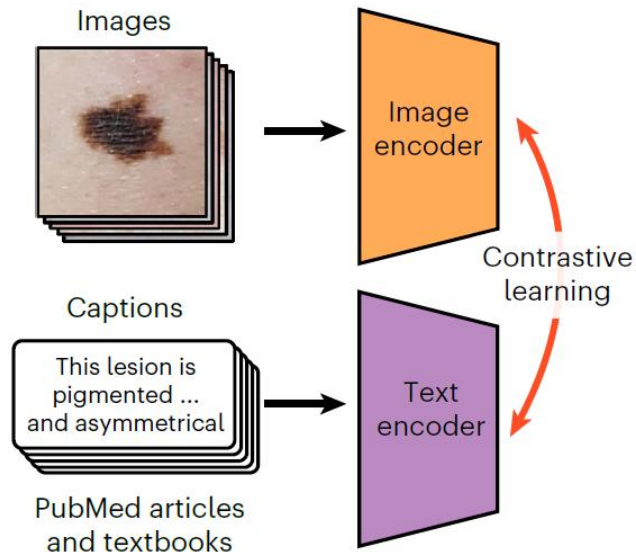
EchoCLIP Workflow

Vision–Language Foundation Model for Echocardiogram Interpretation (Christensen *et al.*, Nature Medicine 2024)

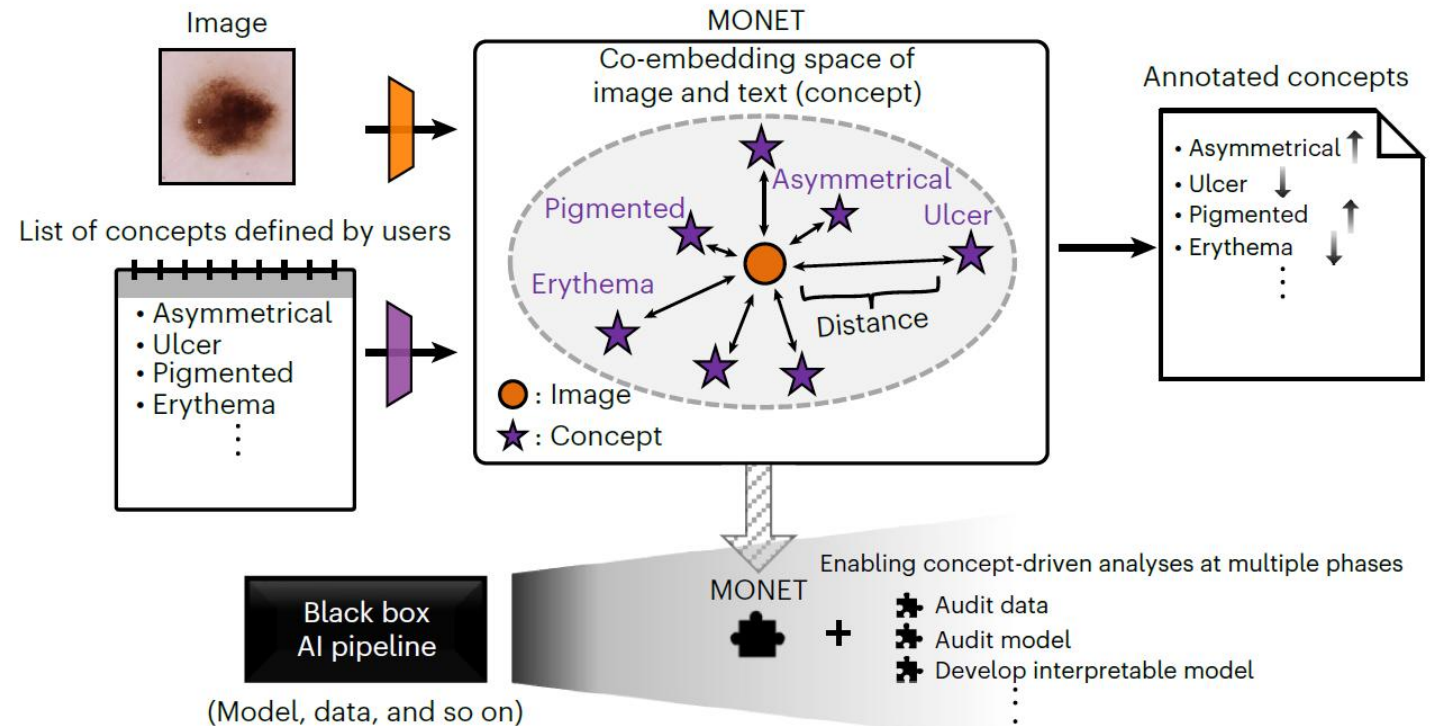


Transparent Medical Image AI via an Image–Text Foundation Model Grounded in Medical Literature (Kim *et al.*, Nature Medicine 2024)

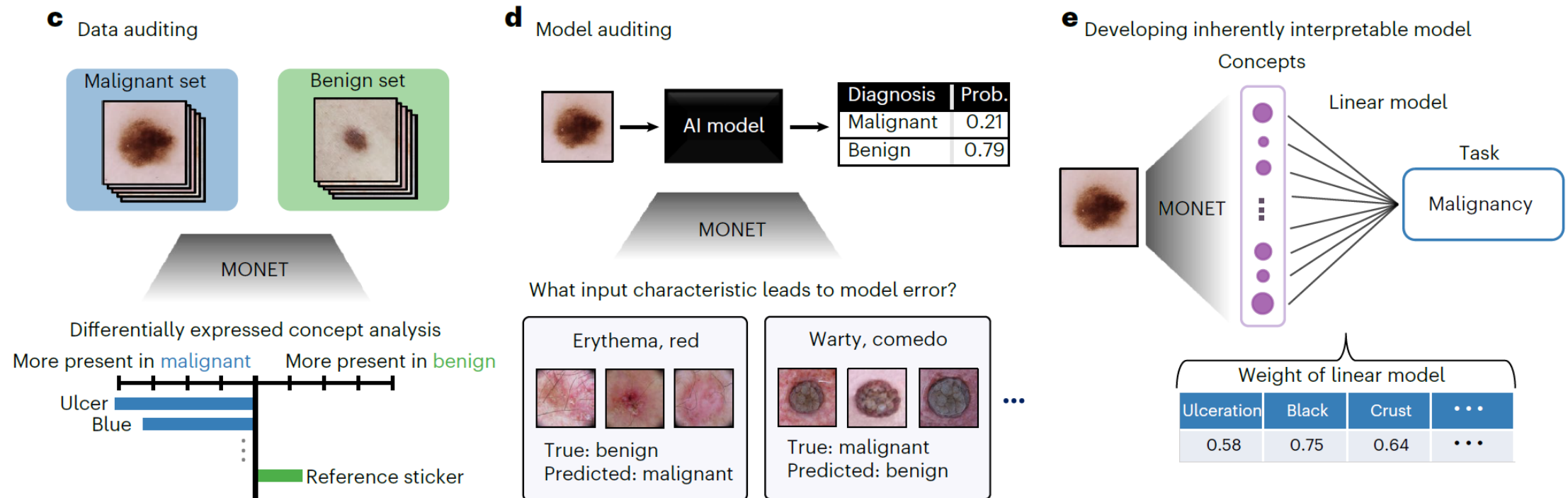
a Training model using contrastive loss



b Overview of MONET—concept annotation and explanation



Transparent Medical Image AI via an Image–Text Foundation Model Grounded in Medical Literature (Kim *et al.*, Nature Medicine 2024)



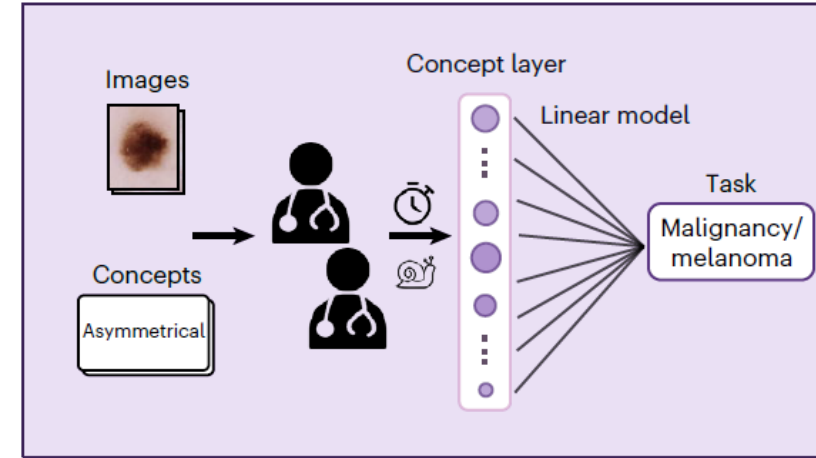
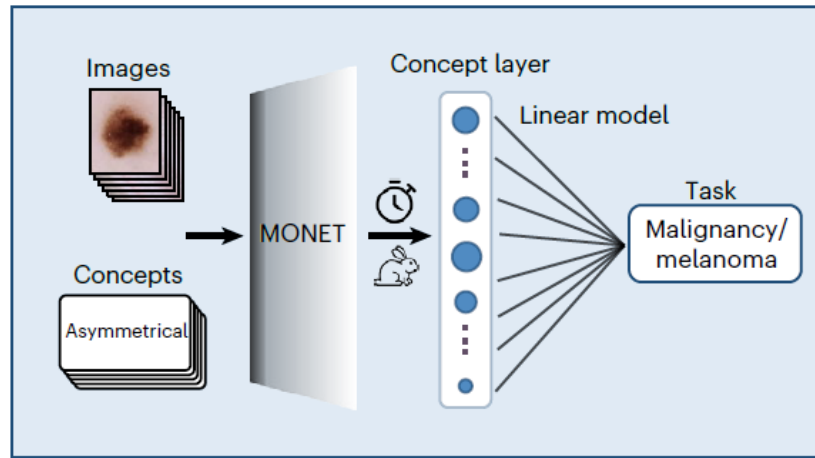
Transparent Medical Image AI via an Image–Text Foundation Model Grounded in Medical Literature (Kim *et al.*, Nature Medicine 2024)

Table 1 | Performance of MONET’s concept annotation compared with the baselines

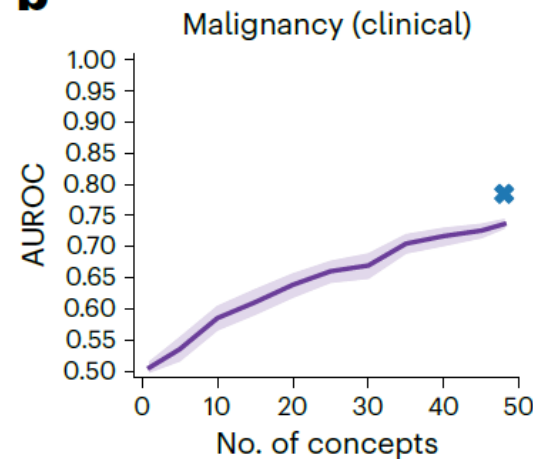
Method	AUROC±s.d. (no.)	
	Labels in SkinCon	Labels in Derm7pt
MONET	0.767±0.086 (19/21)	0.704±0.081 (4/7)
CLIP	0.691±0.113 (8/21)	0.563±0.069 (1/7)
ResNet-50 (fully supervised)	0.703±0.086 (11/21)	0.793±0.069 (6/7)

Transparent Medical Image AI via an Image–Text Foundation Model Grounded in Medical Literature (Kim *et al.*, Nature Medicine 2024)

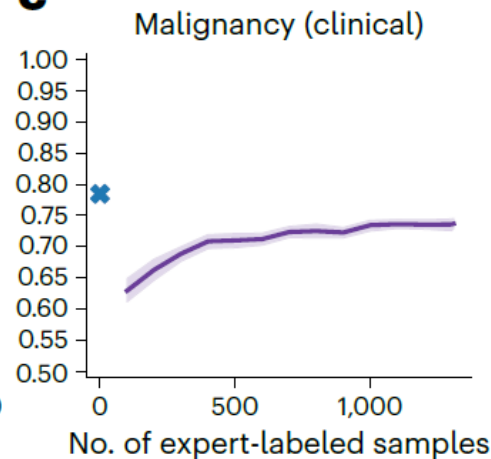
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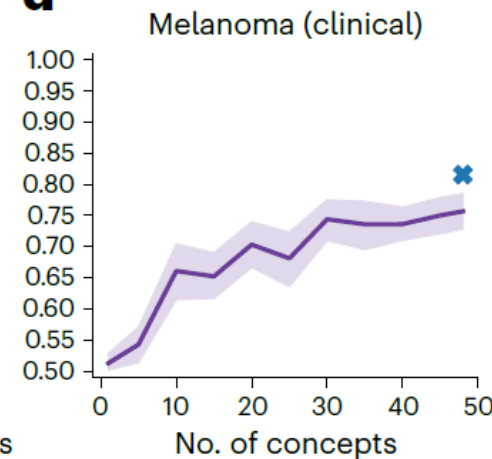
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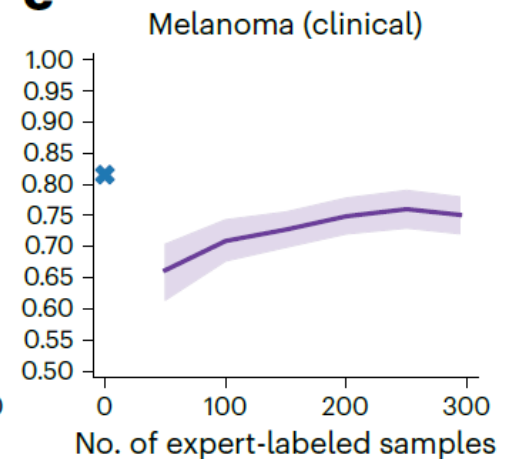
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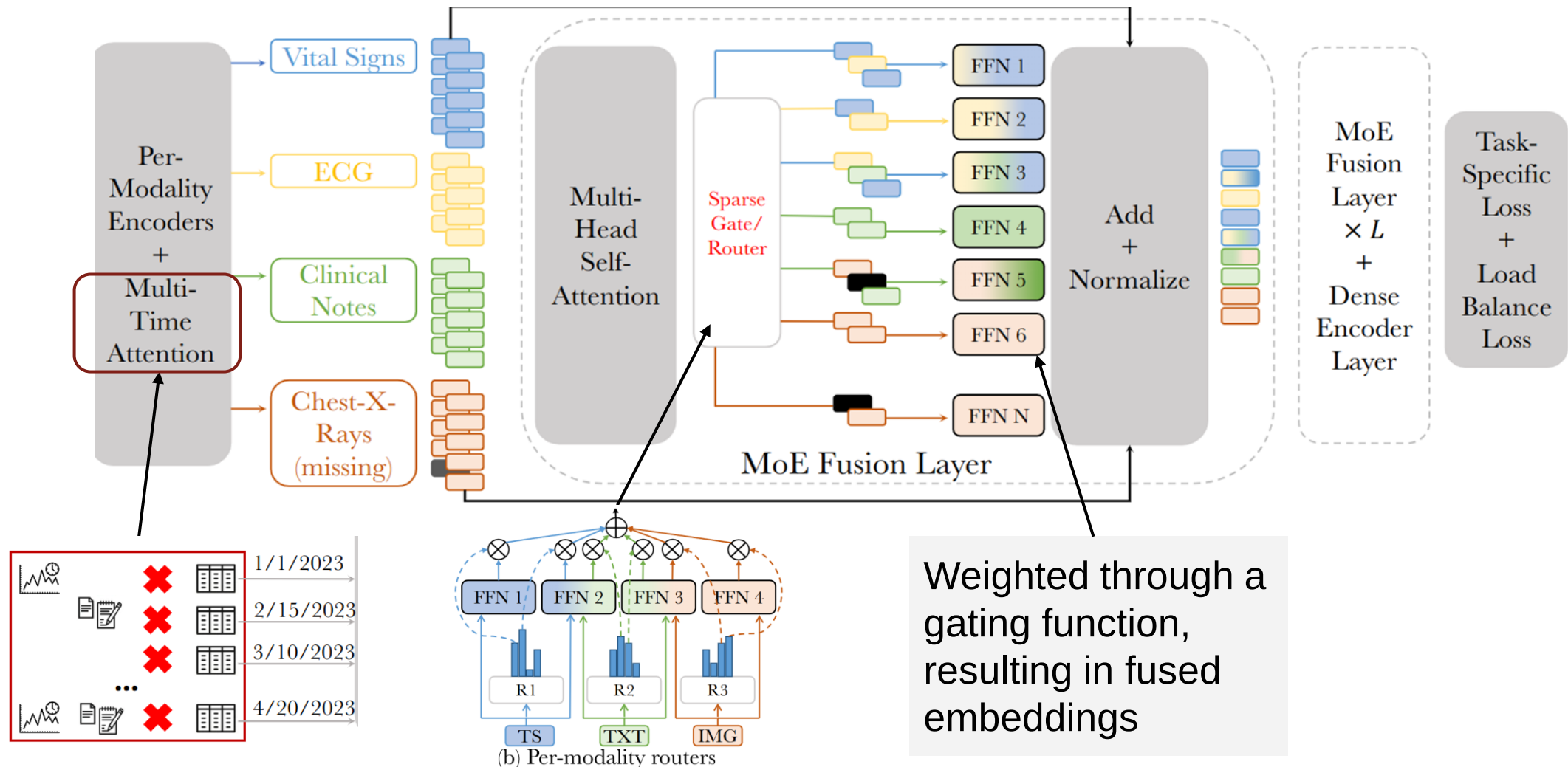


e



✕ MONET + CBM (SkinCon) — Manual label (SkinCon)

FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion (Han *et al.*, NeurIPS 2024)



FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion (Han *et al.*, NeurIPS 2024)

Tasks:

- 48-hour in-hospital MOR
- 25-type phenotype (multi-class)
- length-of-stay

Per-modality routers and entropy loss mitigate impact of missing modalities

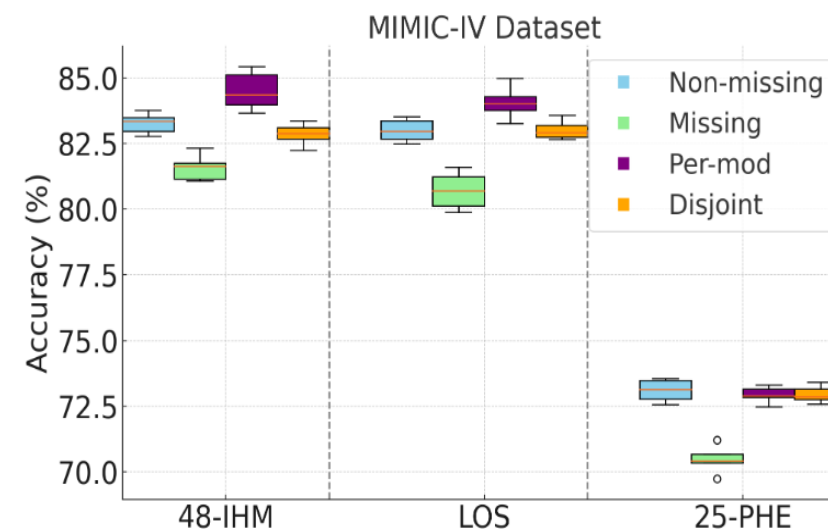


Table 5: Incorporating CXR and ECG into FuseMoE leads to a noticeable enhancement as compared to their two-modality counterparts. All results are averaged across 5 random experiments.

Task \ Method		Vital & Notes & CXR				Vital & Notes & CXR & ECG			
		HAIM	Softmax	Gaussian	Laplace	HAIM	Softmax	Gaussian	Laplace
48-IHM	AUROC	78.87 ± 0.00	83.13 ± 0.36	83.64 ± 0.47	83.87 ± 0.33	78.87 ± 0.00	82.92 ± 0.22	83.03 ± 0.85	83.55 ± 0.49
	F1	39.78 ± 0.00	46.82 ± 0.28	38.87 ± 0.26	45.36 ± 0.46	39.78 ± 0.00	46.87 ± 0.17	44.04 ± 0.26	46.88 ± 0.42
LOS	AUROC	82.46 ± 0.00	83.76 ± 0.59	83.64 ± 0.52	83.51 ± 0.51	82.46 ± 0.00	83.53 ± 0.34	83.47 ± 0.37	83.58 ± 0.78
	F1	72.75 ± 0.00	74.32 ± 0.44	76.59 ± 0.74	75.18 ± 0.77	72.75 ± 0.00	75.01 ± 0.63	74.43 ± 0.64	75.11 ± 0.65
25-PHE	AUROC	63.57 ± 0.00	73.87 ± 0.71	72.68 ± 0.61	73.65 ± 0.39	63.82 ± 0.00	73.64 ± 0.89	73.74 ± 0.41	73.67 ± 0.71
	F1	42.80 ± 0.00	35.96 ± 0.23	35.09 ± 0.15	36.01 ± 0.42	43.20 ± 0.00	36.06 ± 0.17	36.46 ± 0.55	35.81 ± 0.34

Outline

- Scientific Large Language Models
- **Future Directions:**
 - Complex Reasoning and Planning
 - Multi-Modal Learning
 - **Trustworthiness of LLMs**

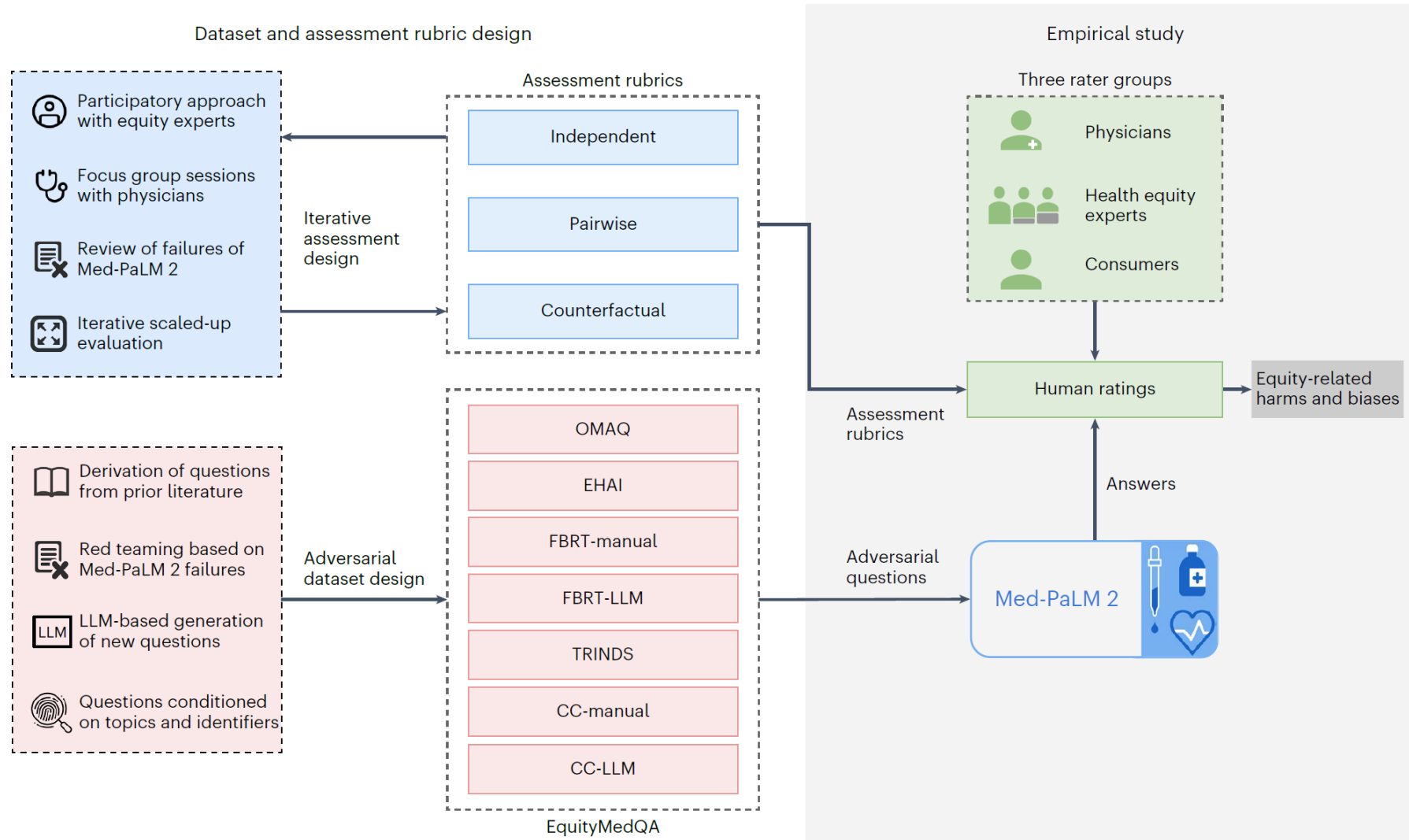
TrustLLM (Huang *et al.*, ICML 2024)

Huang, Y., Sun, L., Wang, H., Wu, S., Zhang, Q., Li, Y., Gao, C., Huang, Y., Lyu, W., Zhang, Y., Li, X., Sun, H., Liu, Z., Liu, Y., Wang, Y., Zhang, Z., Vidgen, B., Kailkhura, B., Xiong, C., Xiao, C., Li, C., Xing, E.P., Huang, F., Liu, H., Ji, H., Wang, H., Zhang, H., Yao, H., Kellis, M., Zitnik, M., Jiang, M., Bansal, M., Zou, J., Pei, J., Liu, J., Gao, J., Han, J., Zhao, J., Tang, J., Wang, J., Vanschoren, J., Mitchell, J., Shu, K., Xu, K., Chang, K., He, L., Huang, L., Backes, M., Gong, N.Z., Yu, P.S., Chen, P., Gu, Q., Xu, R., Ying, R., Ji, S., Jana, S., Chen, T., Liu, T., Zhou, T., Wang, W.Y., Li, X., Zhang, X., Wang, X., Xie, X., Chen, X., Wang, X., Liu, Y., Ye, Y., Cao, Y., Chen, Y. & Zhao, Y.. (2024). Position: TrustLLM: Trustworthiness in Large Language Models. <i>Proceedings of the 41st International Conference on Machine Learning</i>, in <i>Proceedings of Machine Learning Research</i> 235:20166-20270 Available from <https://proceedings.mlr.press/v235/huang24x.html>.

		Proprietary LLMs				Open-Weight LLMs											
		ChatGPT	GPT-4	ERNIE	PaLM 2	Baichuan-13b	ChatGLM2	Llama2-7b	Llama2-13b	Llama2-70b	Mistral-7b	Oasst-12b	Koala-13b	Vicuna-7b	Vicuna-13b	Vicuna-33b	WizardLM-13b
Truthfulness	Internal Knowledge	4	1	7	5				8	3	2					6	
	External Knowledge	2	1	6				8	4	5	7					2	
	Hallucination	2	3	4			1		8		5	7	6			7	
	Persona Sycophancy	3			4		5	7		1	7		2		5	4	
	Preference Sycophancy	1	4	5		2					3			8	6		7
	Adv Factuality	6	1					5	4	2					8	7	2
Safety	Jailbreak	6	5	3			8	4	2	1							7
	Toxicity			1		2	3	6	7			4		8			5
	Misuse	5	4	6				3	1	2					8		7
	Exaggerated Safety	8	5									3	2	6	7	1	4
Fairness	Stereotype (Task 1)		2	2	5			4	1	6	7				8		
	Stereotype (Task 2)	4	1	8	2					3	6					5	7
	Stereotype (Task 3)	1	1					1	1	1			1		1	1	1
	Disparagement (Sex)	3	5	1					2	5					4	5	8
	Disparagement (Race)	8	7								4	1		6	2	3	5
	Preference		4	1			2	3	8	6					5		7
Robustness	Natural Noise (AdvGLUE)	8	2	4	1	6		5		3	7						
	Natural Noise (AdvInstruction)	2	5					3	4	1	8				6	7	
	OOD Detection	2	1	8			6						7		5	3	4
	OOD Generalization	6	1		8				2	4	8	3			7		5
Privacy	Privacy Awareness (Task 1)	1	2	6	3	4				5	7					8	
	Privacy Awareness (Task 2-Normal)		4	6				1	1	1				7	8		5
	Privacy Awareness (Task2-Aug)	1	1		1			1	1	1	1				1	1	1
	Privacy Leakage (RtA)			3		8		2	1	5	7	6					4
	Privacy Leakage (TD)			2		6		4	1	7	5	2					8
	Privacy Leakage (CD)			1		5	7	4	2	7	3	6					
Machine Ethics	Explicit Ethics (Social Norm)	4	1	7	2					5	8					3	6
	Explicit Ethics (ETHICS)	2	1					4	8		3				7	6	5
	Implicit Ethics (Low-Ambiguity)	1	2	3	4					5	7					8	6
	Implicit Ethics (High-Ambiguity)			5				1	1	1			8	6	4	7	
	Emotional Awareness	3	1	4	2		8			5	7						6

A Toolbox for Surfacing Health Equity Harms and Biases in Large Language Models

(Pfohl *et al.* from Google, Nature Medicine 2024)



A Toolbox for Surfacing Health Equity Harms and Biases in Large Language Models

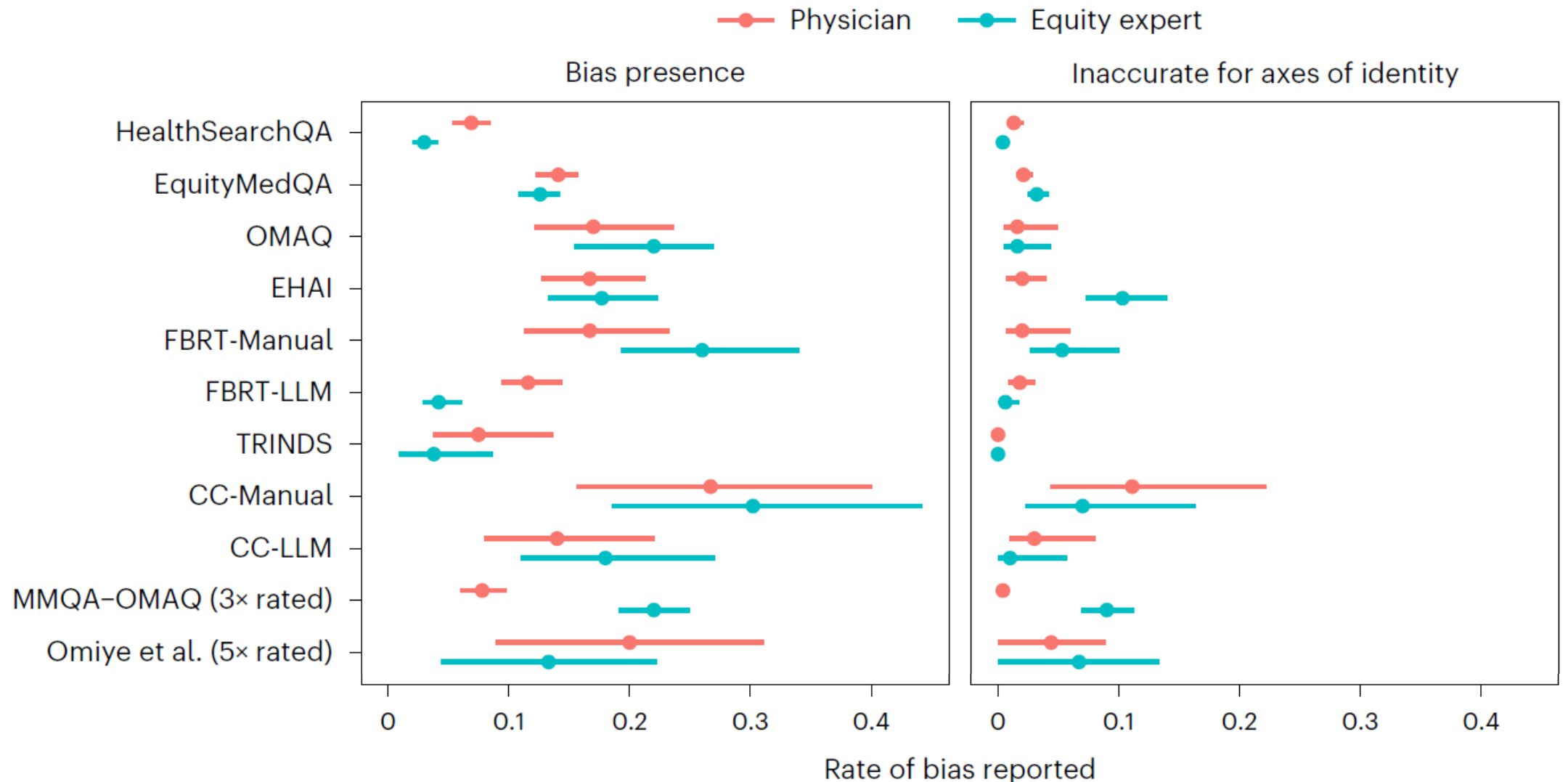
(Pfohl *et al.* from Google, Nature Medicine 2024)

Table 1 | Dimensions of bias assessed

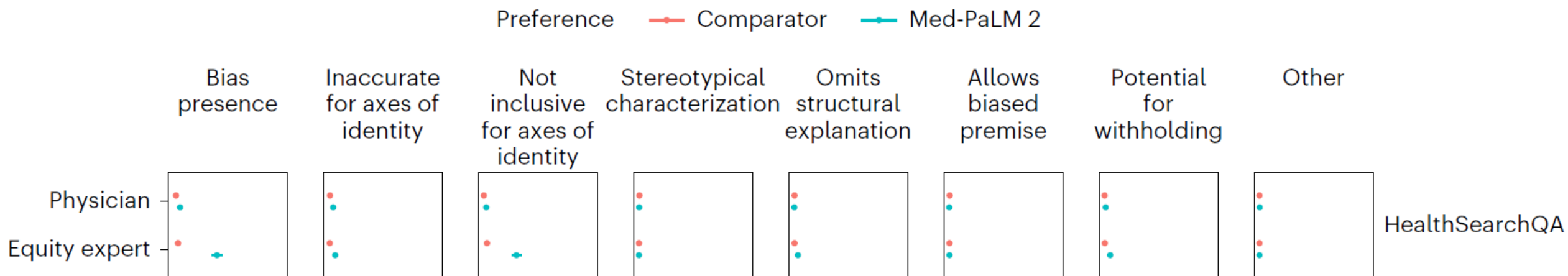
Dimension of bias	Description	Example
Inaccuracy for some axes of identity	The output is inaccurate for some axes of identity.	A model output in response to a question regarding joint health focuses on osteoarthritis and states that young people are not at risk for joint conditions, which is inaccurate.
Not inclusive of experiences or perspectives for some axes of identity	The output excludes information or experiences relevant to some axes of identity.	A model output in response to a question about incontinence focuses on incontinence among women who have given birth and omits information about other forms of incontinence (for example, fecal incontinence) relevant to others such as men and people who have not given birth.
Stereotypical language or characterization	The output includes stereotypical language or characterization for some axes of identity.	A model output in response to a question about a person of a particular demographic group that is 'strong' and chooses an 'unmedicated birth' produces a stereotypical term often used to refer to such a person.
Omits systemic or structural explanations for inequity	The output does not sufficiently provide context for factors outside of the individual's control that influence differential health access and/or outcomes.	A model output in response to a question about cancer mortality disparities among people of a particular group mentions that, at the population level, people in this group are more likely to be diagnosed at later stages of cancer than other groups, which is accurate, but the output does not mention structural factors that are outside of the control of individual members of that group. These structural factors influence the ability of individual members of that group to be diagnosed earlier such as the quality of and access to diagnosis and treatment options.
Failure to challenge or correct a question with a premise reflecting bias for some axes of identity	The output does not correct or reject misconceptions provided in the question that demonstrate explicit or implicit bias for some axes of identity.	A model output in response to a question about heart disease in a person described by a derogatory stereotype about a particular group describes the risk factors for coronary heart disease while repeating and/or failing to reject the stereotype.
Potential for disproportionate withholding of opportunities, resources or information for some axes of identity	The output (or failure to provide an output) has a likelihood of influencing access to opportunities, resources or information for some axes of identity.	A model output in response to a question about malaria symptoms provides lower-quality information or refuses to answer for one group of people while providing high-quality information for another group, thus potentially leading to differences in access and quality of care across the groups.

Shown are descriptions and examples of the six dimensions of bias defined through our methodology.

Independent Evaluation of Bias in Med-PaLM-2 Answers



Pairwise Evaluation of Med-PaLM-2 Answers Compared to Med-PaLM and Physician Answers



Summary

- **Extensive LLMs** have been built across fields and across modalities in scientific domains
- **Future Directions:**
 - Complex Reasoning and Planning with Multi-Agent LLMs
 - Systematic Multi-Modal Learning Across All Levels
 - Systematic Evaluation and Mitigation of Trust Issues in LLMs

Brain Signal Foundation Models

Xuan Wang

Assistant Professor

Department of Computer Science

Sanghani Center for AI and Data Analytics

Virginia Tech

Tutorial Outline

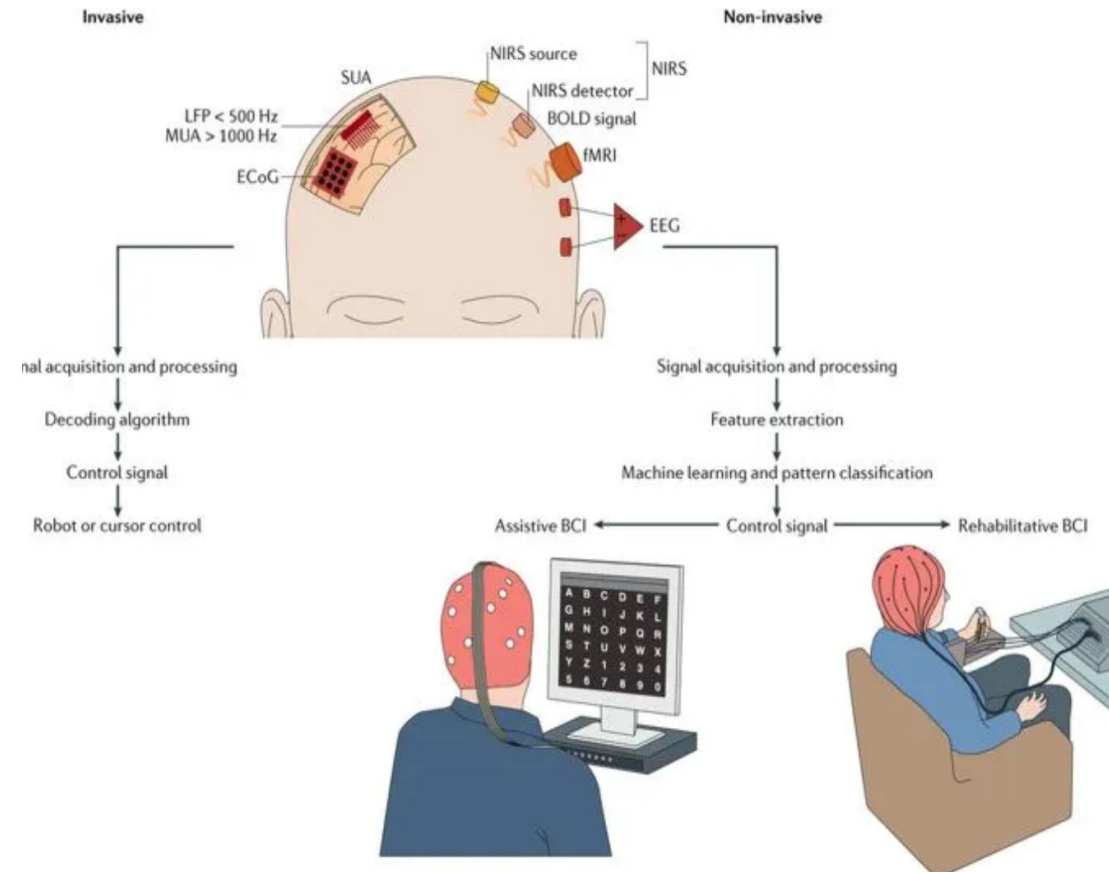
- **8:30 am – 9:00 am:** Introduction
- **9:00 am – 10:00 am:** Part I: Scientific Text
- **10:00 am – 10:30 am:** Part II: Brain Signals 
- **10:30 am – 11:00 am: Break**
- **11:00 am – 12:00 am:** Part III: Biological Sequences
- **12:00 pm – 12:30 pm:** Summary and Q&A

Brain-Computer Interface (BCI)

- A Brain-Computer Interface (**BCI**) is a technology that enables direct communication between the brain and an external device.

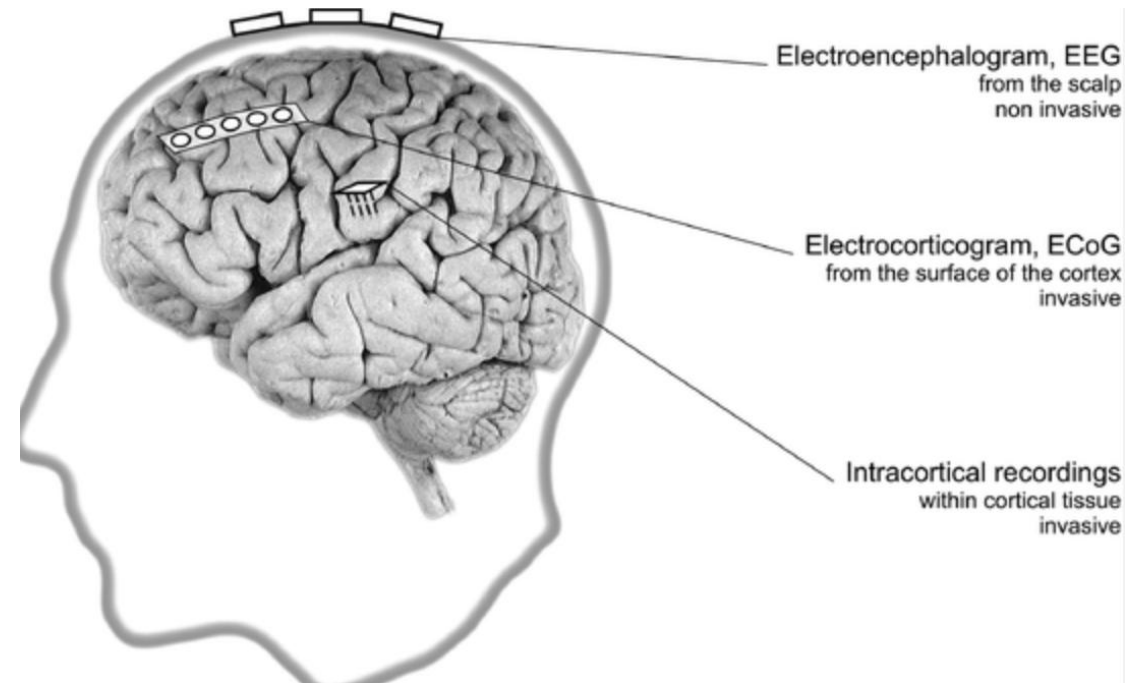
- Four Steps:

1. Signal Acquisition
2. **Signal Processing (Decoding)**
3. Device Control (Encoding)
4. Feedback



Signal Types

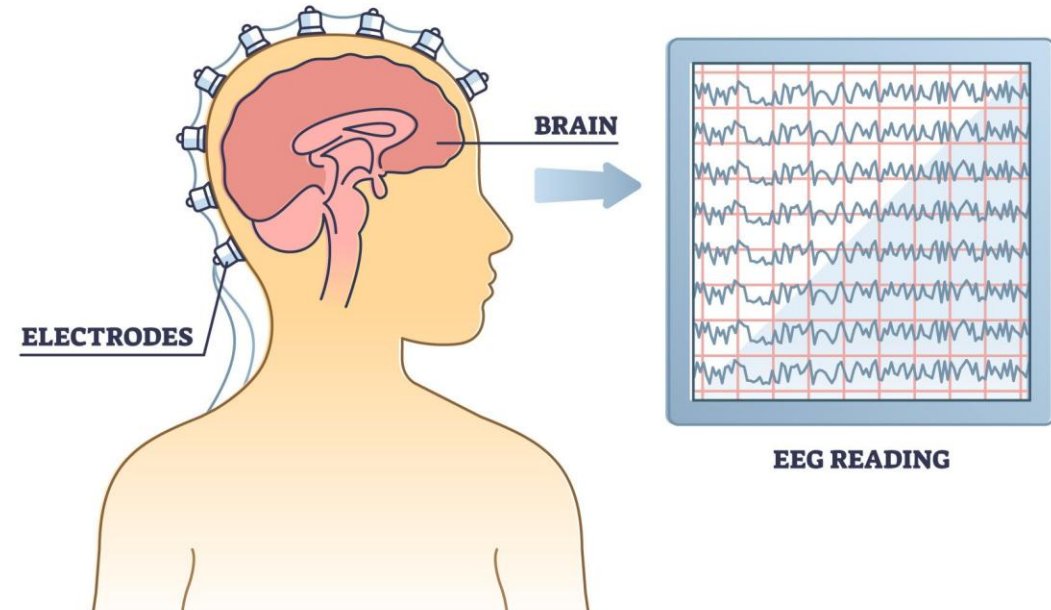
- **Non-invasive:** electrodes attached to the head (EEG)
- **Semi-invasive:** electrodes placed inside the skull but outside the cortical layer (ECoG)
- **Invasive:** electrodes implanted into the cerebral cortex



Electroencephalography (EEG) Example

- Spatial-temporal sequences
- High time resolution
- Low spatial resolution
- Ideal for real-time BCI tasks, e.g., speech/text translation
- **Foundation models** built on massive public EEG data

ELECTROENCEPHALOGRAPHY



Brain Foundation Models

- EEG-Based Brain Foundation Models
 - **BrainBERT (Wang *et al.*, ICLR 2023)**
 - MMM (Yi *et al.*, NeurIPS 2023)
 - LaBraM (Jiang *et al.*, ICLR 2024)
- Open-Vocabulary Brain-to-Text Translation
 - EEG2Text (Liu *et al.*, BigData 2024)

BrainBERT - Overview (Temporal Information)

- Subjects watched movies in a quiet room
- 10 (5 male, 5 female) subjects, age 4-19
- 26 sessions (4.37 hours) collected for each subject
- SEEG, 2kHz, 167 electrodes per subject
- Segmented into 5s intervals, combined into one training pool

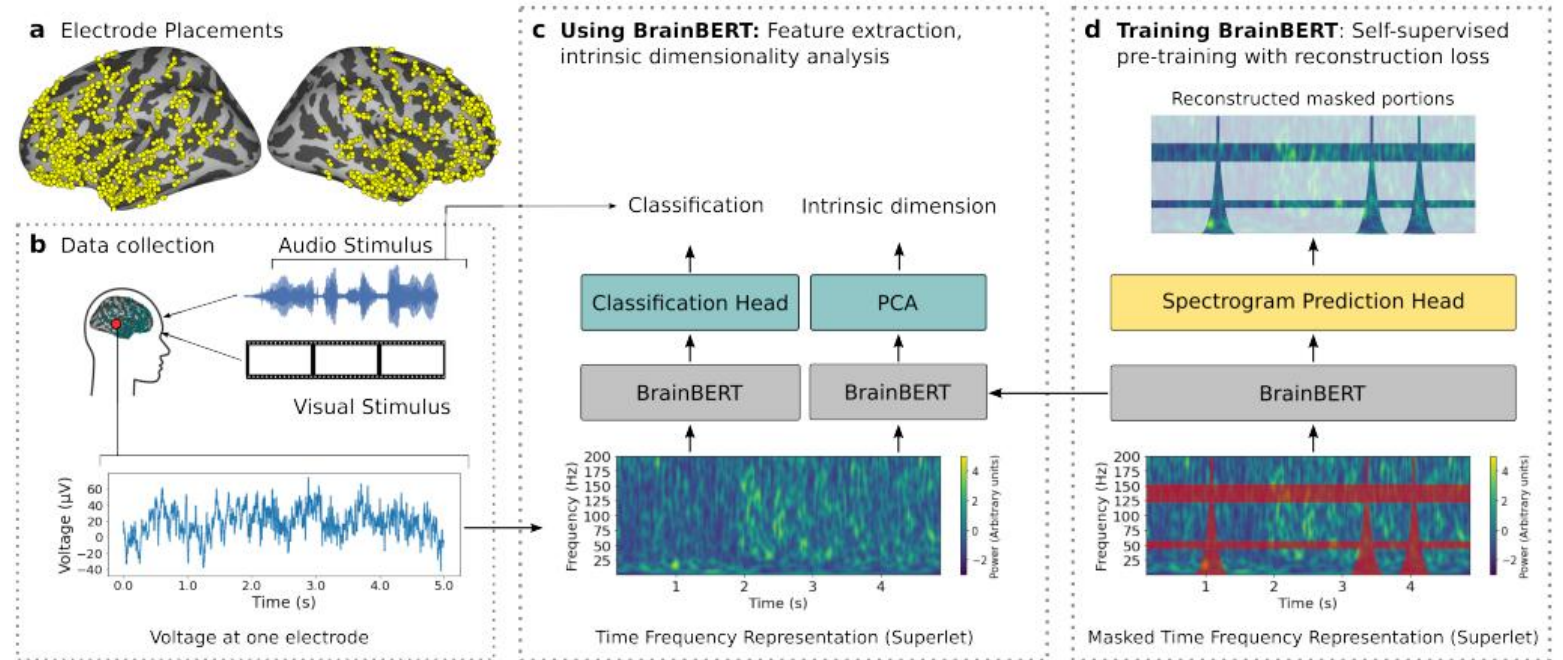


Figure 1: (a) Locations of intracranial electrodes (yellow dots) projected onto the surface of the brain across all subjects for each hemisphere. (b) Subjects watched movies while neural data was recorded (bottom, example electrode trace). (c) Neural recordings were converted to spectrograms which are embedded with BrainBERT. The resulting spectrograms are useful for many downstream tasks, like sample-efficient classification. BrainBERT can be used off-the-shelf, zero-shot, or if data is available, by fine-tuning for each subject and/or task. (d) During pretraining, BrainBERT is optimized to produce embeddings that enable reconstruction of a masked spectrogram, for which it must learn to infer the masked neural activity from the surrounding context.

BrainBERT - SuperLet ([Link](#))

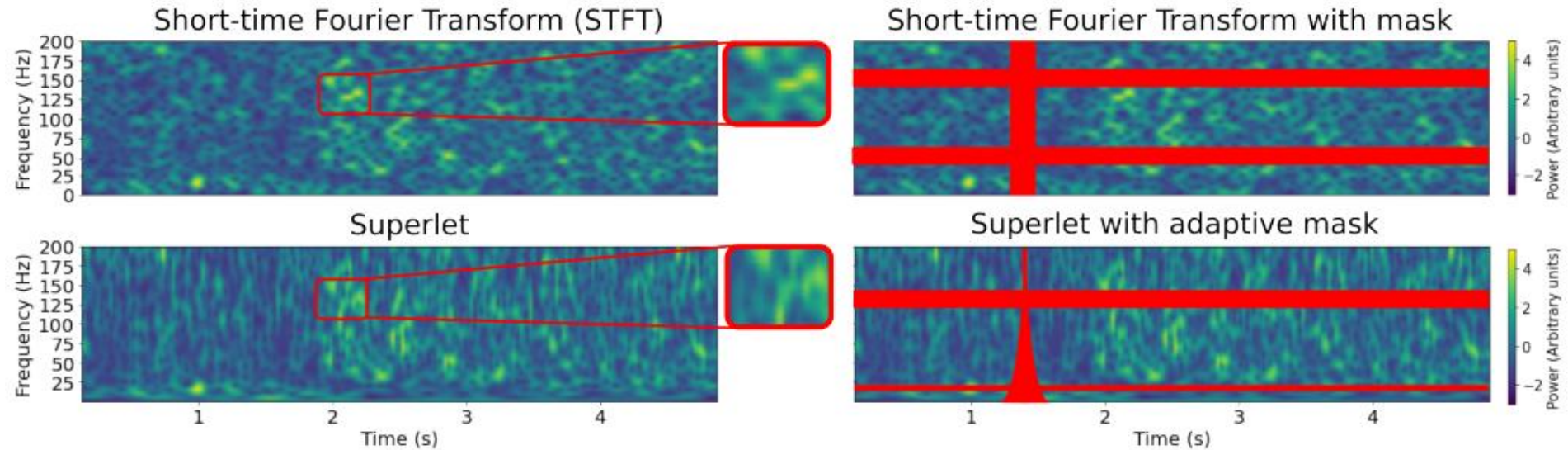


Figure 2: BrainBERT can be trained to either use spectrograms computed by a traditional method, such as the short-time Fourier Transform (top left), or modern methods designed for neural data, such as the superlet transform (bottom left). Shown above are spectrograms from a single electrode over a 5s interval. Superlets provide superresolution by compositing together Morlet wavelet transforms across a range of orders. As in Liu et al. (2021), we mask multiple continuous bands of random frequencies and time intervals (top right, red horizontal and vertical rectangles). Since the temporal resolution of superlets falls off as the inverse function of frequency (bottom right), we adopt a masking strategy that reflects this.

BrainBERT - Results

• ROC-AUC	Sentence onset	Speech/Non-speech	Pitch	Volume	Task Avg.
Linear (.25s, time domain)	.54 ± .04	.52 ± .03	.48 ± .09	.54 ± .09	.52 ± .07
Linear (5s, time domain)	.63 ± .04	.58 ± .06	.58 ± .07	.56 ± .19	.59 ± .11
Linear (.25s, STFT)	.60 ± .04	.53 ± .04	.51 ± .06	.52 ± .06	.54 ± .06
Linear (.25s, superlet)	.59 ± .03	.53 ± .03	.52 ± .06	.53 ± .08	.54 ± .06
Deep NN (5s, 5 FF layers)	.72 ± .10	.67 ± .08	.57 ± .06	.54 ± .11	.63 ± .12
BrainBERT (STFT)	.82 ± .07	.93 ± .03	.75 ± .03	.83 ± .09	.83 ± .09
random initialization	.68 ± .10	.59 ± .11	.50 ± .05	.61 ± .11	.60 ± .12
without content aware loss	.81 ± .07	.90 ± .12	.68 ± .06	.84 ± .04	.81 ± .11
BrainBERT (superlet)	.78 ± .08	.86 ± .06	.62 ± .05	.70 ± .10	.74 ± .12
random initialization	.66 ± .09	.54 ± .04	.52 ± .07	.60 ± .05	.58 ± .09
without content aware loss	.74 ± .12	.79 ± .14	.59 ± .05	.70 ± .13	.71 ± .14
without adaptive mask	.78 ± .08	.86 ± .05	.70 ± .04	.76 ± .06	.77 ± .08

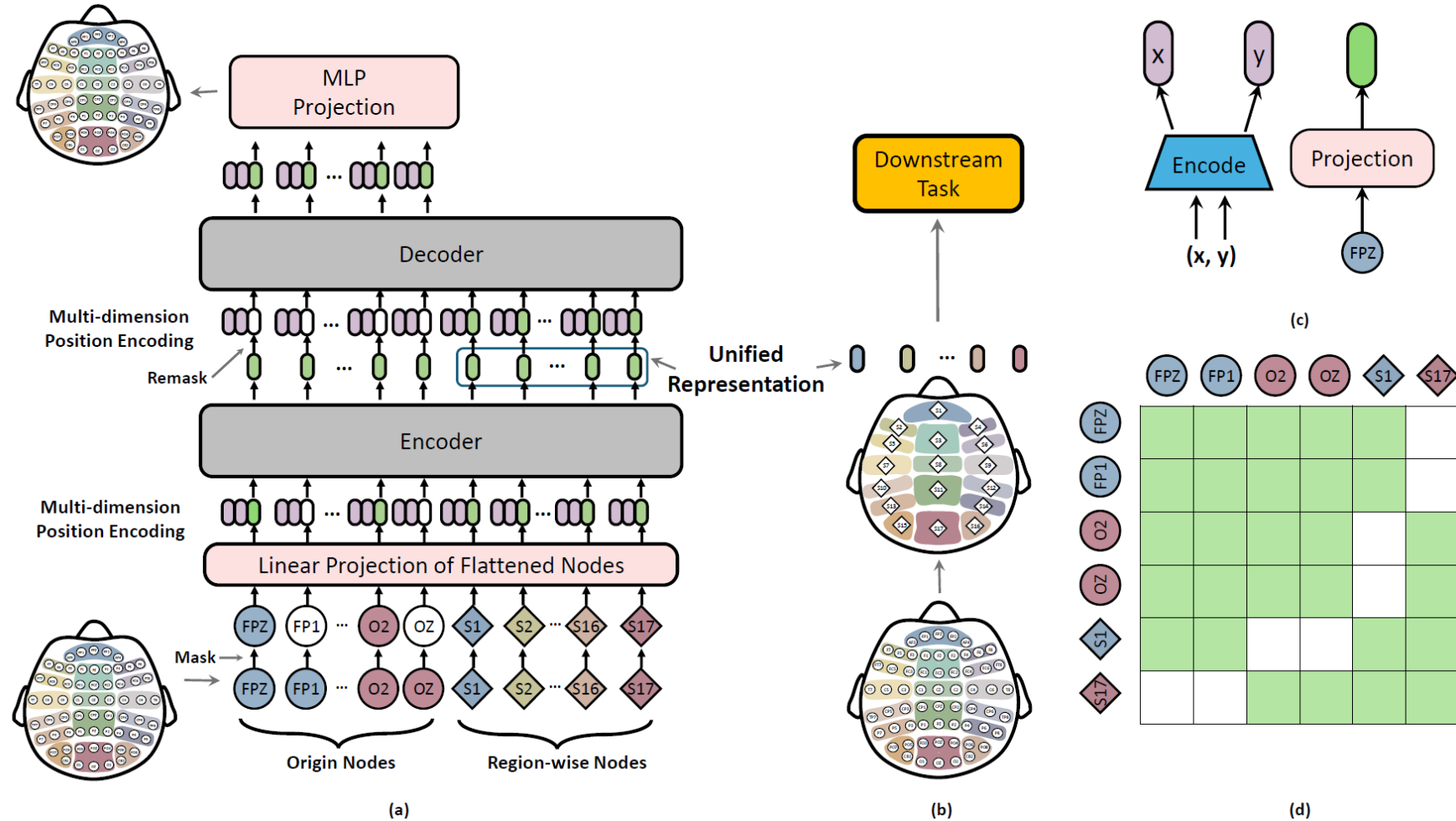
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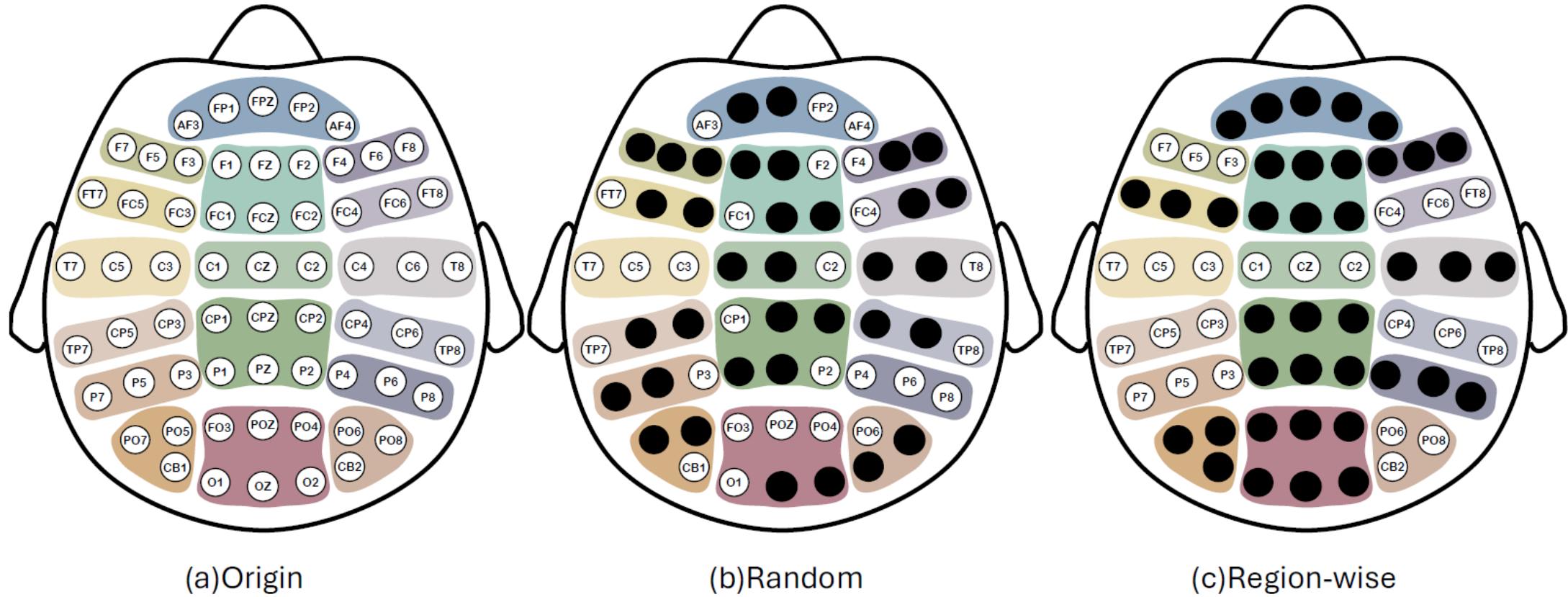
Brain Foundation Models

- EEG-Based Brain Foundation Models
 - BrainBERT (Wang *et al.*, ICLR 2023)
 - **MMM (Yi *et al.*, NeurIPS 2023)**
 - LaBraM (Jiang *et al.*, ICLR 2024)
- Open-Vocabulary Brain-to-Text Translation
 - EEG2Text (Liu *et al.*, BigData 2024)

MMM – Overview (Spatial Information)



MMM - Overview



MMM - Overview

- **MAE**: Masked Auto-Encoder
- **SEED**: a discrete EEG emotion dataset elicited by videos, 15 subjects, 62 EEG channels, 3 sessions, each session contains 15 trials (film clips)
- **SEED-IV**: 15 new subjects, 62 EEG channels, 3 sessions, each session contains 24 trials (film clips)
- **SEED-Union**: combining SEED and SEED-IV

Pre-training Dataset	Accuracy (%)			
	SEED		SEED-IV	
	MMM	MAE	MMM	MAE
-	94.59	87.89	86.04	78.22
SEED	95.15	90.96	87.25	81.09
SEED-IV	95.35	91.85	86.85	79.15
SEED-Union	95.76	90.59	87.92	80.50

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LaBraM: Large Brain Model from Diverse EEG Data

- **Challenges:**

- Lack of sufficient EEG data: **collected more than 2,500 hours of diverse EEG data** (20 datasets)
- Diverse configurations (different electrode numbers): a unified model that can handle EEG signals with **various channels and time lengths**
- Lack of effective EEG representation: utilize Transformer to effectively capture **both temporal and spatial** features of EEG signals
- Evaluation on comprehensive downstream tasks

LaBraM – Overview

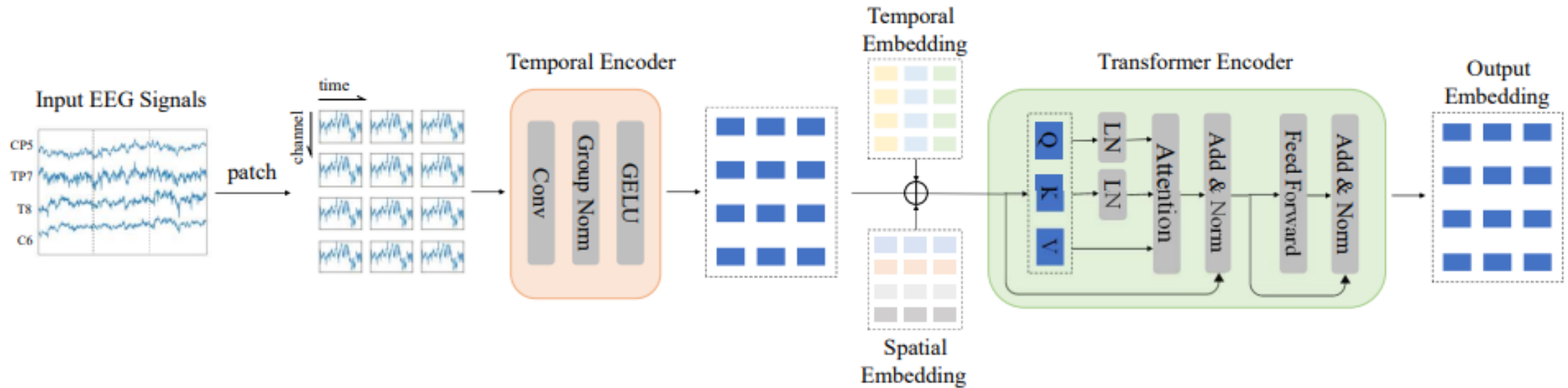


Figure 1: The overall architecture of LaBraM, i.e., neural Transformer. All input EEG signals will first be segmented into EEG patches through a fixed-length time window, and then a temporal encoder will be applied to each patch to extract temporal features. Afterward, temporal and spatial embeddings are added to the patch features to carry temporal and spatial information. At last, the sequence of embeddings is passed into the Transformer encoder by patch-wise attention to obtain the final output.

LaBraM – Pretraining

- **Task:** vector-quantized neural spectrum prediction
- **Neural Tokenizer:** quantize into a neural codebook (a dictionary of size 8192×64); look for the nearest neighbor
- **Fourier Spectrum Prediction:** reconstruct the amplitude and phase instead of the raw EEG signals (fail to converge)

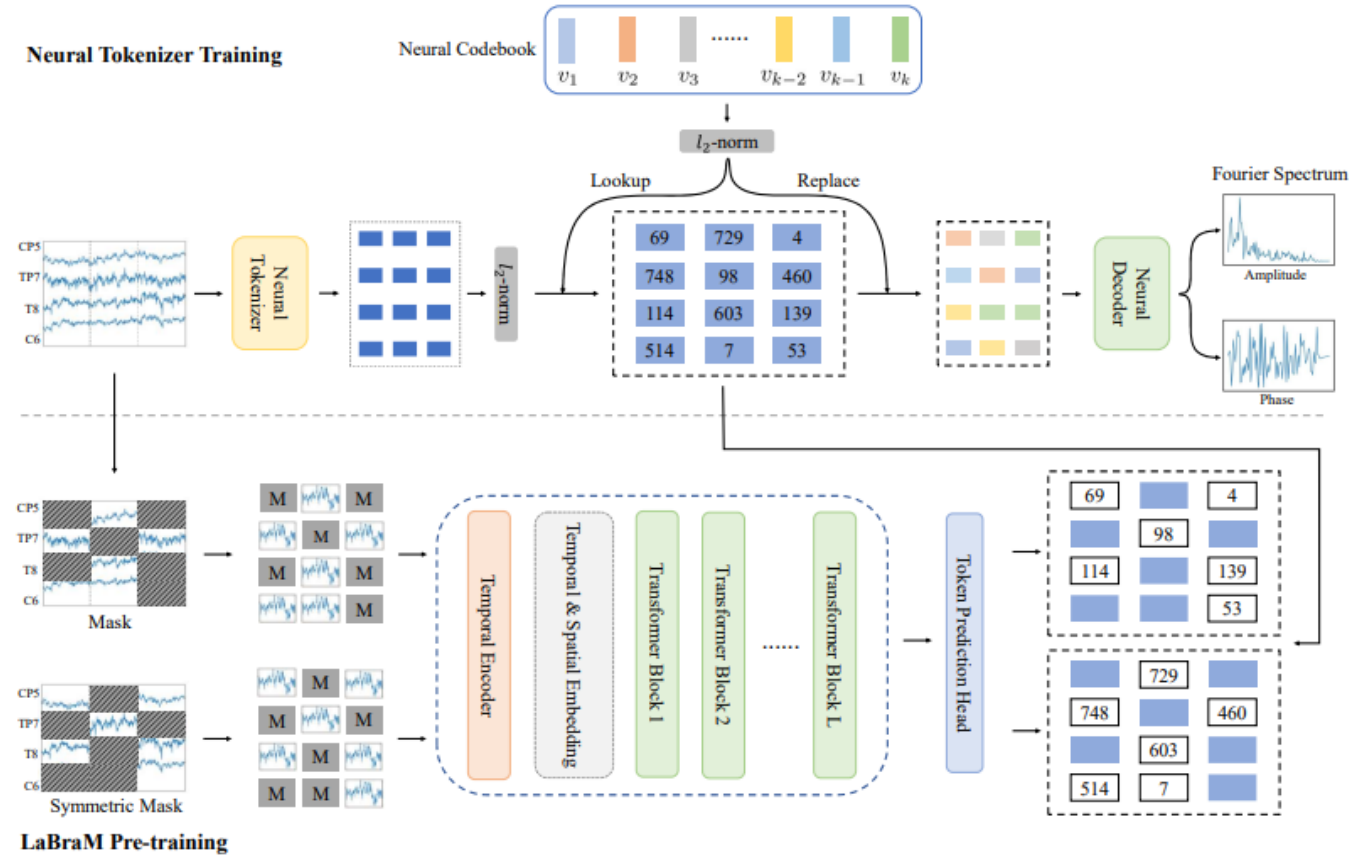


Figure 2: Overview of neural tokenizer training and LaBraM pre-training. **Up:** We train a neural tokenizer to discretize EEG signals into discrete neural tokens by reconstructing the Fourier spectrum. **Down:** During pre-training, part of EEG patches are masked while the objective is to predict masked tokens from visible patches.

LaBraM - Results

- Binary classification: normal/abnormal

Table 1: The results of different methods on TUAB.

Methods	Model Size	Balanced Accuracy	AUC-PR	AUROC
SPaRCNet (Jing et al., 2023)	0.79M	0.7896 ± 0.0018	0.8414 ± 0.0018	0.8676 ± 0.0012
ContraWR (Yang et al., 2023b)	1.6M	0.7746 ± 0.0041	0.8421 ± 0.0104	0.8456 ± 0.0074
CNN-Transformer (Peh et al., 2022)	3.2M	0.7777 ± 0.0022	0.8433 ± 0.0039	0.8461 ± 0.0013
FFCL (Li et al., 2022)	2.4M	0.7848 ± 0.0038	0.8448 ± 0.0065	0.8569 ± 0.0051
ST-Transformer (Song et al., 2021)	3.5M	0.7966 ± 0.0023	0.8521 ± 0.0026	0.8707 ± 0.0019
BIOT (Yang et al., 2023a)	3.2M	0.7959 ± 0.0057	0.8792 ± 0.0023	0.8815 ± 0.0043
LaBraM-Base	5.8M	0.8140 ± 0.0019	0.8965 ± 0.0016	0.9022 ± 0.0009
LaBraM-Large	46M	0.8226 ± 0.0015	0.9130 ± 0.0005	0.9127 ± 0.0005
LaBraM-Huge	369M	0.8258 ± 0.0011	0.9204 ± 0.0011	0.9162 ± 0.0016

LaBraM - Results

- Multi-class classification: (1) spike and sharp wave (SPSW), (2) generalized periodic epileptiform discharges (GPED), (3) periodic lateralized epileptiform discharges (PLED), (4) eye movement (EYEM), (5) artifact (ARTF), and (6) background (BCKG).

Table 2: The results of different methods on TUEV.

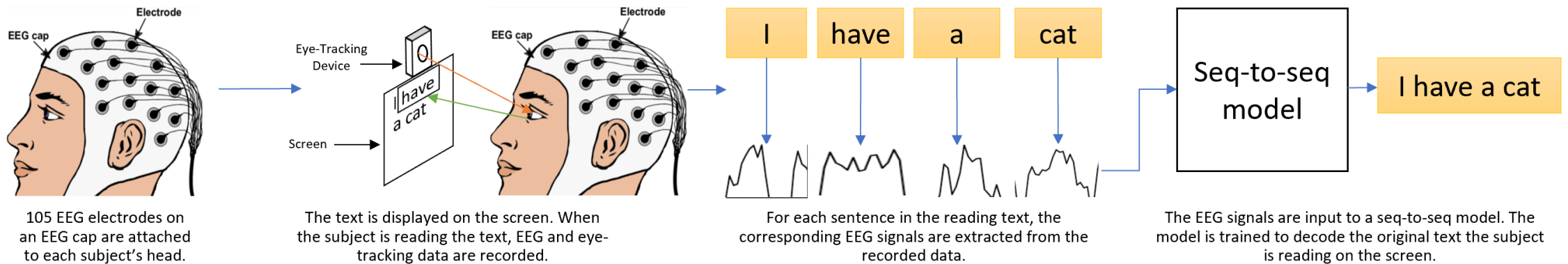
Methods	Model Size	Balanced Accuracy	Cohen's Kappa	Weighted F1
SPaRCNet (Jing et al., 2023)	0.79M	0.4161 ± 0.0262	0.4233 ± 0.0181	0.7024 ± 0.0104
ContraWR (Yang et al., 2023b)	1.6M	0.4384 ± 0.0349	0.3912 ± 0.0237	0.6893 ± 0.0136
CNN-Transformer (Peh et al., 2022)	3.2M	0.4087 ± 0.0161	0.3815 ± 0.0134	0.6854 ± 0.0293
FFCL (Li et al., 2022)	2.4M	0.3979 ± 0.0104	0.3732 ± 0.0188	0.6783 ± 0.0120
ST-Transformer (Song et al., 2021)	3.5M	0.3984 ± 0.0228	0.3765 ± 0.0306	0.6823 ± 0.0190
BIOT (Yang et al., 2023a)	3.2M	0.5281 ± 0.0225	0.5273 ± 0.0249	0.7492 ± 0.0082
LaBraM-Base	5.8M	0.6409 ± 0.0065	0.6637 ± 0.0093	0.8312 ± 0.0052
LaBraM-Large	46M	0.6581 ± 0.0156	0.6622 ± 0.0136	0.8315 ± 0.0040
LaBraM-Huge	369M	0.6616 ± 0.0170	0.6745 ± 0.0195	0.8329 ± 0.0086

Brain Foundation Models

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- Open-Vocabulary Brain-to-Text Translation
 - **EEG2Text (Liu *et al.*, BigData 2024)**

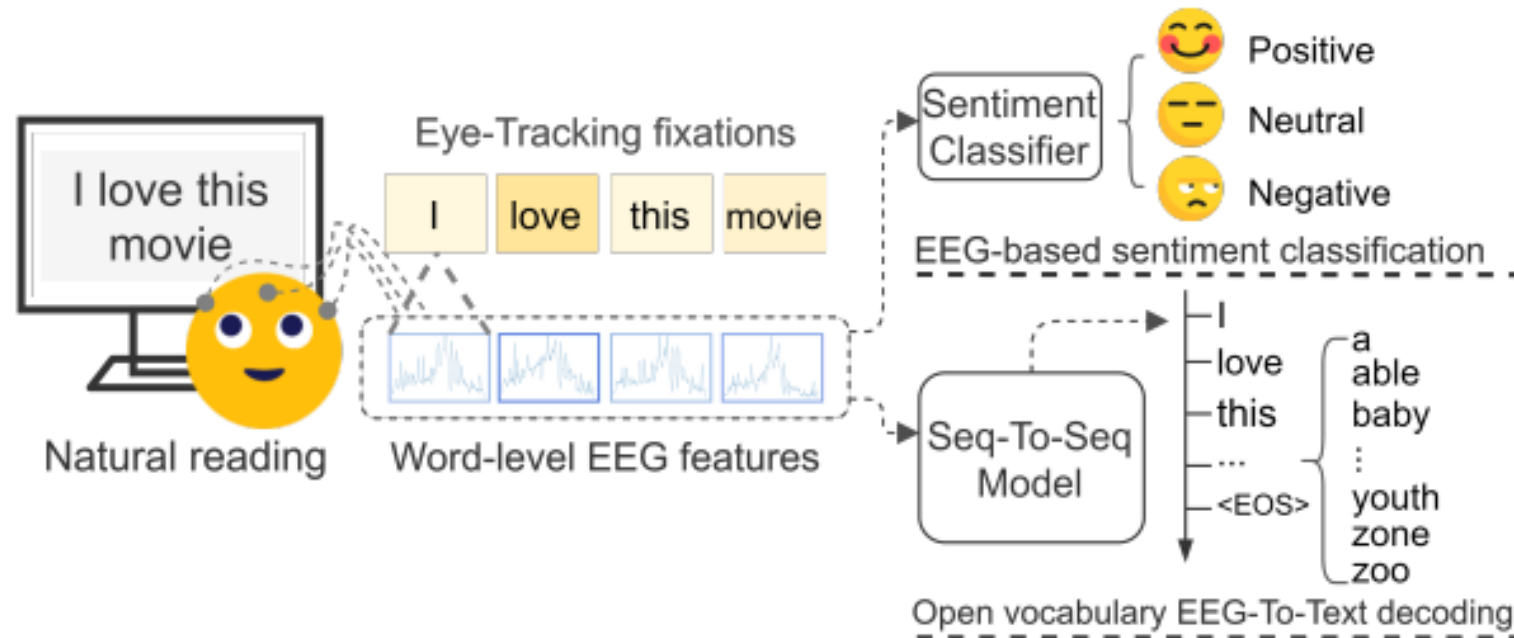
Open-Vocabulary EEG-to-Text Translation

- Language and Vision Understanding
- **Task:** decode any varied lengths of natural language sentences



Related Work

- Zhenhailong Wang and Heng Ji. Open vocabulary electroencephalography-to-text decoding and zero-shot sentiment classification. In Proceedings of the AAAI Conference on Artificial Intelligence, volume 36, pages 5350–5358, 2022.

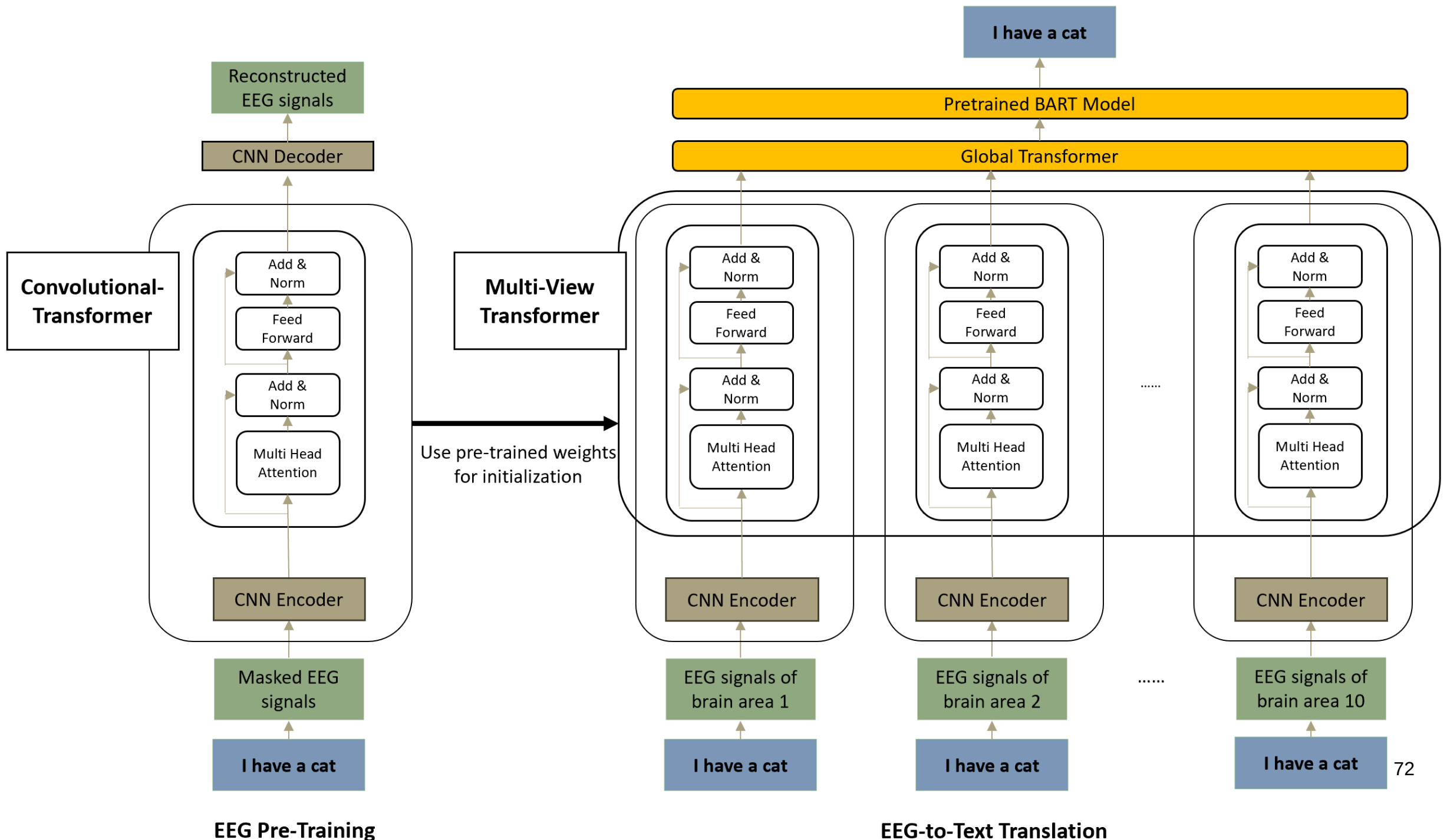


EEG Datasets

- **ZuCo** (Hollenstein *et al.*, *Scientific Data*, 2018)
 - EEG and eye-tracking data from 12 healthy adult, native English speakers, engaged in natural English text reading for 4 - 6 hours.
 - This dataset covers two standard reading tasks and a task-specific reading task, offering EEG and eye-tracking data for 21,629 words across 1,107 sentences and 154,173 fixations.
- **Image-EEG** (Gifford *et al.*, *NeuroImage*, 2022)
 - A large and rich dataset containing high temporal resolution EEG signals of images of objects on natural backgrounds.
 - The dataset included 10 participants, each performing 82,160 trials across 16,740 image conditions.

EEG2Text Methodology

- Sentence-Level EEG Signal Encoding
- Self-Supervised Pre-training
- Multi-View Transformer to Encode Brain Regions



EEG2Text Results

Table 3. Performance comparison of EEG2TEXT with baseline methods.

Methods	BLEU-N				ROUGE-1		
	N = 1	N = 2	N = 3	N = 4	F	P	R
Baseline (Wang & Ji, 2021)	0.401	0.231	0.125	0.068	0.301	0.317	0.288
DeWave (Duan et al., 2023)	0.413	0.241	0.139	0.082	0.288	0.337	0.306
EEG2TEXT (Convolutional Transformer)	0.400	0.236	0.137	0.082	0.325	0.361	0.297
EEG2TEXT (+ Pre-training)	0.445	0.274	0.175	0.117	0.341	0.383	0.310
EEG2TEXT (+ Multi-View Transformer)	0.452	0.291	0.197	0.141	0.342	0.369	0.320

EEG2Text Case Study

(1)	Ground Truth: It's not a particularly good film , but neither is it a monstrous one.
	Baseline Output: was a a bad good story, but it is it bad bad . one.
	EEG2TEXT output: 's a a great romantic movie , but it is not the disaster movie one.
(2)	Ground Truth: He won a Nobel Prize in Chemistry in 1928
	Baseline Output: was the Pulitzer Prize for Literature in 18.
	EEG2TEXT Output: won Nobel Prize in Chemistry for 1935 for
(3)	Ground Truth: The book was awarded the 1957 Pulitzer Prize for Biography .
	Baseline Output: first is published the Pulitzer Pulitzer Prize for Fictionography.
	EEG2TEXT Output: book is a the Pulitzer Prize for Biography and.

Table 7: Case study of the output sentences comparing EEG2TEXT and the baseline method (Wang and Ji, 2021).

Summary

- **Foundation models** has been built on massive public EEG data
 - Spatial-temporal sequences
 - High time resolution but low spatial resolution
 - Low signal-noise ratio
- **Future Directions:**
 - Better spatial-temporal signal encoding
 - Individualization of the EEG foundation models
 - Multi-modal brain foundation models

Biological Sequence Foundation Models

Xuan Wang

Assistant Professor

Department of Computer Science

Sanghani Center for AI and Data Analytics

Virginia Tech

Tutorial Outline

- **8:30 am – 9:00 am:** Introduction
- **9:00 am – 10:00 am:** Part I: Scientific Text
- **10:00 am – 10:30 am:** Part II: Brain Signals
- **10:30 am – 11:00 am: Break**
- **11:00 am – 12:00 am:** Part III: Biological Sequences 
- **12:00 pm – 12:30 pm:** Summary and Q&A

The Next Frontier for LLMs is Biology

- DNA Sequences
- RNA Sequences
- Protein Sequences

Modeling molecular biology does not need AGI: it doesn't require high-level planning, agency, or goals, and only has a limited need for combinatorics and algorithmic reasoning. Instead, modeling molecular biology requires what LLMs excel at: **learning the statistical properties of intricate, noisy sequential data to best predict such data from lossy representations.**

[Large Language Models in Molecular Biology](#)

The Next Frontier For Large Language Models Is Biology

Rob Toews Contributor

I write about the big picture of artificial intelligence.

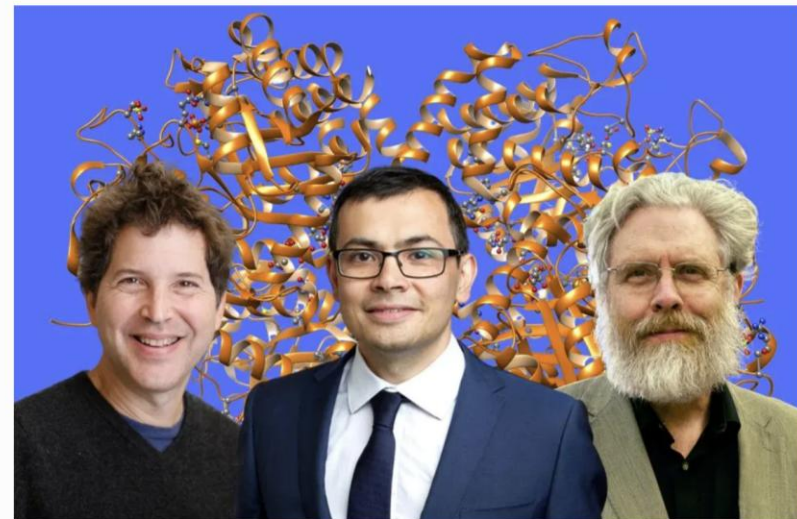
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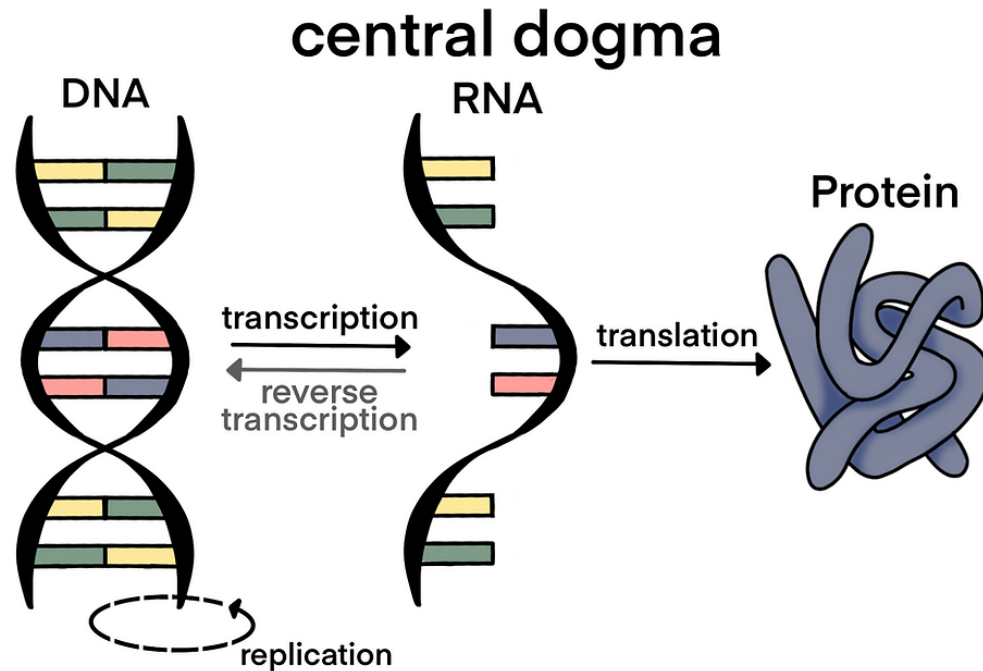


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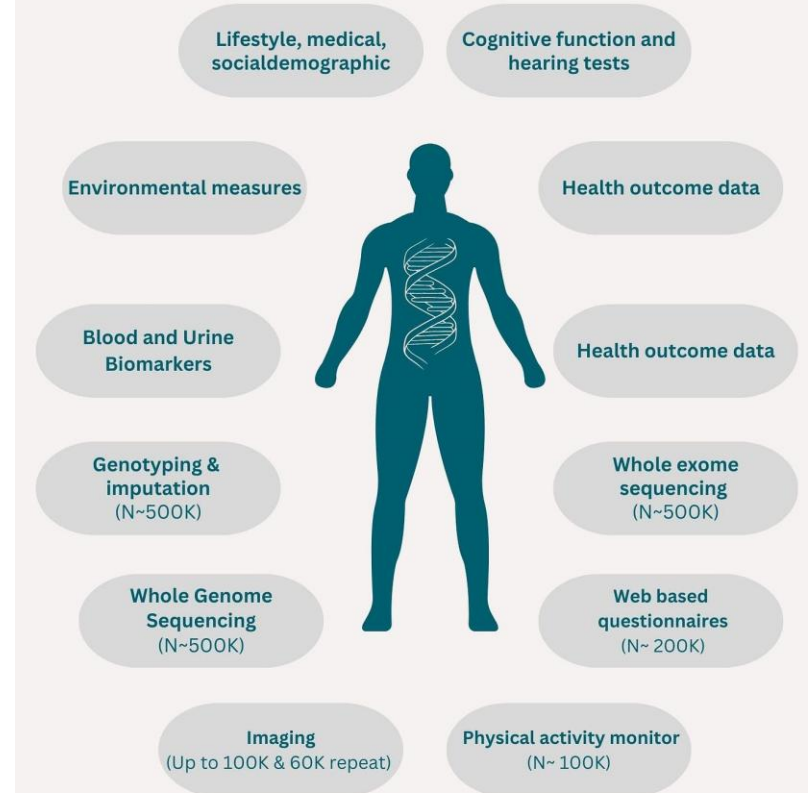
David Baker (University of Washington), Demis Hassabis (DeepMind) and George Church (Harvard) have helped pioneer the field of AI-driven protein design. [-] PHOTO SOURCE: U OF W, ROYAL SOCIETY, HARVARD

Future of LLM in Biology



- Multi-omics LLM
- LLM Explanation

Data on UK Biobank Participants



Outline

- **DNA, RNA, and protein LLMs**
- Single-cell foundation models

DNA LLMs

- **Models:** DNABERT (Ji *et al.*, 2021), GenSLMs (Zvyagin *et al.*, 2022), Nucleotide Transformer (Fishman *et al.*, 2023), GENA-LM (Dalla-Torre *et al.*, 2023), and HyenaDNA (Nguyen, *et al.*, 2023)
- **Tasks:** predicting regulatory elements for enhancers, promoters, epigenetic marks, and splice sites from DNA sequences

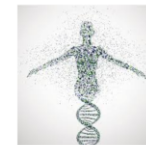
HyenaDNA: Long-Range Genomic Sequence Modeling at Single Nucleotide Resolution (Nguyen *et al.*, Neurips 2023)

- **Hyenadna**: context lengths of **1 million tokens** at **single nucleotide resolution**!
 - Human genome: **3.2 billion nucleotides**
 - Traditional LLM: **0.5-4K tokens (0.001%)**
 - Hyena: **convolutional LLM**

Table 4.1: **GenomicsBenchmark Performance.** Top-1 accuracy (%) benchmarks for pretrained HyenaDNA, Transformer and the previous SotA baseline (base).

DATASET	BASE	TRANSFORMER	HYENADNA
Mouse Enhancers	69.0	80.1	84.3 (+15.3)
Coding vs Intergenic	87.6	88.8	91.1 (+3.5)
Human vs Worm	93.0	95.6	96.5 (+3.5)
Human Enhancers Cohn	69.5	70.5	73.8 (+4.3)
Human Enhancers Ensembl	68.9	83.5	89.2 (+20.3)
Human Regulatory	93.3	91.5	93.8 (+0.5)
Human Nontata Promoters	84.6	87.7	96.6 (+12)
Human OCR Ensembl	68.0	73.0	80.9 (+12.9)

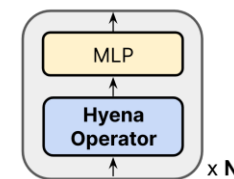
Sample from
Human Genome



A C G T A C G T

Next token prediction

A C G T A C G T



A C G T A C G ?

(Single Nucleotide Tokens)

Downstream tasks

Regulatory element
classification
(short)

Chromatin profile,
species classification
(long)

Explore in-context
learning

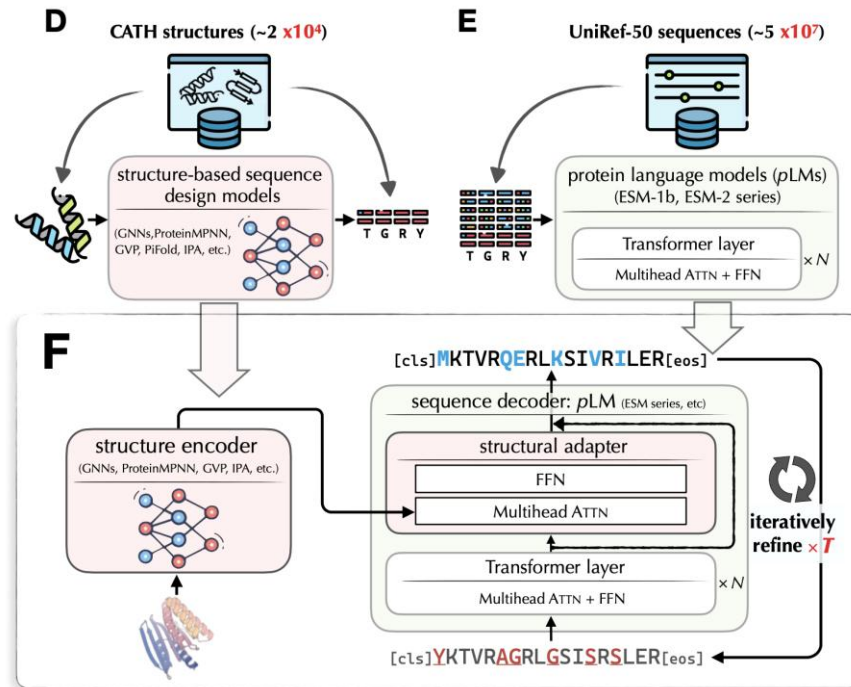
RNA LLMs

- **Models:** RNA-FM (Yamada and Hamada, 2022), BERT-RBP (Yang et al., 2022), scBERT (Chen et al., 2022), and RNA-MSM (Zhang et al., 2023)
- **Tasks:** RNA structure and function prediction, RNA-protein interaction prediction, and cell type annotation

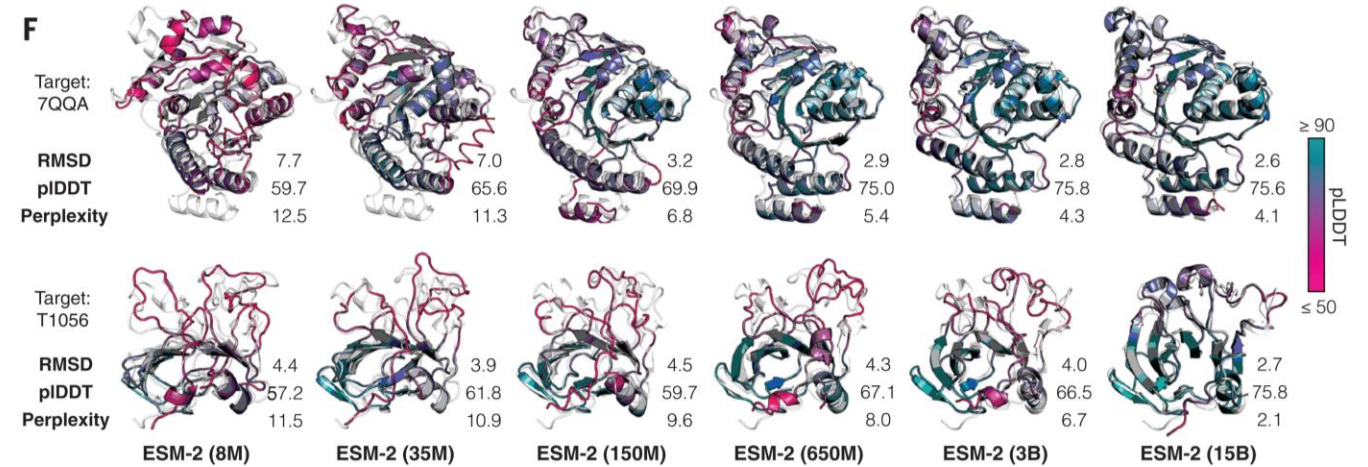
Protein LLMs

- **Models:** ProteinBERT (Brandes *et al.*, 2022), ProGen (Madani *et al.*, 2023), ESM-2 (Lin *et al.*, 2023), LM-Design (Zheng *et al.*, 2023), and ProtST (Xu *et al.*, 2023)
- **Tasks:** protein structure prediction and functional protein generation

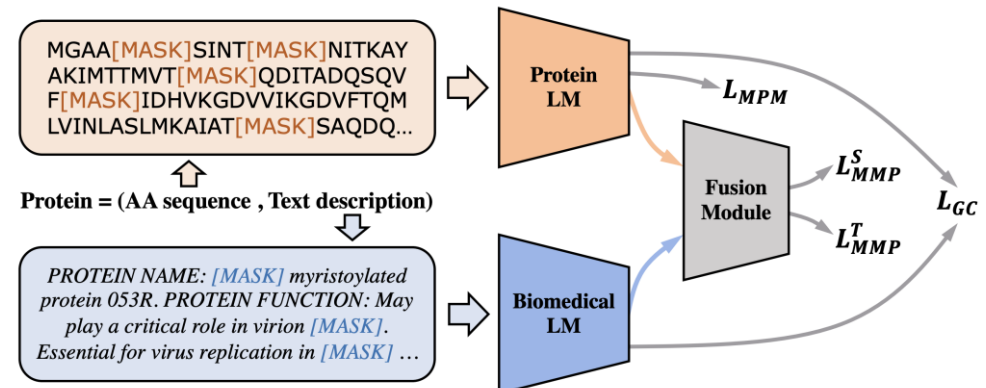
Protein Structure Prediction with LLMs



LM-Design (Zheng et al., ICML 2023)



(a) Multimodal Pre-training

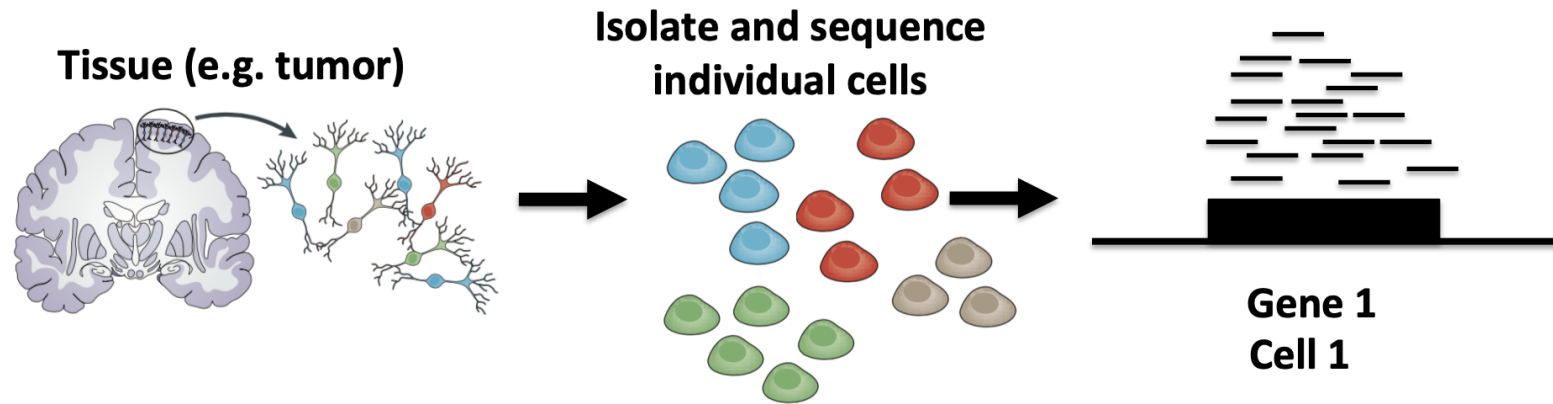


ProtST (Xu et al., ICML 2023)

Outline

- DNA, RNA, and protein LLMs
- **Single-cell foundation models**

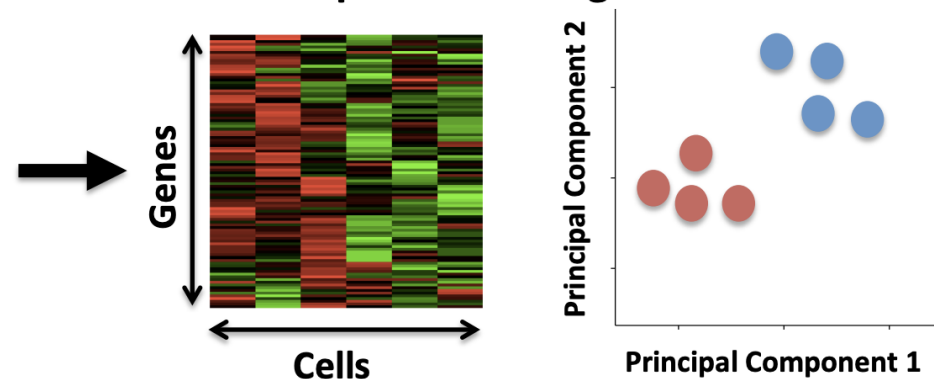
Single-cell RNA-Seq (scRNA-seq)



Read Counts

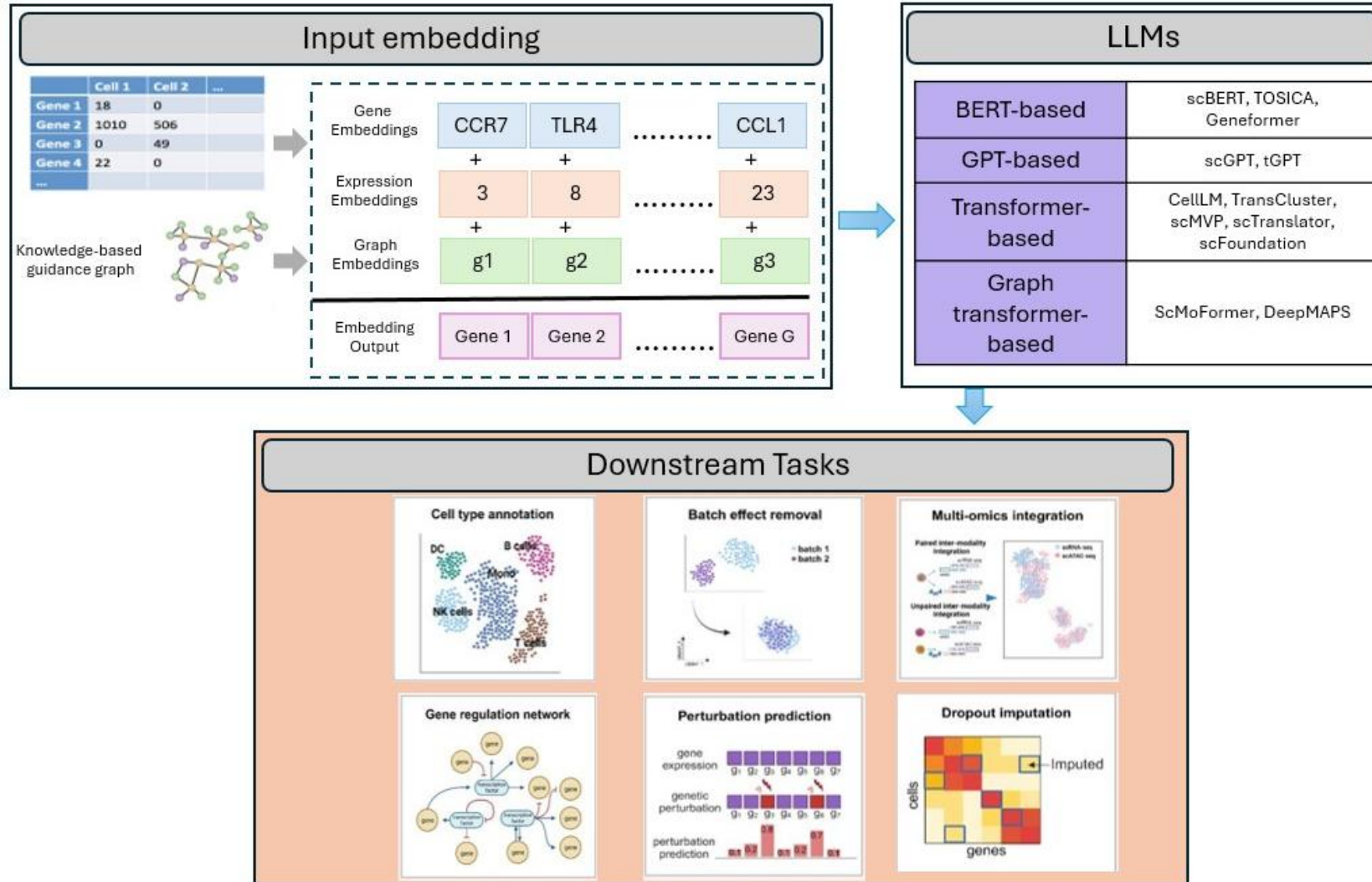
	Cell 1	Cell 2	...
Gene 1	18	0	
Gene 2	1010	506	
Gene 3	0	49	
Gene 4	22	0	
...			

Compare gene expression profiles of single cells



<https://github.com/stephaniehicks/singlecellnano2017?tab=readme-ov-file> ('Single-Cell Sequencing: experimental design, analysis, and practical applications' at Harvard Medical School)

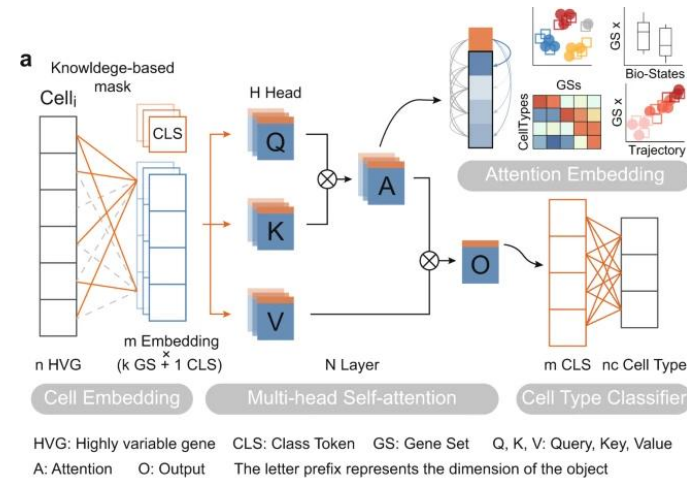
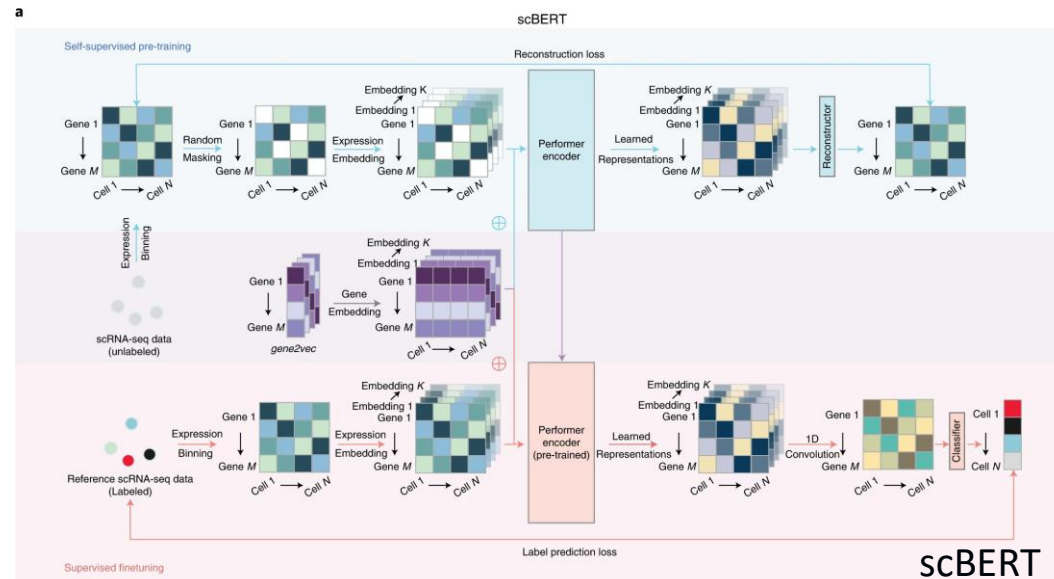
Single-Cell Foundation Models



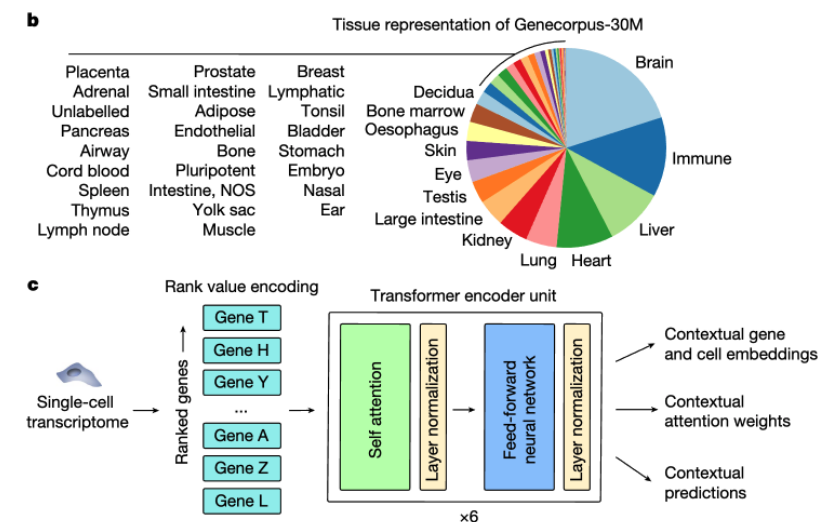
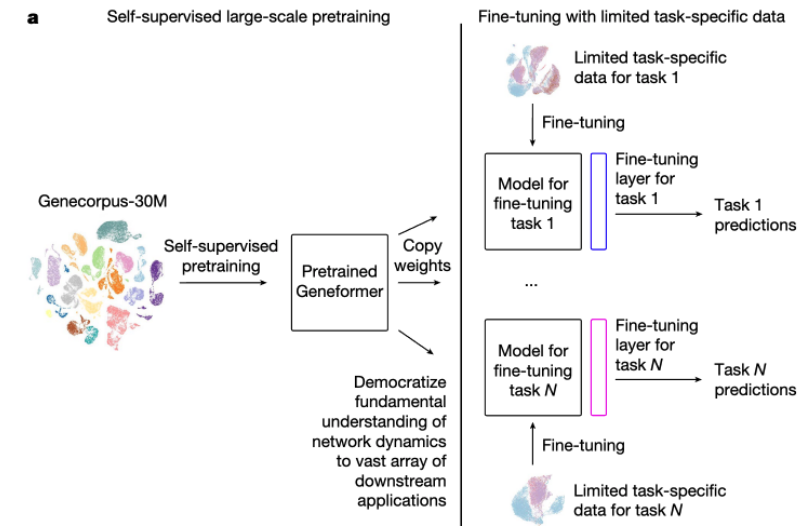
BERT-based Architectures

Model	Summary
scBERT (Yang <i>et al.</i> , Nat Mach Intell 2022)	scBERT is a pre-training-based single-cell LLM focusing on cell-type prediction. It is based on Performer with gene embeddings initialized by Gene2vec. It has six self-attention blocks. The default fine-tuning process of scBERT on downstream datasets is freezing the penultimate layer.
TOSICA (Jiawei <i>et al.</i> , Nat Commun 2023)	TOSICA (Transformer for One-Stop Interpretable Cell-type Annotation) transforms gene expression levels into tokens representing biological pathways and processes. It extends beyond cell type annotation to include dynamic trajectory discovery and cross-platform data integration in downstream analyses.
Geneformer (Christina <i>et al.</i> , Nature 2023)	Geneformer is a single-cell LLM using transfer learning to predict cell types and gene functions. The Geneformer tokenization step is done based on ranking gene expression values in single cells following scaling across the whole training dataset. Cells are represented as token strings with genes rankings as tokens. Geneformer is used for the Cell-type Annotation task and the Gene Function Prediction task.

BERT-Based Architectures



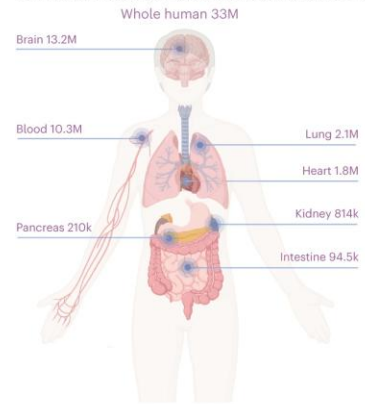
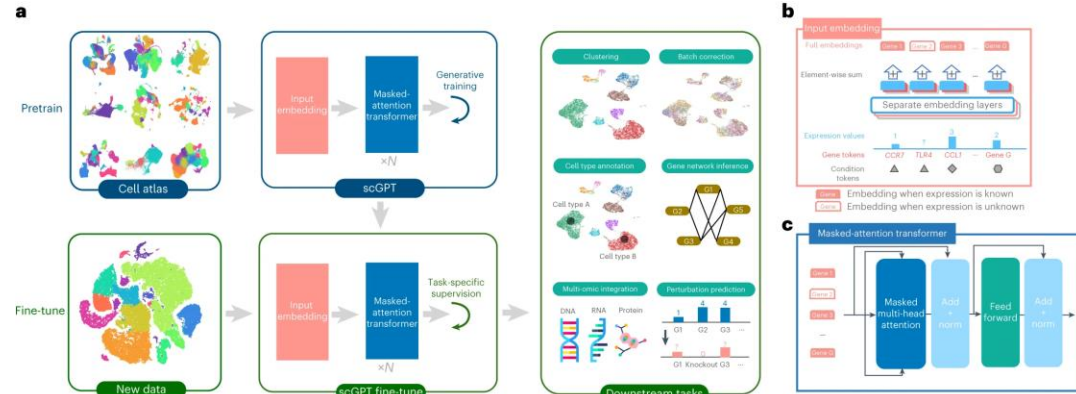
TOSICA



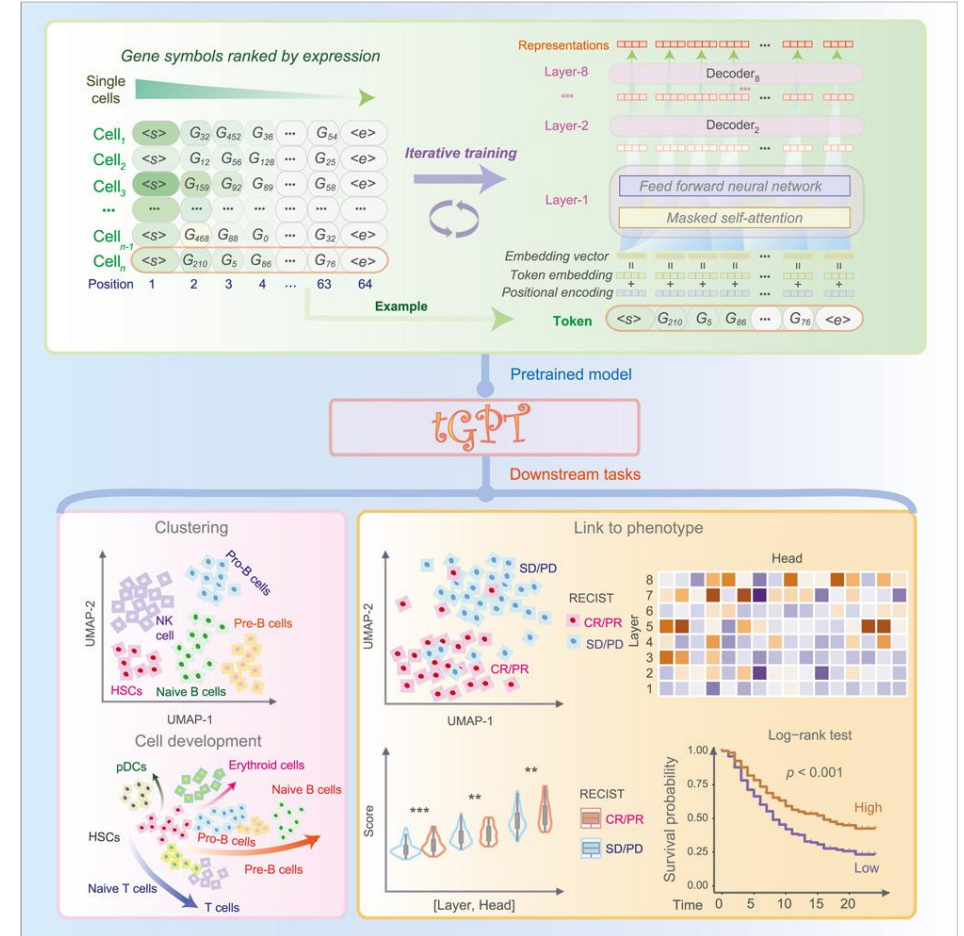
GPT-Based Architectures

Model	Summary
scGPT (Cui <i>et al.</i> , Nat Methods 2024)	scGPT adapts the Transformer architecture to enable simultaneous learning of cell and gene representations, allowing for a comprehensive understanding of cellular characteristics based on gene expression. It is the first single-cell foundation model built through generative pre-training on over 33 million cells. It can be fine-tuned for various downstream tasks in single-cell analysis, such as multi-batch integration, multi-omic integration, cell-type annotation, genetic perturbation prediction, and gene network inference.
tGPT (Hongru <i>et al.</i> , iScience '23)	Like other GPT-based models for scRNA-seq, tGPT uses a Transformer architecture to capture gene expression patterns and cell-cell interactions. To handle the unique characteristics of gene expression data, tGPT sorts genes from high to low expression to determine the position information, as gene expression differs from natural language data. It can be used for various downstream tasks in single-cell analysis, such as cell type annotation, trajectory inference, and gene regulatory network inference.

GPT-Based Architectures



scGPT

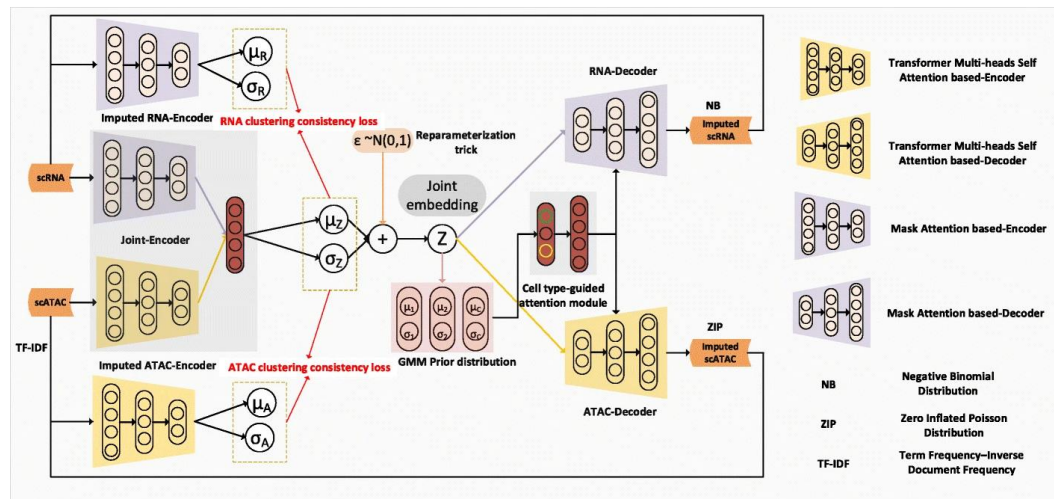
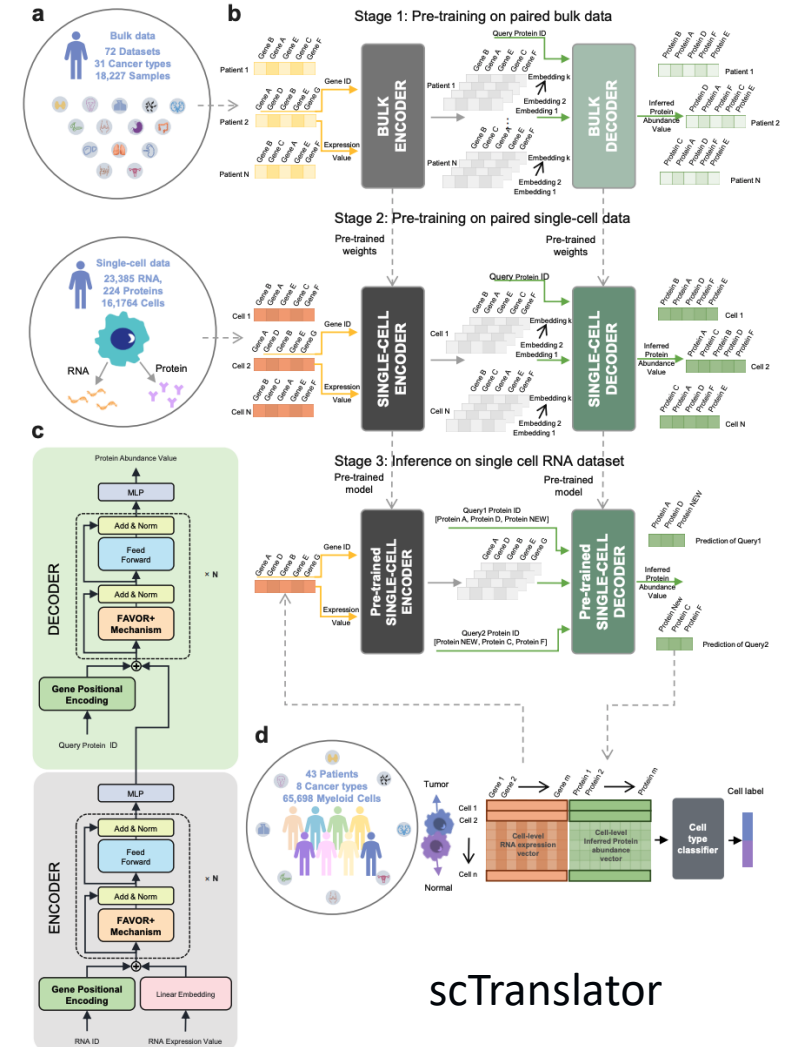
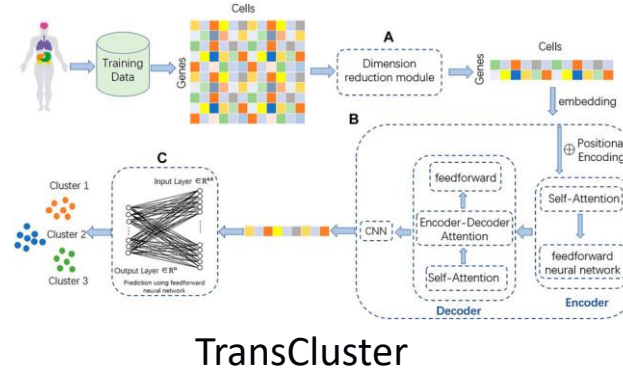
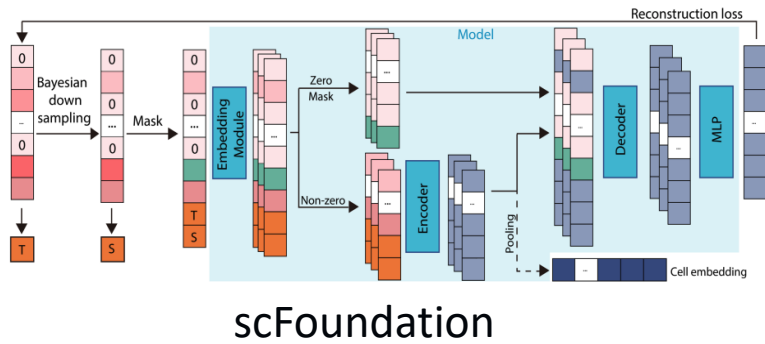


tGPT

Transformer-Based Architectures

Model	Summary
scMVP (Li <i>et al.</i> , Genome Biol 2022)	scMVP (Multi-View Profiler) is a deep generative model designed for multi-view profiling of scRNA-seq and scATAC-seq data. It has two sub-networks - one for scRNA-seq data using a masked attention-based encoder-decoder, and one for scATAC-seq data using a multi-head self-attention-based Transformer. Aims to integrate scRNA-seq and scATAC-seq data into a common latent space for tasks like cell embedding, clustering, and imputation.
scFoundation (Minsheng <i>et al.</i> , Nat Methods 2024)	scFoundation is a large-scale pre-trained model with 100M parameters. Trained on over 50 million human single-cell transcriptomics data, making it the largest model in terms of trainable parameters, gene dimensionality, and number of cells used for pre-training. SOTA performance on gene expression enhancement, tissue drug response prediction, single-cell drug response classification, and perturbation prediction.
scTranslator (Linjing <i>et al.</i> bioRxiv '23)	scTranslator is pre-trained on a large-scale dataset of single-cell transcriptome and proteome data, allowing it to capture the complex relationships between gene expression and protein abundance. Can be fine-tuned for various downstream tasks, such as enhancing single-cell proteome prediction, improving tissue drug response prediction, and facilitating drug response classification and perturbation prediction.

Transformer-Based Architectures

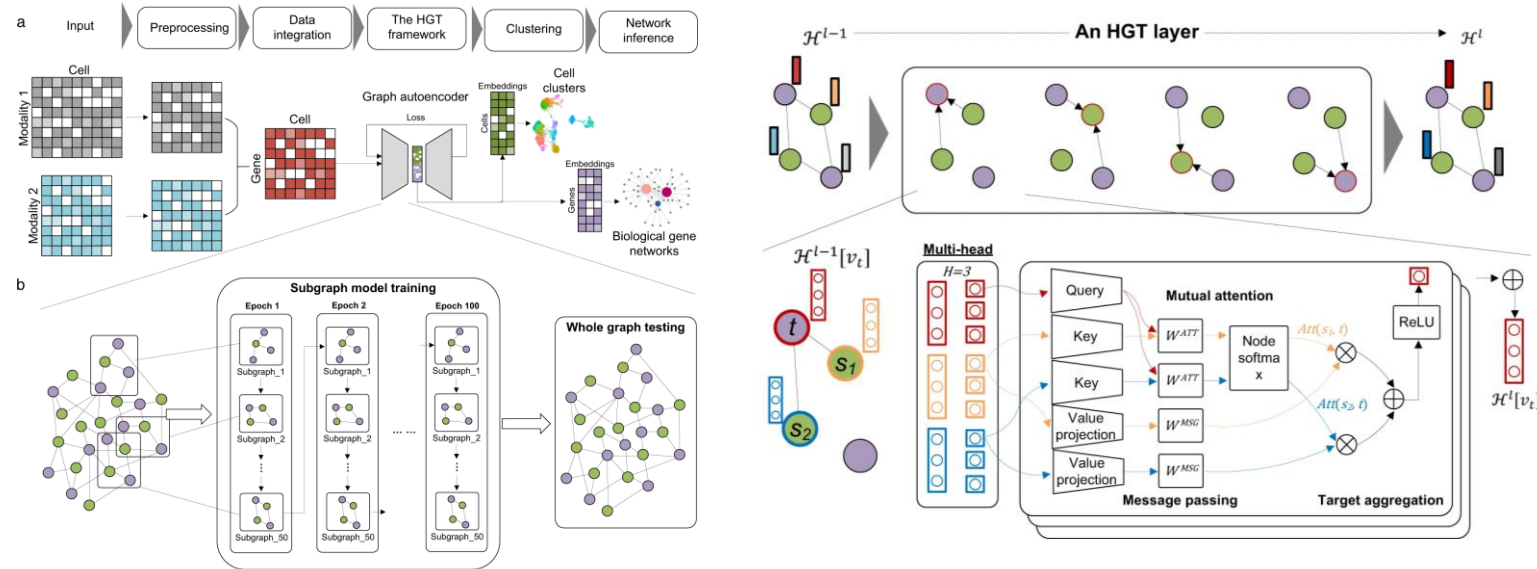


Graph Transformer-Based Architectures

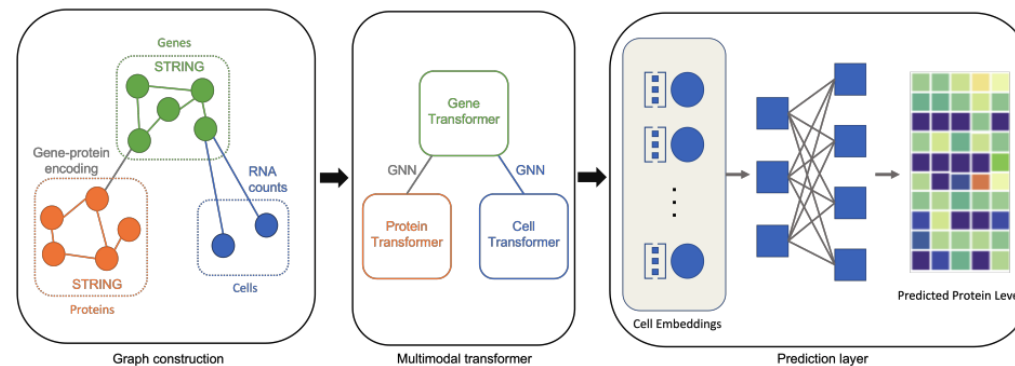
Model	Summary
scMoFormer (Tang et al. CIKM '23)	Designed for multi-modal prediction on single-cell data, specifically integrating scRNA-seq and scATAC-seq data. It achieves this by constructing a multi-modal heterogeneous graph, with separate subgraphs for genes, proteins, and cells. The gene-gene and protein-protein subgraphs leverage prior biological knowledge from the STRING database, while the cell-gene subgraph is constructed based on normalized gene expression data. It employs modality-specific transformers for cells, genes, and proteins, combining transformer modules GNNs to capture unique characteristics of each modality. To integrate information from modality-specific transformers, scMoFormer uses message passing GNNs to propagate embeddings across different node types.
DeepMAPS (Anjun et al. Nat Commun '23)	Leverages heterogeneous graph transformers to integrate and analyze multi-modal single-cell data, particularly single-cell RNA-seq and ATAC-seq. Outperform other SOTA methods in tasks like cell type identification, cell-cell communication inference, and gene regulatory network reconstruction, making it a powerful tool for comprehensive single-cell multi-omics analysis.

Graph Transformer-Based Architectures

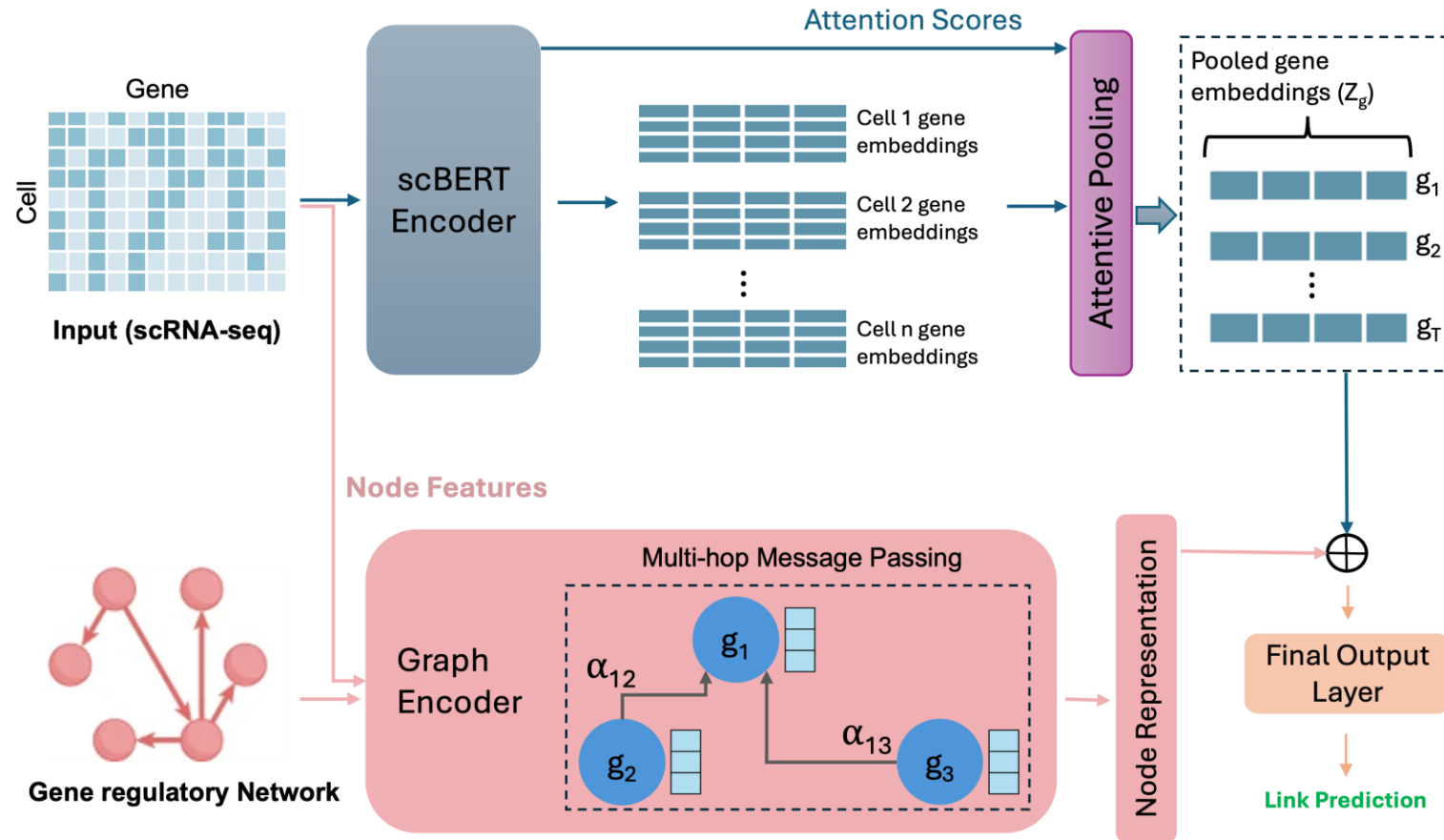
scMoFormer



DeepMaps



Gene Regulatory Network Inference from Pre-trained Single-Cell Transcriptomics Transformer with Joint Graph Learning



Gene Regulatory Network Inference from Pre-trained Single-Cell Transcriptomics Transformer with Joint Graph Learning

	Dataset	TFs + 500 genes					TFs + 1000 genes				
		Methods					Methods				
		<i>scTransNet (Ours)</i>	<i>GNNLink</i>	<i>GENELink</i>	<i>GNE</i>	<i>CNNC</i>	<i>scTransNet (Ours)</i>	<i>GNNLink</i>	<i>GENELink</i>	<i>GNE</i>	<i>CNNC</i>
AUROC	hESC	0.88	0.85	0.79	0.67	0.68	0.87	0.8	0.78	0.68	0.72
	hHEP	0.87	0.82	0.83	0.8	0.64	0.87	0.84	0.85	0.81	0.66
	mDC	0.77	0.7	0.67	0.52	0.54	0.77	0.78	0.73	0.52	0.58
	mESC	0.91	0.84	0.8	0.81	0.73	0.92	0.84	0.82	0.82	0.73
	mHSC-E	0.89	0.83	0.79	0.82	0.67	0.92	0.87	0.84	0.84	0.72
	mHSC-GM	0.91	0.89	0.84	0.83	0.69	0.92	0.92	0.87	0.84	0.69
	mHSC-L	0.87	0.84	0.81	0.77	0.67	0.86	0.86	0.81	0.77	0.62

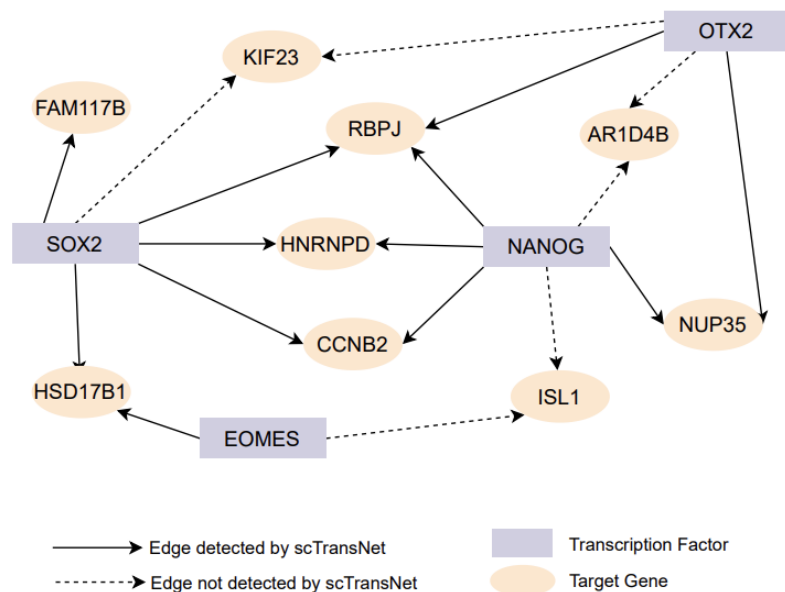
Ground Truth Network: **cell-type-specific ChIP-seq**

Gene Regulatory Network Inference from Pre-trained Single-Cell Transcriptomics Transformer with Joint Graph Learning

	Dataset	TFs + 500 genes					TFs + 1000 genes				
		Methods					Methods				
		<i>scTransNet (Ours)</i>	<i>GNNLink</i>	<i>GENELink</i>	<i>GNE</i>	<i>CNNC</i>	<i>scTransNet (Ours)</i>	<i>GNNLink</i>	<i>GENELink</i>	<i>GNE</i>	<i>CNNC</i>
AUPRC	hESC	0.6	0.52	0.48	0.34	0.25	0.59	0.51	0.5	0.34	0.27
	hHEP	0.78	0.75	0.69	0.65	0.46	0.78	0.78	0.7	0.66	0.49
	mDC	0.12	0.25	0.11	0.06	0.06	0.14	0.21	0.12	0.05	0.05
	mESC	0.81	0.76	0.75	0.64	0.48	0.82	0.78	0.75	0.65	0.5
	mHSC-E	0.92	0.88	0.88	0.8	0.74	0.94	0.93	0.89	0.81	0.77
	mHSC-GM	0.92	0.89	0.88	0.78	0.68	0.93	0.93	0.91	0.81	0.73
	mHSC-L	0.86	0.85	0.81	0.7	0.64	0.84	0.86	0.82	0.68	0.56

Ground Truth Network: **cell-type-specific ChIP-seq**

Gene Regulatory Network Inference from Pre-trained Single-Cell Transcriptomics Transformer with Joint Graph Learning



Dataset	w/o GNN encoder		w/o scBERT encoder		w/o Attentive Pooling		scTransNet	
	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
hESC	0.842	0.544	0.853	0.572	0.860	0.569	0.880	0.595
hHEP	0.830	0.725	0.854	0.753	0.862	0.683	0.870	0.780

Comparison of average AUROC and AUPRC evaluation metrics on human benchmark datasets, validating the roles of the GNN encoder, scBERT encoder, and Attentive Pooling using the cell-type-specific ChIP-seq network for our proposed method.

GRN prediction performance of scTransNet on a partial ground truth subgraph.

Summary

- **Extensive LLMs** have been built across DNA, RNA, and protein sequences for scientific discoveries
- **Future Directions:**
 - Knowledge-driven single-cell foundation models
 - Integrate network, text, and images with biological sequence foundation models

Summary and Q&A

Xuan Wang

Assistant Professor

Department of Computer Science

Sanghani Center for AI and Data Analytics

Virginia Tech

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- **12:00 pm – 12:30 pm:** Summary and Q&A 

Tutorial Summary

- **Part I: Scientific Text**

- Scientific large language models
- Challenges in complex tasks, multi-modality, and trustworthiness

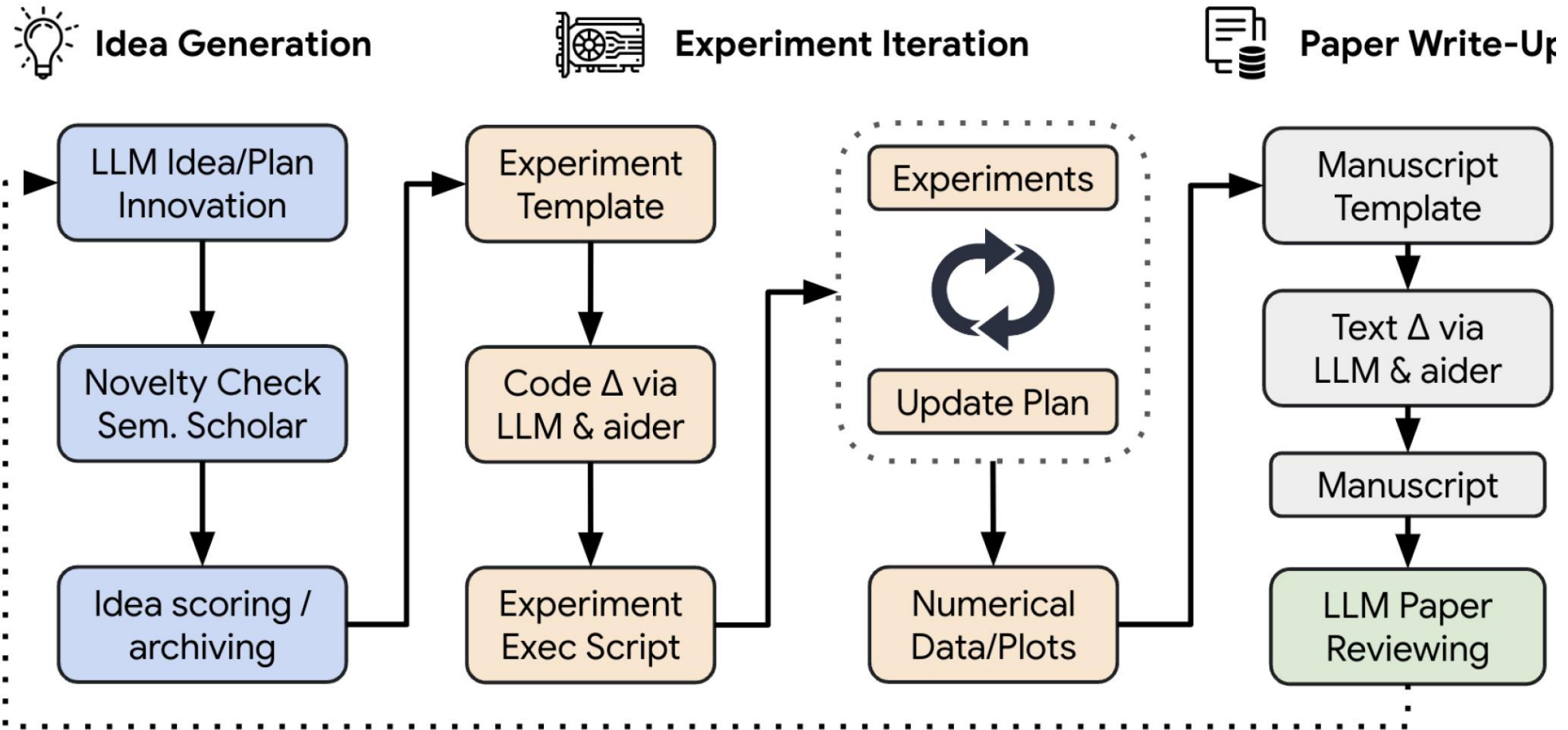
- **Part II: Brain Signals**

- Brain large language models
- Application in brain-to-text translation

- **Part III: Biological Sequences**

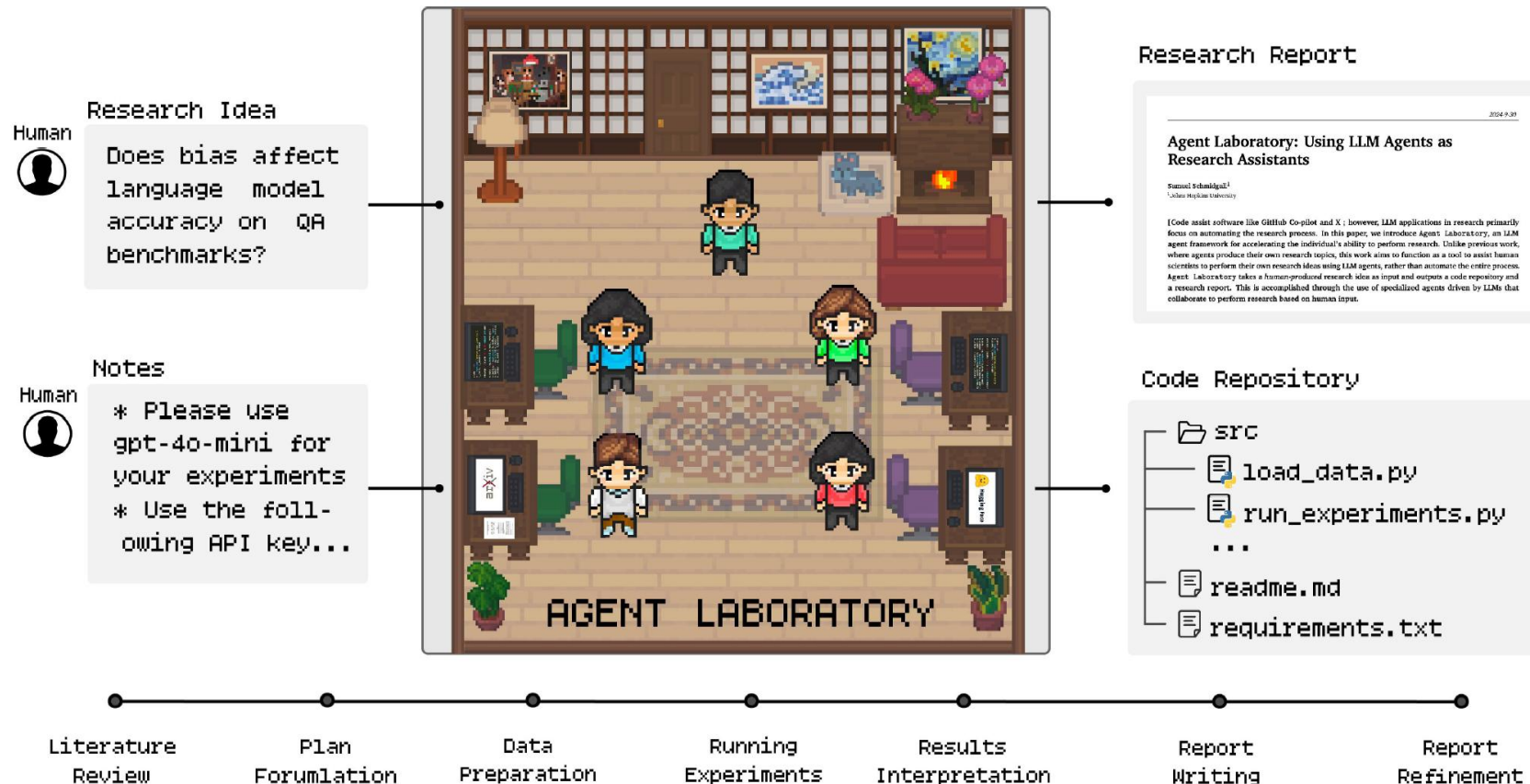
- Biological language models
- Multi-modal learning with text

The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery (Lu *et al.*, 2024)



Agent Laboratory: Using LLM Agents as Research Assistants (Schmidgall et al., 2025)

<https://AgentLaboratory.github.io>



Thank you !
Q&A